

Mind the rank! Finite-resource effects in quantum linear regression

Surprises in Quantum Linear Regression with Finite Resources

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Quantum linear regression offers a non-variational approach to quantum machine learning, yet its behavior under realistic measurement constraints remains poorly understood. Focusing on quantum extreme learning machines, we analyze their performance when the feature probabilities of each training state are estimated from only N measurements. A key observation is that the ideal probability matrix has rank at most the square of the Hilbert space dimension of the measured states, whereas its finite-shot counterpart is full rank with high probability. As a result, the pseudoinverse diverges for large N , and in particular the finite-shot pseudoinverse does not converge to the ideal Moore–Penrose pseudoinverse. Leveraging tools from random matrix theory and concentration inequalities, we show that these divergences can be controlled and derive the leading behavior of the test error, up to corrections that are small with high probability. We prove that the test error contains a bias term scaling as N^{-2} , while fluctuations over realizations of the training shot noise are characterized by a training variance scaling as $(Nn_{\text{tr}})^{-1}$. Although the additional singular directions are projected out when the training targets and test probabilities are exact, we show that noise in the targets features can couple to these directions and amplify the variance arising from finite test statistics. These results establish a resource tradeoff between the dataset size and the number of measurements per state, and provide a baseline for resource-aware training and regularization strategies.

A central question in quantum machine learning (QML) [1, 2] is how the resources required for training and inference scale with the desired prediction accuracy. In the context of variational quantum algorithms [3], phenomena such as barren plateaus [4–7] and, more broadly, exponential concentration [8–12], have been shown to severely hinder trainability and limit the potential practical advantage of models based on parametrized quantum circuits. By contrast, comparatively less attention has been devoted to the resource requirements of non-variational QML architectures, in which the quantum processing is fixed rather than optimized during training. A common structure underlying many such approaches is that predictions are constructed as linear combinations of expectation values or measurement probabilities, or, equivalently, from linear models in quantum feature space [13–15]. Prominent examples include quantum reservoir computing [16, 17], quantum extreme learning machines (QELMs) [15, 18–22], and quantum kernel methods [13, 23–28]. The same structure underlies the training of recently proposed quantum Gaussian processes [29, 30]. The linearity of the relation between input density matrices and measured observables implies that the achievable targets are representable by some effective observable [15]. In these models the training process itself is particularly easy, and amounts

to solving a classical linear regression problem on data obtained from the quantum experiment. Nonetheless, as we will show, the quantum nature of the data used at training and inference time in these models makes for some surprising departures from what one might naively expect from classical linear regression theory. ~~By contrast, comparatively less attention has been devoted to quantum linear regression models, such as quantum extreme learning machines (QELMs) [15, 18–22], quantum fidelity kernels [24–28] and quantum Gaussian processes [29, 31]. In these approaches, the map from the input quantum states to the targets is assumed to be linear, $y^T = WP$, where W is a weight matrix to be learned and P is a matrix collecting the expectation values of a family of operators μ over the training states p . The linearity implies that the achievable targets are likewise expectation values of a fixed observable: $y_i = \text{tr}(\mathcal{O}p_i)$ [15]. In particular, the canonical minimum norm weights are obtained with the Moore–Penrose pseudoinverse [32] as $W = yP^\dagger$. In the QELM setting, the operators μ are the elements of a POVM and P collects the probabilities to observe any measurement outcome on each input state.~~

We study the general setting of *quantum linear regression* problems, by which we mean any setting in which inference consists of taking a linear combination of estimated expectation values measured on the output of a fixed quantum device. The resulting predictor thus takes the form $f_w(\sigma) = \sum_b w_b \text{tr}(\mu_b \sigma)$, where σ is the input states at inference time, $\{\mu_b\}$ is a fixed col-

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lection of observables characterising the protocol, and w_b are the coefficients learnt from the training. For QELMs, the operators μ_b are the elements of some effective POVM, whereas for quantum fidelity kernels they are the training states themselves. Training consists of estimating the expectation values of the observables μ_b on a set of n_{tr} training states using N measurements per state, and collecting the resulting data into a finite-shot feature matrix $\hat{P}_N \in \mathbb{R}^{n_{\text{out}} \times n_{\text{tr}}}$. In the infinite-shot limit, $\hat{P}_N \rightarrow P$, where P denotes the corresponding ideal feature matrix. The resulting linear regression problem thus takes the form: given $\hat{P}_N$ and the matrix Y collecting the training features, find W such that $Y \simeq W\hat{P}_N$. The associated minimum-norm least-squares estimator is then $W = Y\hat{P}_N^+$. Crucially, the ideal and empirical feature matrices have different rank structure, which makes for some surprising departures from what one might expect from classical linear regression theory. In quantum models, P is often rank deficient, with $\text{rank}(P) \leq d^2$ if d is the Hilbert-space dimension of the input states. However, stochastic fluctuations make its noisy counterpart $\hat{P}_N$, which is the one available in any realistic implementation, full rank with high probability. Consequently, even though $\hat{P}_N \rightarrow P$ as $N \rightarrow \infty$, the empirical pseudoinverse $\hat{P}_N^+$ does not generically converge to P^+ [33, 34]. ~~The key distinction between classical and quantum settings lies in the structure of the feature matrix. In the idealized limit of infinite resources, P is known exactly, and is generally rank-deficient: its rank is upper bounded by the square of the Hilbert space dimension of the input states. However, in practice, P can only be estimated with finite precision, using a finite number of measurements N for each training state, thus in a realistic scenario we only have access to the noisy counterpart of the probability matrix, which we shall denote with $\hat{P}_N$. In particular, stochastic noise makes $\hat{P}_N$ be full rank with high probability. This rank increase makes the empirical pseudoinverse $\hat{P}_N^+$ not converge to P^+ as $N \rightarrow \infty$ [33, 34], which in turn makes analyzing the prediction performances harder, as these are calculated through the noisy estimator $\hat{W} = Y\hat{P}_N^+$. This discontinuity of the pseudoinverse is a key technical obstacle which our results seek to overcome.~~

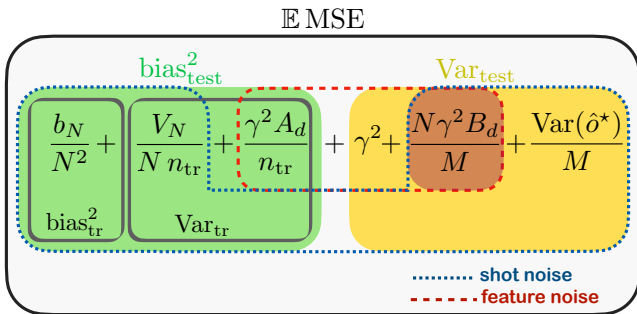


Figure 1.

While the statistical properties of classical linear regression have been studied extensively [35–44], particularly in high-dimensional regimes where both the feature dimension and the dataset size grow large, much less is known rigorously when the feature matrix itself is subject to the sampling noise typical of quantum models [45–47]. Here, we address this gap by initiating a systematic study of the effects of shot noise on quantum linear regression, focusing in particular on the QELM paradigm. We develop two complementary approaches to overcome the technical difficulties associated with the rank discontinuity: first, we construct a perturbative expansion adapted to rank-changing fluctuations, which isolates the singular part of the empirical pseudoinverse and identifies the blocks that affect physical predictions. Although the pseudoinverse contains terms that diverge as $N \rightarrow \infty$, we show that these singular contributions do not affect physical predictions when the training targets are known exactly. We then derive, within a homogeneous Gaussian model of the training shot noise, the scaling of the average mean-squared prediction error (MSE) on N and n_{tr} , identifying two main regimes:

- for $N \gtrsim n_{\text{tr}}$, the MSE is dominated by stochastic fluctuations due to finite training statistics and scales as $(N n_{\text{tr}})^{-1}$;
- for $N \lesssim n_{\text{tr}}$, these fluctuations become subleading and the MSE is dominated by a quantum-induced systematic bias, arising from the rank increase and associated ill-conditioning produced by finite-shot noise, and scales as N^{-2} .

Consequently, at fixed N , increasing the number of training states eventually ceases to improve the prediction error: beyond $n_{\text{tr}} \sim N$, the MSE saturates at a floor set by the finite measurement statistics available for each training state. These results highlight a genuine tradeoff between dataset size and per-state measurement statistics. Thus, unlike in standard linear regression intuition, increasing the size of the training set does not by itself improve predictive performance unless the per-state measurement budget is increased accordingly. We support our analytical results with ample numerical evidence showing excellent agreement with the theoretical estimates.

We complement this structural analysis with a rigorous treatment of multinomial shot noise. In particular, we derive an exact decomposition of the prediction error into a deterministic bias, a centered fluctuation, and a non-linear remainder. Using standard concentration and moment bounds [48, 49], we show that the remainder is subleading with high probability in most regimes of interest. This analysis allows us to prove the scaling results unconditionally, without relying on Gaussian noise models, and further enables us to characterize the behaviour of the MSE even in the very low statistics regime.

We then proceed to also study the effect of noise on the training targets themselves, modelling scenarios in which preparation errors cause the training states to not be precisely characterized. We find that this noise causes unexpected interactions with the training and test statistics, and creates a further contributions to the MSE that scale as N/M . This means that increasing the training statistics without correspondingly increasing the statistics used at inference time can counterintuitively worsen performances. ~~The singular directions created by shot noise do not affect predictions when the training targets and test probabilities lie exactly in the physical subspaces determined by the ideal probability matrix. This protection is lost, however, when the training targets are themselves noisy. In that case, target fluctuations can couple to the noise-induced directions and amplify the finite test statistics variance. We characterize this mechanism in a Gaussian reference model and show, both analytically and numerically, that it can be mitigated by increasing either the number of test measurements or the size of the training set.~~

Overall, by combining ideas from quantum machine learning and random matrix theory, our results indicate that familiar intuitions from classical learning must be revised to account for constraints imposed by the geometry of quantum state space. More broadly, our work highlights that finite statistics can qualitatively alter conclusions drawn from idealized, infinite-sample analyses, motivating a more statistics-aware approach to quantum machine learning.

The rest of the paper is organized as follows. [Section I](#) reviews classical linear regression, and [section II](#) introduces its quantum counterpart. [Section III](#) identifies the rank-changing instability, and [section IV](#) gives a complementary structural analysis of its Schur-complement mechanism. [Section V](#) recovers the resource laws in a homogeneous Gaussian reference model. The main bias-variance result and its resource scaling are presented in [section VI](#), and finally [section VII](#) discusses the amplification mechanism that appears when the training targets are noisy, before we conclude with the resulting resource-allocation perspective in [section VIII](#).

I. CLASSICAL LINEAR REGRESSION THEORY

In the standard treatment of linear regression [\[43, 50, 51\]](#), the relation between an input $x \in \mathbb{R}^d$ and a response $y \in \mathbb{R}$ is assumed to be linear: $y = \beta^T x + \eta$, where $\beta \in \mathbb{R}^d$ is the regression vector we want to estimate, and η is stochastic noise. We assume that the sample noises are conditionally independent and homoscedastic, $\mathbb{E}[\eta | x] = 0$ and $\mathbb{E}[\eta^2 | x] = \sigma^2$. Notably, this setting accounts for noise in y but not in x . The input-output pairs (x, y) are drawn from a distribu-

tion $\mathcal{D}$. We quantify predictive performance through the population mean-squared error

$$\text{MSE}(\hat{\beta}) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\left(y - \hat{\beta}^T x \right)^2 \right]. \quad (1)$$

For any estimator $\hat{\beta}$ this risk decomposes as

$$\text{MSE}(\hat{\beta}) = \sigma^2 + \|\beta - \hat{\beta}\|_{\Sigma}^2, \quad (2)$$

where $\Sigma = \mathbb{E}_{\mathcal{D}}[xx^T]$ is the second moment matrix, and $\|v\|_{\Sigma}^2 \equiv v^T \Sigma v$ denotes the induced quadratic form (a norm when Σ is positive definite). The first term in [eq. \(2\)](#) is the irreducible noise, whereas the second is the excess risk. In practice, the expectation in [eq. \(1\)](#) is estimated on a finite test set sampled from $\mathcal{D}$.

Direct minimization of the population risk is usually impossible. The standard approach is to sample a training set $\mathcal{E}_{\text{tr}} = \{(x_i, y_i)\}_{i=1}^{n_{\text{tr}}}$ from $\mathcal{D}$ and minimize the *empirical risk*

$$R_{\text{tr}}(\mathbf{b}) = \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \left(y_i - \mathbf{b}^T x_i \right)^2. \quad (3)$$

This least-squares problem can be solved explicitly. Let $\mathbf{X} \in \mathbb{R}^{d \times n_{\text{tr}}}$ be the training matrix whose i -th column is x_i , and let $\mathbf{y}, \boldsymbol{\eta} \in \mathbb{R}^{n_{\text{tr}}}$ collect the training outputs and noise variables, respectively. Up to the factor $1/n_{\text{tr}}$, the empirical risk is $\|\mathbf{y}^T - \mathbf{b}^T \mathbf{X}\|^2$, whose canonical minimum-norm solution is

$$\hat{\beta}_{\text{tr}}^T = \mathbf{y}^T \mathbf{X}^+ = \beta^T \mathbf{X} \mathbf{X}^+ + \boldsymbol{\eta}^T \mathbf{X}^+, \quad (4)$$

where $\mathbf{X}^+$ denotes the Moore–Penrose pseudoinverse of a matrix $\mathbf{X}$.

The MSE in [eq. \(2\)](#) depends on the training set $\mathcal{E}_{\text{tr}}$ and is therefore random with respect to both $\mathbf{X}$ and $\boldsymbol{\eta}$. At fixed $\mathbf{X}$, its average over the noise decomposes as

$$\mathbb{E}_{\boldsymbol{\eta}} \|\hat{\beta}_{\text{tr}} - \beta\|_{\Sigma}^2 = \text{bias}^2(\mathbf{X}) + \text{Var}(\mathbf{X}), \quad (5)$$

where $\text{Var}(\mathbf{X}) \equiv \mathbb{E}_{\boldsymbol{\eta}} \|\hat{\beta}_{\text{tr}} - \mathbb{E}_{\boldsymbol{\eta}}[\hat{\beta}_{\text{tr}}]\|_{\Sigma}^2$ measures the sensitivity of the estimator with respect to noise realizations, and $\text{bias}^2(\mathbf{X}) \equiv \|\beta - \mathbb{E}_{\boldsymbol{\eta}}[\hat{\beta}_{\text{tr}}]\|_{\Sigma}^2$ measures the average error for a given $\mathbf{X}$. For the empirical least-squares estimator in [eq. \(4\)](#), these equal

$$\begin{aligned} \text{bias}^2(\mathbf{X}) &= \|(\mathbf{X} \mathbf{X}^+ - \mathbf{I}) \beta\|_{\Sigma}^2, \\ \text{Var}(\mathbf{X}) &= \mathbb{E}_{\boldsymbol{\eta}} \|(\mathbf{X}^+)^T \boldsymbol{\eta}\|_{\Sigma}^2 = \sigma^2 \text{Tr}[\mathbf{X}^+ \Sigma (\mathbf{X}^+)^T]. \end{aligned} \quad (6)$$

The bias is exactly zero in the *underparametrized regime*, $n_{\text{tr}} \geq d$, provided $\text{rank}(\mathbf{X}) = d$, since then $\mathbf{X} \mathbf{X}^+ = \mathbf{I}_d$.

II. QUANTUM LINEAR REGRESSION: GENERAL SETUP

Consider a linear-regression task in which the input features x are estimates of probabilities obtained by

measuring quantum states with finite resources. The training data consist of pairs $(\rho, y) \sim \mathcal{D}$, where each state ρ is evolved under fixed quantum dynamics and subsequently measured. Describing the overall measurement procedure by an effective POVM $\mu = \{\mu_a\}_{a=1}^{n_{\text{out}}}$, the associated probability vector is $\mathbf{p}(\rho) \equiv \langle \mu, \rho \rangle \in \mathbb{R}^{n_{\text{out}}}$. Here, $\langle \mu_a, \rho \rangle \equiv \text{tr}(\mu_a \rho)$ is the probability of observing outcome a , and the POVM elements μ_a are taken to incorporate both the quantum dynamics and the measurement apparatus.

We study supervised quantum machine-learning models with a linear readout trained on these measurement data, as in QELMs and memoryless quantum reservoir-computing architectures. The learning problem is therefore to find a matrix $\mathbf{W}$ such that $\mathbf{y} \simeq \mathbf{W}\langle \mu, \rho \rangle$. For a scalar target, $\mathbf{W}$ is a row vector; for a vector-valued target $\mathbf{y} \in \mathbb{R}^{n_f}$, one has $\mathbf{W} \in \mathbb{R}^{n_f \times n_{\text{out}}}$.

Henceforth, we focus on scalar targets that are linear functionals of the input state: $y = \langle \mathcal{O}, \rho \rangle$ for some observable $\mathcal{O}$. In this case, the condition $\langle \mathcal{O}, \rho \rangle = \mathbf{W}\mathbf{p}(\rho)$ for all ρ is equivalent to $\mathcal{O} = \mathbf{W}\mu \equiv \sum_a w_a \mu_a$. Thus, learnability is determined by whether the target observable lies in the real linear span of the effective POVM elements.

In what follows, we assume perfect knowledge of the targets $\mathbf{y}$. By contrast, the input features are affected by statistical noise because the probabilities $\langle \mu, \rho \rangle$ must be estimated on a quantum device using finitely many shots. More precisely, suppose that ρ is measured N times, and let $\hat{\mathbf{n}}(\rho) = \{\hat{n}_a(\rho)\}_{a=1}^{n_{\text{out}}}$ be the count vector, whose a -th component records how many times outcome a is observed. The empirical probability vector is then

$$\hat{\mathbf{p}}_N(\rho) = \hat{\mathbf{n}}(\rho)/N, \quad \hat{\mathbf{n}}(\rho) \sim \text{Mult}(N, \mathbf{p}(\rho)), \quad (7)$$

where $\text{Mult}(N, \mathbf{p}(\rho))$ denotes the multinomial distribution with N trials and probability vector $\mathbf{p}(\rho)$. By construction,

$$\begin{aligned} \mathbb{E}[\hat{\mathbf{p}}_N(\rho)] &= \mathbf{p}(\rho), \\ \text{Cov}[\hat{\mathbf{p}}_N(\rho)] &= N^{-1}\mathbf{\Sigma}(\mathbf{p}(\rho)), \end{aligned} \quad (8)$$

where $\mathbf{\Sigma}(\mathbf{p}) \equiv \text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^T$ is the categorical covariance matrix. In what follows, we use $\hat{\mathbf{p}}_\sigma$ as shorthand for the empirical frequency $\hat{\mathbf{p}}_N(\sigma)$, leaving its dependence on the statistics implicit. Similarly, we use $\hat{\mathbf{p}}_\sigma$ for the exact probability vector $\mathbf{p}(\sigma)$ and $\mathbf{\Sigma}_\sigma$ for the corresponding categorical covariance matrix.

To understand how finite measurement resources affect the performance of quantum linear regression, we proceed in two steps. We first consider an arbitrary estimator $\hat{\mathbf{W}}$, irrespective of how it was obtained, and study the estimation error associated with a generic test state σ . Let $\mathbf{W}$ denote an ideal readout satisfying $\langle \mathcal{O}, \rho \rangle = \mathbf{W}\mathbf{p}(\rho)$ for every state ρ , and suppose that

the empirical probability vector $\hat{\mathbf{p}}_\sigma$ is estimated using M test shots. Averaging the squared prediction error with respect to the possible realizations of the empirical probability vector $\hat{\mathbf{p}}_\sigma$, we define the *finite-statistics MSE* as

$$\text{MSE}(\hat{\mathbf{W}}, M, \sigma) = \mathbb{E}_{\hat{\mathbf{p}}_\sigma} [(\hat{\mathbf{W}}\hat{\mathbf{p}}_\sigma - \mathbf{W}\mathbf{p}_\sigma)^2]. \quad (9)$$

The dependence of the MSE on M is hidden in the frequency vector $\hat{\mathbf{p}}_\sigma = \hat{\mathbf{p}}_M(\sigma)$, which is estimated using a budget of M shots. In the next sections, $\hat{\mathbf{W}}$ will also be obtained from a training procedure involving a finite training statistics budget N . However, for the purpose of the decomposition discussed here, $\hat{\mathbf{W}}$ is left as completely general. The finite-statistics MSE admits the usual bias–variance decomposition:

$$\text{MSE}(\hat{\mathbf{W}}, M, \sigma) = \text{bias}_{\text{test}}^2(\hat{\mathbf{W}}, \sigma) + \text{Var}_{\text{test}}(\hat{\mathbf{W}}, M, \sigma), \quad (10)$$

where the *test bias* and *test variance* are defined by

$$\begin{aligned} \text{bias}_{\text{test}}^2(\hat{\mathbf{W}}, \sigma) &\equiv (\mathbb{E}_{\hat{\mathbf{p}}_\sigma} [\hat{\mathbf{W}}\hat{\mathbf{p}}_\sigma] - \mathbf{W}\mathbf{p}_\sigma)^2, \\ \text{Var}_{\text{test}}(\hat{\mathbf{W}}, M, \sigma) &\equiv \mathbb{E}_{\hat{\mathbf{p}}_\sigma} [(\hat{\mathbf{W}}\hat{\mathbf{p}}_\sigma)^2] - (\hat{\mathbf{W}}\mathbf{p}_\sigma)^2. \end{aligned} \quad (11)$$

Using eq. (8), these two quantities can be evaluated explicitly:

$$\begin{aligned} \text{bias}_{\text{test}}^2(\hat{\mathbf{W}}, \sigma) &= [(\hat{\mathbf{W}} - \mathbf{W})\mathbf{p}_\sigma]^2, \\ \text{Var}_{\text{test}}(\hat{\mathbf{W}}, M, \sigma) &= M^{-1}\hat{\mathbf{W}}\mathbf{\Sigma}_\sigma\hat{\mathbf{W}}^T. \end{aligned} \quad (12)$$

The interpretation of the two terms in eq. (12) is straightforward. The test bias is the squared systematic prediction error due to the discrepancy between the ideal readout $\mathbf{W}$ and its estimate $\hat{\mathbf{W}}$. The test variance quantifies fluctuations in the prediction caused by shot noise at test time. Indeed, the test variance vanishes as $M \rightarrow \infty$, whereas the test bias is independent of M and survives in this limit. Thus, increasing the number of test shots cannot reduce the bias caused by the choice of $\hat{\mathbf{W}}$. We therefore refer to the test bias as the *infinite-statistics MSE* and denote it by $\text{MSE}(\hat{\mathbf{W}}, \infty, \sigma)$. We obtain the following useful decomposition:

$$\text{MSE}(\hat{\mathbf{W}}, M, \sigma) = \text{MSE}(\hat{\mathbf{W}}, \infty, \sigma) + \frac{\hat{\mathbf{W}}\mathbf{\Sigma}_\sigma\hat{\mathbf{W}}^T}{M}. \quad (13)$$

Having separated these two sources of test error, we can focus on the infinite-statistics MSE. For fixed $\hat{\mathbf{W}}$, the test variance can be reduced by increasing M , whereas the infinite-statistics MSE depends entirely on the choice of $\hat{\mathbf{W}}$. It is therefore natural to examine how this choice is affected by training with finite statistics.

Let $\mathbf{P} \in \mathbb{R}^{n_{\text{out}} \times n_{\text{tr}}}$ denote the matrix of exact measurement probabilities for the n_{tr} training states, with i -th column $\mathbf{p}(\rho_i)$. Its rank satisfies

$$\text{rank}(\mathbf{P}) \leq \min(d^2, n_{\text{out}}, n_{\text{tr}}), \quad (14)$$

because the real vector space of Hermitian operators on a d -dimensional Hilbert space has dimension d^2 . If the training states span this operator space and the effective POVM is informationally complete, then $\text{rank}(\mathbf{P}) = d^2$. We focus on the overcomplete regime $n_{\text{tr}}, n_{\text{out}} > d^2$, under these spanning and informational-completeness assumptions. In this regime, $\mathbf{P}$ is necessarily rank deficient. We show in [sections III and IV](#) how this rank constraint controls the stability of quantum linear regression.

In realistic settings, only finite measurement resources are available, so one must work with a noisy empirical version of $\mathbf{P}$. If N measurements are performed independently on each training state, we define the empirical frequency matrix $\hat{\mathbf{P}}_N$ by taking its i -th column to be

$$\hat{\mathbf{p}}_i = \hat{\mathbf{n}}^{(i)} / N, \quad \hat{\mathbf{n}}^{(i)} \sim \text{Mult}(N, \mathbf{p}_i). \quad (15)$$

Under the assumptions above, when $n_{\text{tr}}, n_{\text{out}} > d^2$, the regression problem is exactly solvable for any target observable $\mathcal{O}$ [15]. Let $\mathbf{y} = (y_1, \dots, y_{n_{\text{tr}}})^T$ be the vector of training targets. Among the generally nonunique exact solutions, the canonical minimum-norm solution is $\mathbf{W} = \mathbf{y}^T \mathbf{P}^+$. Shot-noise fluctuations in $\hat{\mathbf{P}}_N$ generally prevent the exact recovery of these ideal weights from finite data. The standard unregularized pseudoinverse training rule instead yields the estimator

$$\hat{\mathbf{W}}_N = \mathbf{y}^T \hat{\mathbf{P}}_N^+. \quad (16)$$

The estimator $\hat{\mathbf{W}}_N$ is a random variable that depends nonlinearly on the particular realization of the training shot noise through the Moore–Penrose pseudoinverse $\hat{\mathbf{P}}_N^+$.

To characterize the effect of shot noise during the training stage, we average the infinite-statistics MSE over realizations of the training noise and define

$$\text{MSE}(N, \infty, \sigma) \equiv \mathbb{E}_{\hat{\mathbf{P}}_N} [\text{MSE}(\hat{\mathbf{W}}_N, \infty, \sigma)]. \quad (17)$$

Here and below, we suppress the dependence on the exact design matrix $\mathbf{P}$ and the fixed POVM μ . As before, the averaged MSE can be decomposed into a *training bias* and a *training variance*:

$$\begin{aligned} \text{MSE}(N, \infty, \sigma) &= \text{bias}^2(N, \sigma) + \text{Var}_{\text{tr}}(N, \sigma), \\ \text{bias}^2(N, \sigma) &\equiv \left[\left(\mathbf{W} - \mathbb{E}_{\hat{\mathbf{P}}_N} [\hat{\mathbf{W}}_N] \right) \mathbf{p}_\sigma \right]^2, \\ \text{Var}_{\text{tr}}(N, \sigma) &\equiv \mathbb{E}_{\hat{\mathbf{P}}_N} \left[\left((\hat{\mathbf{W}}_N - \mathbb{E}_{\hat{\mathbf{P}}_N} [\hat{\mathbf{W}}_N]) \mathbf{p}_\sigma \right)^2 \right]. \end{aligned} \quad (18)$$

The training variance $\text{Var}_{\text{tr}}(N, \sigma)$ quantifies the fluctuations of the prediction around its mean over realizations of the training noise. The training bias $\text{bias}^2(N, \sigma)$

is the squared difference between the target prediction and this mean prediction. Its source is explicit in [eq. \(18\)](#): in general, $\mathbb{E}_{\hat{\mathbf{P}}_N} [\hat{\mathbf{W}}_N] \neq \mathbf{W}$ because the pseudoinverse depends nonlinearly on the empirical frequency matrix. The same nonlinearity makes the dependence of the bias and variance on both N and the number of training states n_{tr} highly nontrivial.

Our main objective is to show that, despite the nonlinear dependence of bias^2 and Var_{tr} on the training noise, their scaling with N and n_{tr} can be controlled and characterized.

III. QUANTUM LINEAR REGRESSION IS ILL-CONDITIONED

Before deriving the MSE scaling, we isolate the geometric obstruction created by the physical perturbation $\hat{\mathbf{P}}_N - \mathbf{P} = N^{-1/2} \hat{\mathbf{E}}_N$. The Moore–Penrose pseudoinverse is discontinuous under perturbations that change the rank. More precisely, if $\text{rank}(\mathbf{P} + \delta \mathbf{P}) \neq \text{rank}(\mathbf{P})$, then $\lim_{\delta \mathbf{P} \rightarrow 0} (\mathbf{P} + \delta \mathbf{P})^+$ does not generally converge to $\mathbf{P}^+$ [33, 34, 52].

Some intuition about the behavior of the Moore–Penrose pseudoinverse under perturbations can be obtained from standard matrix perturbation theory [34], and in particular from the notion of *acute perturbations*. Let $\mathbf{A} \in \mathbb{C}^{n \times m}$ and $\hat{\mathbf{A}}_\tau = \mathbf{A} + \tau \hat{\mathbf{E}}$, where $\tau > 0$ is a generic perturbation parameter. We say that $\tau \hat{\mathbf{E}}$ is an *acute perturbation* for $\mathbf{A}$ if

$$\|\hat{\mathbf{A}}_\tau \hat{\mathbf{A}}_\tau^+ - \mathbf{A} \mathbf{A}^+\|_{\text{op}} < 1 \quad \text{and} \quad \|\hat{\mathbf{A}}_\tau^+ \hat{\mathbf{A}}_\tau - \mathbf{A}^+ \mathbf{A}\|_{\text{op}} < 1$$

. Here $\mathbf{A} \mathbf{A}^+$ is the projector onto the image of $\mathbf{A}$, while $\mathbf{A}^+ \mathbf{A}$ is the projector onto the row space of $\mathbf{A}$.

The acuteness of a perturbation tells us whether the inverse of a matrix is stable under perturbation. In particular, the instability for non-acute perturbations can be quantified as follows:

Theorem 1 ([33]). *If $\tau \hat{\mathbf{E}}$ is not an acute perturbation for $\mathbf{A}$,*

$$\|\hat{\mathbf{A}}_\tau^+ - \mathbf{A}^+\|_{\text{op}} \geq \tau^{-1} \|\hat{\mathbf{E}}\|_{\text{op}}^{-1}. \quad (19)$$

If, further, $\text{rank}(\hat{\mathbf{A}}_\tau) \geq \text{rank}(\mathbf{A})$, then

$$\|\hat{\mathbf{A}}_\tau^+\|_{\text{op}} \geq \tau^{-1} \|\hat{\mathbf{E}}\|_{\text{op}}^{-1}. \quad (20)$$

For a fixed rank-increasing direction $\hat{\mathbf{E}}$ for which nonacuteness persists as $\tau \rightarrow 0$, these bounds exhibit the resulting τ^{-1} divergence. The following rank criterion is also useful:

Theorem 2 ([53]). *If two matrices $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{n \times m}$ are acute with respect to one another, then $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{B})$. If $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{B})$ and $\|\mathbf{B} - \mathbf{A}\|_{\text{op}} \leq \|\mathbf{A}^+\|_{\text{op}}^{-1}$, then they are acute with respect to one another.*

In the overcomplete regime considered here, shot noise generically increases the rank:

$$\text{rank}(\mathbf{P}) = d^2 < \text{rank}(\hat{\mathbf{P}}_N) = \min(n_{\text{out}}, n_{\text{tr}}). \quad (21)$$

On this full-rank event the perturbation is nonacute, and

$$\|\hat{\mathbf{P}}_N^+\|_{\text{op}} \geq \sqrt{N} \|\hat{\mathbf{\Xi}}\|_{\text{op}}^{-1}. \quad (22)$$

This growth is intrinsic to the unregularized pseudoinverse estimator in the overcomplete setting.

Equation (22) concerns the norm of the full pseudoinverse and does not by itself imply that physical predictions diverge. We will show in section IV that its singular block is annihilated by left and right multiplication by $\mathbf{y}^T$ and $\mathbf{p}_\sigma$. Nevertheless, this does not mean that the singularity has no effect at all: it becomes relevant when the target $\mathbf{y}$ is noisy, as we will discuss in section VII.

IV. PERTURBATIVE ANALYSIS OF NOISE PERTURBATIONS

This section provides intuition for the instability in section III, while the fully rigorous proof of the resource-scaling is deferred to section VI. The purpose of this section is to identify, at a fixed noise realization, the blocks through which a nonacute perturbation affects the pseudoinverse and to show why generalized Schur complements replace the coefficients of an ordinary Taylor expansion.

Recall that, in the regime $n_{\text{tr}}, n_{\text{out}} > d^2$, the exact probability matrix $\mathbf{P}$ has rank d^2 , while its finite-shot estimate $\hat{\mathbf{P}}_N$ has rank $\min(n_{\text{out}}, n_{\text{tr}})$ with high probability (we refer to section SM3 for the rigorous version of these statements). Under these conditions, the standard Taylor expansion of the Moore–Penrose pseudoinverse cannot be applied directly, and the calculation requires a perturbative approach tailored to this setting.

We decompose the noisy probability matrix as $\hat{\mathbf{P}}_N = \mathbf{P} + N^{-1/2} \hat{\mathbf{\Xi}}$, where the columns $\hat{\xi}_i$ of $\hat{\mathbf{\Xi}}$ are the centered, rescaled multinomial fluctuations. Each column is independent but not identically distributed, as $\text{Cov}(\hat{\xi}_i, \hat{\xi}_i^T) = \Sigma_i$. The $N^{-1/2}$ rescaling makes $\hat{\mathbf{\Xi}}$ only depend on N via its third and higher moments, and we thus suppress its explicit N dependence for sake of notation here. We also fix the regime $n_{\text{tr}} > n_{\text{out}}$. With high probability, $\hat{\mathbf{P}}_N$ is surjective, and its pseudoinverse can be written as $\hat{\mathbf{P}}_N^+ = \hat{\mathbf{P}}_N^T (\hat{\mathbf{P}}_N \hat{\mathbf{P}}_N^T)^{-1}$. The discontinuity of the pseudoinverse therefore arises from the inverse of the $n_{\text{out}} \times n_{\text{out}}$ matrix

$$\hat{\mathbf{P}}_N \hat{\mathbf{P}}_N^T = \mathbf{P} \mathbf{P}^T + \frac{1}{\sqrt{N}} (\mathbf{P} \hat{\mathbf{\Xi}}^T + \hat{\mathbf{\Xi}} \mathbf{P}^T) + \frac{1}{N} \hat{\mathbf{\Xi}} \hat{\mathbf{\Xi}}^T. \quad (23)$$

Rank deficiency gives $\mathbf{P}$ a nontrivial kernel and cokernel. Choose isometries V_1, V_2 in the training space and U_1, U_2 in the outcome space such that the corresponding orthogonal projectors are

$$\begin{aligned} \Pi_1 &\equiv \Pi_{\text{supp}(\mathbf{P})} = \mathbf{P}^+ \mathbf{P} = V_1 V_1^T, \\ \Pi_1 &\equiv \Pi_{\text{im}(\mathbf{P})} = \mathbf{P} \mathbf{P}^+ = U_1 U_1^T. \end{aligned} \quad (24)$$

Write $\Pi_2 \equiv \mathbf{I} - \Pi_1 = V_2 V_2^T$ for the projector onto $\ker(\mathbf{P})$ and $\Pi_2 \equiv \mathbf{I} - \Pi_1 = U_2 U_2^T$ for the projector onto $\text{coker}(\mathbf{P})$. By suitable changes of basis in the input and output spaces, we can always write $\hat{\mathbf{P}}_N$ in block form as

$$\hat{\mathbf{P}}_N = \begin{pmatrix} \mathbf{P}_{11} + N^{-1/2} \hat{\mathbf{\Xi}}_{11} & N^{-1/2} \hat{\mathbf{\Xi}}_{12} \\ N^{-1/2} \hat{\mathbf{\Xi}}_{21} & N^{-1/2} \hat{\mathbf{\Xi}}_{22} \end{pmatrix}, \quad (25)$$

where $\hat{\mathbf{\Xi}}_{mn} \equiv U_m^T \hat{\mathbf{\Xi}} V_n$ and $\mathbf{P}_{11} \equiv U_1^T \mathbf{P} V_1$. By definition, $\mathbf{P} = \Pi_1 \mathbf{P} \Pi_1$ and $\mathbf{P}_{11} \in \mathbb{R}^{d^2 \times d^2}$ is invertible.

Using eq. (25), one can find a fixed-realization asymptotic expansion for $\hat{\mathbf{P}}_N^+$. We show in section SM2 that $\Delta \hat{\mathbf{P}}_N^+ \equiv \hat{\mathbf{P}}_N^+ - \mathbf{P}^+$ has leading terms

$$\begin{aligned} (\Delta \hat{\mathbf{P}}_N^+)_{11} &= -N^{-1/2} \mathbf{P}_{11}^+ (\hat{\mathbf{\Xi}} / \hat{\mathbf{\Xi}}_{22}) \mathbf{P}_{11}^+ + O(N^{-1}), \\ (\Delta \hat{\mathbf{P}}_N^+)_{12} &= -\mathbf{P}_{11}^+ \hat{\mathbf{\Xi}}_{12} \hat{\mathbf{\Xi}}_{22}^+ + O(N^{-1/2}), \\ (\Delta \hat{\mathbf{P}}_N^+)_{21} &= -\hat{\mathbf{\Xi}}_{22}^+ \hat{\mathbf{\Xi}}_{21} \mathbf{P}_{11}^+ + O(N^{-1/2}), \\ (\Delta \hat{\mathbf{P}}_N^+)_{22} &= \sqrt{N} \hat{\mathbf{\Xi}}_{22}^+ + O(1). \end{aligned} \quad (26)$$

where we introduced the *generalized Schur complement*

$$\hat{\mathbf{\Xi}} / \hat{\mathbf{\Xi}}_{22} \equiv \hat{\mathbf{\Xi}}_{11} - \hat{\mathbf{\Xi}}_{12} \hat{\mathbf{\Xi}}_{22}^+ \hat{\mathbf{\Xi}}_{21}. \quad (27)$$

It is evident from eq. (26) that $\Delta \hat{\mathbf{P}}_N^+$ does not converge to zero as $N \rightarrow \infty$: the off-diagonal blocks approach finite stochastic limits, whereas the (2, 2) block diverges as $\sqrt{N}$. Notably, the discontinuity is primarily controlled by $\hat{\mathbf{\Xi}}_{22}^+$.

Using the pseudoinverse estimator $\hat{\mathbf{W}}_N$ as given in eq. (16), the MSE for a given training noise realization and test state σ , at infinite test statistics, is

$$\|(\hat{\mathbf{W}}_N - \mathbf{W}) \mathbf{p}_\sigma\|^2 = [\mathbf{y}^T \Delta \hat{\mathbf{P}}_N^+ \mathbf{p}_\sigma]^2 \equiv (\Delta \delta)^2. \quad (28)$$

By construction, $\mathbf{p}_\sigma \equiv \langle \mu, \sigma \rangle$ satisfies $\Pi_2 \mathbf{p}_\sigma = 0$, and $\mathbf{y}^T \equiv \langle \mathcal{O}, \rho \rangle$ satisfies $\mathbf{y}^T \Pi_2 = 0$. In words, these amount to saying that $\mathbf{p}_\sigma$ is in the column space of $\mathbf{P}$ while $\mathbf{y}^T$ is in its row space. Thus the divergent (2, 2) block in eq. (26) does not enter in $\text{MSE}(\hat{\mathbf{W}}_N, \infty)$. Nonetheless, it can amplify the finite-test-statistics variance when the training targets contain components outside the row space of $\mathbf{P}$. Expanding eq. (28) via eq. (26) gives

$$\Delta \delta = \mathbf{w}_\sigma^T U_1 \left(\frac{\mathbf{A}_1}{\sqrt{N}} + \frac{\mathbf{A}_2}{N} \right) V_1^T \langle \bar{\rho}, \sigma \rangle + O\left(\frac{1}{N^{3/2}}\right), \quad (29)$$

where we kept the first two nontrivial orders in N of $(\Delta\hat{P}_N^+)_{11}$, and used $w_{\mathcal{O}} \equiv \langle \tilde{\mu}, \mathcal{O} \rangle$. Here $\tilde{\mu}$ and $\tilde{\rho}$ are, respectively, the canonical dual measurement and training frames defined in [section SM1.5](#). The operators A_1, A_2 are given by

$$\begin{aligned} A_1 &= -\hat{\mathbb{E}}/\hat{\mathbb{E}}_{22}, \\ A_2 &= (\hat{\mathbb{E}}/\hat{\mathbb{E}}_{22})P_{11}^+(\hat{\mathbb{E}}/\hat{\mathbb{E}}_{22}) - \hat{\mathbb{E}}_{12}\hat{Q}\hat{\mathbb{E}}_{12}^T P_{11}^{+T}\hat{Y}, \end{aligned} \quad (30)$$

where $\hat{Q} \equiv I - \hat{\mathbb{E}}_{22}^+\hat{\mathbb{E}}_{22}$ is the projector onto $\ker(\hat{\mathbb{E}}_{22})$, and $\hat{Y} \equiv I + \hat{\mathbb{E}}_{21}^T(\hat{\mathbb{E}}_{22}\hat{\mathbb{E}}_{22}^T)^{-1}\hat{\mathbb{E}}_{21}$. When $n_{\text{tr}} = n_{\text{out}}$, the block $\hat{\mathbb{E}}_{22}$ is square; if it is nonsingular, then $\hat{Q} = 0$. Thus $\hat{Q}$ captures the additional kernel present in the rectangular case. Here $O(N^{-3/2})$ denotes the next term in the formal expansion at a fixed noise realization for which the required block inverses exist; it is not a uniform probabilistic remainder bound. In particular, the nonlinear random coefficients $\hat{\mathbb{E}}_{22}^+$, $\hat{Q}$, and $\hat{Y}$ cannot be averaged term by term without additional conditioning estimates.

[Equation \(29\)](#) exposes the nontrivial consequences of computing the pseudoinverse in a non-acute perturbation regime, but the resulting random coefficients A_1, A_2 are nonlinear functions of $\hat{\mathbb{E}}$ and depend in particular on the diverging term $\hat{\mathbb{E}}_{22}^+$, and therefore cannot be averaged term by term without additional control. We will show how to handle this issue first in [section V](#), in a homogeneous centered Gaussian noise model, and then present in [section VI](#) an alternative proof strategy that directly operates on the multinomial noise.

V. HOMOGENEOUS GAUSSIAN APPROXIMATION

The perturbative expression in [eq. \(29\)](#) contains a generalized Schur complement and the projector $\hat{Q}$, both of which are nonlinear functions of the shot-noise matrix. We now show that it is possible to average those quantities in a homogeneous reference model where the shot-noise is taken to be Gaussian. The purpose is to expose the origin of the N^{-2} squared bias and the $(Nn_{\text{tr}})^{-1}$ training variance in a setting where computation can be performed explicitly with standard techniques and that provide a result in great agreement with that obtained within the fully multinomial model. In fact, in [section SM6](#) we collect several simulations that provide evidences of such agreement. The treatment of the true multinomial noise requires different techniques and results from random matrix theory. For this reason, it is established separately in [section VI](#).

In the homogeneous Gaussian model, the shot-noise matrix $\hat{\mathbb{E}}$ is assumed to follow a matrix-normal distribution:

$$\hat{\mathbb{E}} \sim \mathcal{MN}(0, \bar{\Sigma}, I_{n_{\text{tr}}}), \quad \bar{\Sigma} \equiv \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \Sigma^{(i)}. \quad (31)$$

Thus the columns of $\hat{\mathbb{E}}$ are independent centered Gaussian vectors with the same covariance. We use the block coordinates of [eq. \(25\)](#), so $\bar{\Sigma}_{ab} \equiv U_a^T \bar{\Sigma} U_b$, and assume $\bar{\Sigma}_{22}$ is positive definite. Its Schur complement is

$$\bar{\Sigma}_{\text{eff}} \equiv \bar{\Sigma}_{11} - \bar{\Sigma}_{12}\bar{\Sigma}_{22}^{-1}\bar{\Sigma}_{21}. \quad (32)$$

Fix d and n_{out} and let $n_{\text{tr}} \rightarrow \infty$. Since $n_{\text{tr}} \geq n_{\text{out}}$, $\hat{\mathbb{E}}_{22}$ has full row rank almost surely, so $\hat{Q}$ is the orthogonal projector onto $\ker(\hat{\mathbb{E}}_{22})$ and has rank $n_{\text{tr}} - n_{\text{out}}$. Moreover, the generalized Schur complement $\hat{\mathbb{E}}/\hat{\mathbb{E}}_{22}$ turns out also to follow a matrix-normal distribution, conditionally on $\hat{Y}$:

$$\hat{\mathbb{E}}/\hat{\mathbb{E}}_{22} \mid (\hat{\mathbb{E}}_{21}, \hat{\mathbb{E}}_{22}) \sim \mathcal{MN}(0, \bar{\Sigma}_{\text{eff}}, \hat{Y}). \quad (33)$$

Under the same conditioning, $\hat{\mathbb{E}}_{12}\hat{Q}\hat{\mathbb{E}}_{12}^T$ is a random Wishart matrix with scale $\bar{\Sigma}_{\text{eff}}$ and $n_{\text{tr}} - n_{\text{out}}$ degrees of freedom, and it is independent of the generalized Schur complement. This means that we can perform all the averages in the following way: first we average any polynomial in the Schur complement and $\hat{\mathbb{E}}_{12}\hat{Q}\hat{\mathbb{E}}_{12}^T$, using standard formulas for Gaussian and Wishart distributions [\[54\]](#). The momenta of the former are expressed in terms of $\hat{Y}$ than we can in turn average. In fact, let us remember that $\hat{Y} = I + \hat{\mathbb{E}}_{21}^T(\hat{\mathbb{E}}_{22}\hat{\mathbb{E}}_{22}^T)^{-1}\hat{\mathbb{E}}_{21}$ and that $\hat{\mathbb{E}}_{21}, \hat{\mathbb{E}}_{22}$ are independent. Moreover, $(\hat{\mathbb{E}}_{22}\hat{\mathbb{E}}_{22}^T)^{-1}$ is known to be distributed as an inverse Wishart matrix and momenta are explicitly known. For instance, it can be seen that, for $n_{\text{tr}} > n_{\text{out}} + 1$,

$$\mathbb{E}[\hat{Y}] = \frac{n_{\text{tr}} - d^2 - 1}{n_{\text{tr}} - n_{\text{out}} - 1} I. \quad (34)$$

The conditional Gaussian calculation in [lemma 8](#) yields, for $n_{\text{tr}} > n_{\text{out}} + 3$,

$$\begin{aligned} \mathbb{E}[w_{\mathcal{O}}^T U_1 A_1 V_1^T \langle \tilde{\rho}, \sigma \rangle] &= 0, \\ \text{Var}[w_{\mathcal{O}}^T U_1 A_1 V_1^T \langle \tilde{\rho}, \sigma \rangle] &= \frac{V}{n_{\text{tr}}} \frac{n_{\text{tr}} - d^2 - 1}{n_{\text{tr}} - n_{\text{out}} - 1}, \\ \mathbb{E}[w_{\mathcal{O}}^T U_1 A_2 V_1^T \langle \tilde{\rho}, \sigma \rangle] &= - \left(1 - \frac{d^2 + 1}{n_{\text{tr}}} \right) b, \\ \text{Var}[w_{\mathcal{O}}^T U_1 A_2 V_1^T \langle \tilde{\rho}, \sigma \rangle] &= O(n_{\text{tr}}^{-1}). \end{aligned} \quad (35)$$

where we defined the coefficients:

$$\begin{aligned} b &\equiv n_{\text{tr}} w_{\mathcal{O}}^T U_1 \bar{\Sigma}_{\text{eff}} P_{11}^{-T} V_1^T \langle \tilde{\rho}, \sigma \rangle, \\ V &\equiv n_{\text{tr}} \|V_1^T \langle \tilde{\rho}, \sigma \rangle\|_2^2 w_{\mathcal{O}}^T U_1 \bar{\Sigma}_{\text{eff}} U_1^T w_{\mathcal{O}}. \end{aligned} \quad (36)$$

These coincide with the homogeneous-covariance specializations of the general coefficients derived in [section VI](#). The precise value of the right-hand side in [eq. \(35\)](#) is particularly involved and not useful, for this reason we only keep track of its order in

n_{tr} . Computing its order requires an extra assumption on the spectrum of the exact probability matrix, $\lambda_{\min}(\mathbf{P}_{11}/\sqrt{n_{\text{tr}}}) \geq k_{11} > 0$ for some constant k_{11} independent on n_{tr} . For a fixed informationally complete POVM, balanced tight designs obey, exactly such condition, see eqs. (SM15) and (SM16); independent Haar-random training states also obey this condition with high probability for large n_{tr} by the fixed-dimensional law of large numbers, as made explicit in eq. (SM133).

Substituting eq. (35) in eq. (29) identifies the $N^{-1/2}$ term as a centered fluctuation and the N^{-1} term as a mean shift. Define the second-order truncation

$$\Delta\delta^{(2)} \equiv \mathbf{w}_{\mathcal{O}}^T U_1 \left(\frac{\mathbf{A}_1}{\sqrt{N}} + \frac{\mathbf{A}_2}{N} \right) V_1^T \langle \tilde{\rho}, \sigma \rangle. \quad (37)$$

Its expectation and variance are

$$\begin{aligned} \mathbb{E}[\Delta\delta^{(2)}] &= -\frac{b}{N} + O\left(\frac{1}{Nn_{\text{tr}}}\right), \\ \text{Var}(\Delta\delta^{(2)}) &= \frac{V}{Nn_{\text{tr}}} + O\left(\frac{1}{N^2n_{\text{tr}}} + \frac{1}{N^2n_{\text{tr}}^2}\right). \end{aligned} \quad (38)$$

Consequently,

$$\mathbb{E}[(\Delta\delta^{(2)})^2] = \frac{b^2}{N^2} + \frac{V}{Nn_{\text{tr}}} + O\left(\frac{1}{N^2n_{\text{tr}}} + \frac{1}{Nn_{\text{tr}}^2}\right). \quad (39)$$

Thus the centered first-order fluctuation produces the $(Nn_{\text{tr}})^{-1}$ training variance, whereas the second-order mean produces the N^{-2} squared bias.

Let us stress once again that the analysis relies on two main assumptions: first, that the Gaussian homogeneous model provides a good approximation, and second, that the remainder of the Moore–Penrose expansion is well behaved. Although the numerical evidence presented in section SM6 suggests that both assumptions hold, proving them rigorously is not straightforward. For instance, attempting to average higher-order terms in the Moore–Penrose expansion reveals that stronger conditions on n_{tr} are required to ensure the existence of the relevant moments of the Wishart distributions. Moreover, finiteness of every coefficient up to a fixed order does not by itself justify averaging the entire perturbative series or its truncated remainder. The fixed-realization expansion has a random radius of convergence controlled by the smallest relevant singular values. Rare realizations in which $\hat{\Sigma}_{22}$ or the effective Schur complement is nearly singular can therefore prevent the interchange of expectation with an infinite series and can invalidate a remainder estimate that is not uniform in the noise realization. This is why the Gaussian calculation provides exact moments for fixed coefficients but is not, by itself, a proof of an all-orders MSE expansion.

For these reasons, we rely on the more rigorous analysis of section VI to circumvent these issues. Neverthe-

less, the Gaussian model and the Moore–Penrose expansion remain useful tools for obtaining accurate estimates efficiently.

VI. BIAS-VARIANCE DECOMPOSITION AND RESOURCE SCALING

We now sketch how the main resource-scaling result for the true multinomial noise model can be derived without relying on the homogeneous Gaussian assumption of section V. The full proof is rather technical and is deferred to section SM3 and section SM4.

The formal expansion developed in section IV is not uniform over noise realizations. In order to determine $\mathbb{E}[\Delta\delta^2]$, one must control the remainder even on rare, nearly singular events, while the nonlinear coefficients appearing in the expansion are difficult to average directly. Moreover, on such singular events the perturbative expansion itself may break down, so a stronger form of remainder control is required.

We therefore adopt a different strategy. We decompose $\Delta\hat{\delta}$ exactly into a deterministic component, a centered fluctuation, and a remainder collecting all nonlinear contributions. We then control this remainder both with high probability and in mean square, working directly with the true multinomial noise model.

More precisely, conditional on the choice of training states, section SM3 establishes the decomposition

$$\begin{aligned} \Delta\hat{\delta} &= -\frac{b_N}{N} + \hat{L} + \hat{R}, \\ \mathbb{E}_{\hat{\Sigma}}[\hat{L}] &= 0, \quad \mathbb{E}_{\hat{\Sigma}}[\hat{L}^2] = \frac{V_N}{Nn_{\text{tr}}} + O\left(\frac{1}{N^2n_{\text{tr}}}\right). \end{aligned} \quad (40)$$

Here b_N is the deterministic bias coefficient, $\hat{L}$ is the centered first-order fluctuation, and $\hat{R}$ contains all higher-order and nonlinear Schur-complement terms. For large N , the coefficients converge to limits independent of both N and n_{tr} :

$$b_N = b + O(N^{-1}), \quad V_N = V + O(N^{-1}), \quad (41)$$

with $b, V = O(1)$ with respect to n_{tr} . The remainder $\hat{R}$ is subleading with high probability. In particular, for fixed d and n_{out} , in the large- n_{tr} limit we show that

$$\hat{R} = O_{\mathbb{P}}\left(\frac{1}{n_{\text{tr}}\sqrt{N}}\right). \quad (42)$$

The decomposition in eq. (40) is not immediately equivalent to that in eq. (29). In particular, $\hat{R}$ is not simply of order $O(N^{-3/2})$; rather, it contains contributions originating from both the $N^{-1/2}$ and N^{-1} terms in eq. (29). The resulting expected infinite-test-statistics

MSE takes the form

$$\mathbb{E}_{\hat{\Delta}}[\Delta\hat{\delta}^2] = \frac{b_N^2}{N^2} + \frac{V_N}{Nn_{\text{tr}}} + O(\text{Err}_{N,n_{\text{tr}}}), \quad (43)$$

where b_N and V_N are $O_{n_{\text{tr}}}(1)$, with explicit expressions

$$\begin{aligned} b_N &= \mathbf{w}_{\mathcal{O}}^T \bar{\mathbf{S}}_{\text{eff}}^{-1} \mathbf{p}_{\sigma}, \\ V_N &= n_{\text{tr}}^{-1} \sum_{i=1}^{n_{\text{tr}}} \left(\mathbf{p}_i^T \mathbf{S}_{\text{eff}}^{-1} \mathbf{p}_{\sigma} \right)^2 (\mathbf{w}_{\text{eff}}^{\mathcal{O}})^T \mathbf{\Sigma}_i \mathbf{w}_{\text{eff}}^{\mathcal{O}}, \end{aligned} \quad (44)$$

where

$$\mathbf{S}_{\text{eff}} = n_{\text{tr}}^{-1} \mathbf{P} \mathbf{P}^T + N^{-1} \bar{\mathbf{S}}_{\text{eff}}, \quad (45)$$

and $\mathbf{w}_{\mathcal{O}}^{\text{eff}}$ is a deformation of $\mathbf{w}_{\mathcal{O}}$ induced by the Schur complement of the noise matrix. The term $\text{Err}_{N,n_{\text{tr}}}$, defined explicitly in eq. (SM111), tends to zero for fixed d , n_{out} , and N in the large- n_{tr} regime.

This asymptotic MSE exhibits several features that are absent from the Gaussian model. First, both the training bias and the training variance depend explicitly on N through the deformed population matrix $\mathbf{S}_{\text{eff}}$, which contains an N^{-1} correction. The general formulas for the bias and variance therefore remain valid even when N is small. Second, we now have explicit control of the remainder error $\text{Err}_{N,n_{\text{tr}}}$, which vanishes in the asymptotic limit $n_{\text{tr}} \rightarrow \infty$ for any fixed finite N .

For large N , the training bias and variance admit a simpler form in terms of coefficients that are independent of N :

$$\mathbb{E}_{\hat{\Delta}}[\Delta\hat{\delta}^2] = \frac{b^2}{N^2} + \frac{V}{Nn_{\text{tr}}} + O(\text{Err}_{N,n_{\text{tr}}} + N^{-3}), \quad (46)$$

where

$$\begin{aligned} b &= \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \text{Cov}_i(\mathbf{w}_{\mathcal{O}}^{\text{eff}}, n_{\text{tr}} \langle \tilde{\mu}, \tilde{\sigma} \rangle), \\ V &= \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} (n_{\text{tr}} \langle \tilde{\rho}_i, \sigma \rangle)^2 \text{Var}_i(\mathbf{w}_{\mathcal{O}}^{\text{eff}}), \end{aligned} \quad (47)$$

with $\text{Cov}_i(\mathbf{a}, \mathbf{b}) \equiv \mathbf{a}^T \mathbf{\Sigma}_i \mathbf{b}$, and $\text{Var}_i(\mathbf{a}) \equiv \text{Cov}_i(\mathbf{a}, \mathbf{a})$. This form of the MSE is derived in section SM4.1, and yields simpler coefficients b and V at the price of an additional $O(N^{-3})$ replacement error.

The coefficients in eq. (47) coincide exactly with those in eq. (36), obtained within the Gaussian approximation, when the homogeneous condition $\mathbf{\Sigma}_i = \bar{\mathbf{S}}$ is imposed for all training states, as shown in section SM5.3. Even under this homogeneity assumption, however, the Gaussian approximation does not reproduce the exact N -dependent coefficients in eq. (44). This indicates, as expected, that the Gaussian approximation ceases to be accurate at small N . The discrepancy is of order $O(N^{-3})$, independently of n_{tr} . Such fixed- N control is a further advantage of the exact decomposition

developed in this section over the large- N expansions of sections IV and V.

The decomposition eq. (40) therefore identifies the two leading contributions to the infinite-test-statistics squared error. The training bias originates from the deterministic N^{-1} covariance shift in the effective Schur complement, whereas the training variance is generated by the centered linear fluctuation. In particular, at fixed N and large n_{tr} , the variance contribution becomes negligible, while the MSE remains bounded below by a bias floor of order $O(N^{-2})$.

It is useful to contrast this behavior with standard classical linear regression. In the classical setting, when the feature vectors are known exactly and the label noise has zero mean, the least-squares estimator is conditionally unbiased in the underparametrized regime. For $\mathbf{X} \in \mathbb{R}^{d \times n_{\text{tr}}}$, this occurs when $n_{\text{tr}} \geq d$, provided that $\text{rank}(\mathbf{X}) = d$, since in this case $\mathbf{X} \mathbf{X}^+ = \mathbf{I}_d$. This conclusion does not carry over to the quantum setting considered here. The crucial difference is that the input vectors themselves are subject to stochastic noise: the ideal feature matrix $\mathbf{P}$ is replaced by the finite-shot frequency matrix $\hat{\mathbf{P}}_N$. Although the rank of $\mathbf{P}$ is constrained by the dimension d^2 of the operator space, the empirical matrix $\hat{\mathbf{P}}_N$ has rank $\min(n_{\text{out}}, n_{\text{tr}})$ with high probability. Finite-shot noise can therefore change the rank of the design matrix and, consequently, induce discontinuous changes in its Moore–Penrose pseudoinverse.

As a result, even in the quantum analogue of the underparametrized regime, $n_{\text{tr}} > n_{\text{out}}$, the pseudoinverse estimator acquires a systematic bias at finite training statistics. Our analysis shows that, whenever the leading coefficients remain nonzero and the stated conditioning and growth assumptions are satisfied, the bias amplitude scales as N^{-1} , its square as N^{-2} , and the training variance as $(Nn_{\text{tr}})^{-1}$. Increasing the size of the training set therefore suppresses fluctuations across shot-noise realizations, but it neither changes the leading N^{-2} scaling of the squared bias nor, in general, produces an additional n_{tr}^{-1} suppression of the bias floor. Once the variance becomes subleading, adding further training states yields no generic leading-order improvement unless the number of shots per training state is increased as well.

The leading coefficients can be evaluated exactly in a symmetric reference model. Consider training states forming a complex projective two-design, a POVM given by a fixed balanced rank-one tight frame, and a test state σ and target projector $\mathcal{O}$ chosen independently according to the Haar measure. As shown in

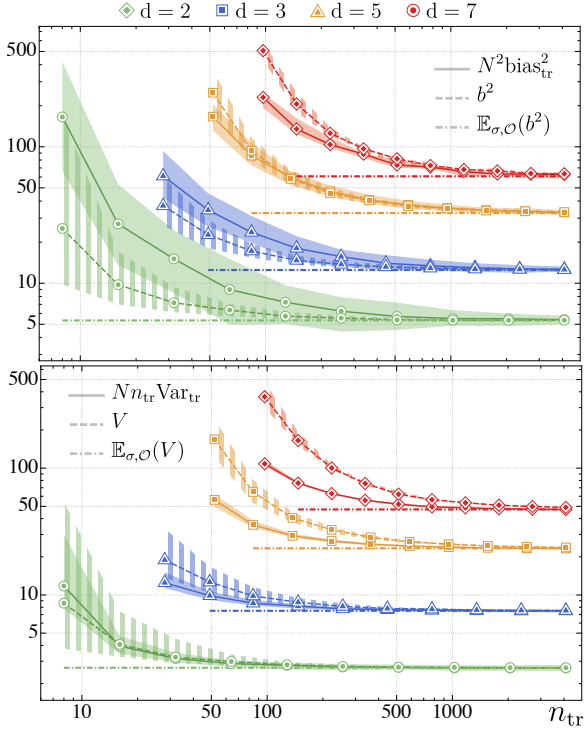


Figure 2. MUB : $N = 10000, n_{\text{test}} = 100, n_{\text{real}} = 100, n_{\text{reps}} = 100$ (LI: *mediane? barre di errore che quantili? quantili fatti su realizzazioni o ripetizioni o entrambi?*)

section SM4.3, the corresponding averages are

$$\begin{aligned} \mathbb{E}_{\sigma, \mathcal{O}}[b^2] &= \frac{(d-1)(d+2)^2}{d+1}, \\ \mathbb{E}_{\sigma, \mathcal{O}}[V] &= \frac{d(d-1)(d+2)}{d+1}. \end{aligned} \quad (48)$$

Both coefficients scale as d^2 for large d . An i.i.d. Haar training sample is not an exact finite two-design, but it approaches the same population frame identities when its empirical frame is sufficiently well conditioned. COMMENT BETTER FIGURES For comparison, in fig. 2 we plot the theoretical behavior against numerical simulations for $d = 2, 4$, and 8 . The POVM is the complete MUB measurement, with $n_{\text{out}} = d(d+1)$ outcomes, and $N = 2048$. Each numerical point averages over 1000 independent choices of the target and training ensemble and, for each choice, over 100 independent shot-noise realizations.

VII. TARGET NOISE AMPLIFICATION

As discussed in section IV, the divergent terms of $\hat{P}_N^+$ do not affect the MSE when considering infinite-test-statistics and exact features. This cancellation relies on the identity $\mathbf{y}^T V_2 = 0$, where the columns of V_2 span $\ker(\mathbf{P})$. However, while this holds when $\mathbf{y} = \langle \rho, \mathcal{O} \rangle$,

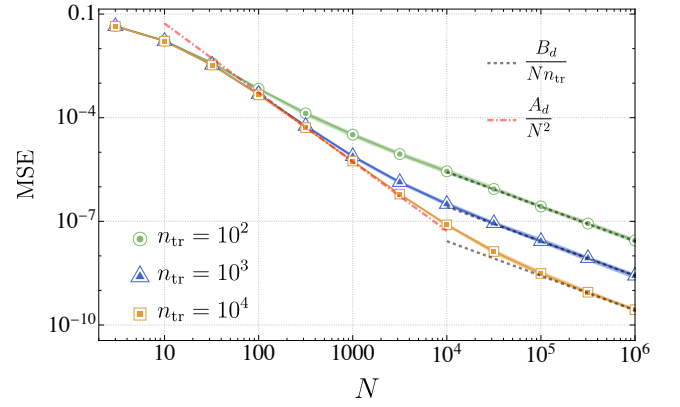


Figure 3. (LI: A_d and B_d are b^2 and V ?)

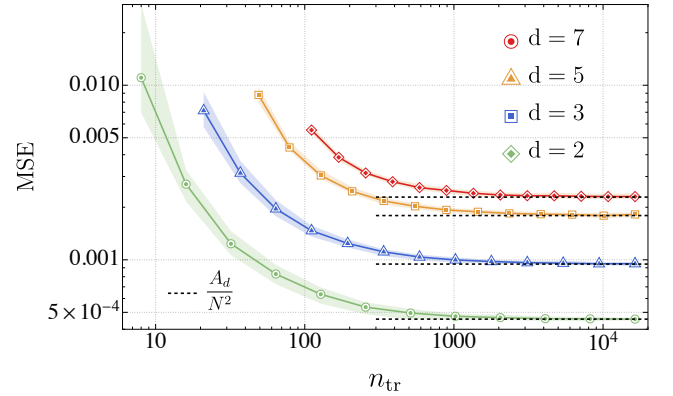


Figure 4.

it fails when $\mathbf{y}$ comes with its own estimation error. Such an error can be due to imperfect preparation of the states on noisy quantum hardware.

Let the target observed for a test state σ be $\hat{y}_\sigma = \mathbf{W} \mathbf{p}_\sigma + \hat{\eta}_\sigma$, where $\hat{\eta}_\sigma \sim \mathcal{N}(0, \gamma^2)$ is independent of the empirical test frequency $\hat{\mathbf{p}}_\sigma$. For any fixed weight vector $\hat{\mathbf{W}}$, the finite-statistics MSE then becomes

$$\begin{aligned} \text{MSE}(\hat{\mathbf{W}}, M, \gamma, \sigma) &\equiv \mathbb{E}_{\hat{\mathbf{p}}_\sigma, \hat{\eta}_\sigma} [(\hat{\mathbf{W}} \hat{\mathbf{p}}_\sigma - \hat{y}_\sigma)^2] \\ &= \gamma^2 + [(\hat{\mathbf{W}} - \mathbf{W}) \mathbf{p}_\sigma]^2 + \frac{1}{M} \hat{\mathbf{W}} \Sigma_\sigma \hat{\mathbf{W}}^T. \end{aligned} \quad (49)$$

The first term is precisely the irreducible noise of classical regression in eq. (2). It survives even when the ideal readout and the exact test probabilities are known. The last term has a related, but operationally distinct, status. At fixed test budget M it is a noise floor that no amount of training data can remove, although it can be suppressed by increasing M . In particular, even the ideal readout has the residual test error

$$\text{MSE}(\mathbf{W}, M, \gamma, \sigma) = \gamma^2 + M^{-1} \mathbf{W} \Sigma_\sigma \mathbf{W}^T. \quad (50)$$

As a consequence, we may interpret the test variance as an additional irreducible source of noise specific to

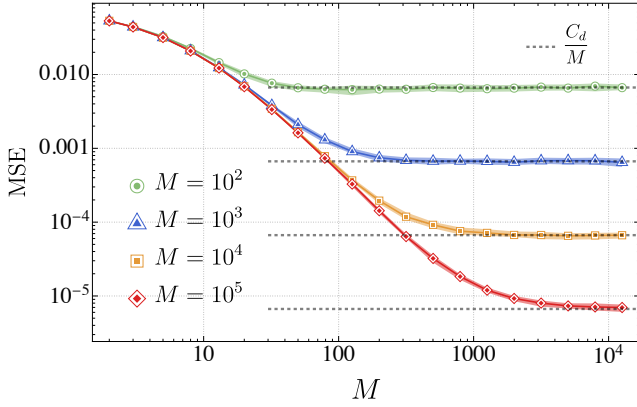


Figure 5. (LI: $\text{Var}(\hat{\sigma}^*)$ for consistency with fig1?)

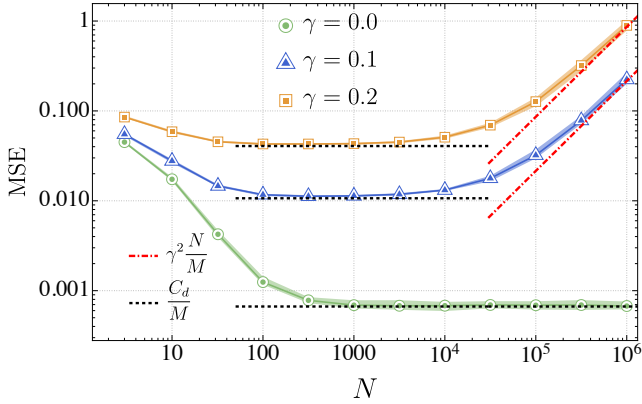


Figure 6. (LI: the $\gamma^2 N/M$ fits are really just this or also include the other base line $\gamma^2 d^2/n_{\text{tr}}$ stuff?)

the quantum setting. Although it can be reduced by increasing the number of measurements during testing, it can never be completely eliminated in realistic scenarios.

Target noise has an additional effect when it is present during training. We model the training targets by

$$\hat{\mathbf{y}} = \mathbf{y} + \hat{\boldsymbol{\eta}}, \quad \hat{\boldsymbol{\eta}} \mid (\hat{\boldsymbol{\Sigma}}, \mathbf{P}) \sim \mathcal{N}(0, \gamma^2 \mathbf{I}_{n_{\text{tr}}}), \quad (51)$$

and take $\hat{\boldsymbol{\eta}}$, $\hat{\boldsymbol{\eta}}_{\sigma}$, and the test sampling to be mutually independent. This homoscedastic Gaussian model isolates the rank effect; it also describes unbiased target-estimation errors when correlations with the feature noise can be neglected. The corresponding estimator is

$$\hat{\mathbf{W}}_{N,\gamma} = \hat{\mathbf{y}}^T \hat{\mathbf{P}}_N^+ = \hat{\mathbf{W}}_N + \hat{\boldsymbol{\eta}}^T \hat{\mathbf{P}}_N^+. \quad (52)$$

Because the second term is conditionally centered, target noise does not change the training bias. It does, however, change both variances in the finite-statistics error decomposition.

Consider first the test variance. Averaging the fixed-readout expression in eq. (12) over the training ran-

domness gives the exact identity

$$\begin{aligned} \mathbb{E}_{\hat{\boldsymbol{\Sigma}}, \hat{\boldsymbol{\eta}}}[\text{Var}_{\text{test}}(\hat{\mathbf{W}}_{N,\gamma}, M, \sigma)] &= \\ &= \mathbb{E}_{\hat{\boldsymbol{\Sigma}}}[\text{Var}_{\text{test}}(\hat{\mathbf{W}}_N, M, \sigma)] + \Delta \text{Var}_{\text{test}}^{(\eta)}(N, M, \sigma), \end{aligned} \quad (53)$$

where we defined extra noise-dependent reads:

$$\Delta \text{Var}_{\text{test}}^{(\eta)}(N, M, \sigma) \equiv \frac{\gamma^2}{M} \mathbb{E}_{\hat{\boldsymbol{\Sigma}}} \text{Tr}[\hat{\mathbf{P}}_N^+ \boldsymbol{\Sigma}_{\sigma} (\hat{\mathbf{P}}_N^+)^T]. \quad (54)$$

Unlike the exact targets $\mathbf{y}$, the isotropic noise $\hat{\boldsymbol{\eta}}$ and the categorical covariance matrix $\boldsymbol{\Sigma}_{\sigma}$ have a component in $\ker(\mathbf{P})$. They therefore probe the divergent (2,2) block of $\hat{\mathbf{P}}_N^+$ in eq. (26).

The same target fluctuations also contribute to the training variance. If $\text{Var}_{\text{tr}}(N, \gamma, \sigma)$ denotes the variance of $\hat{\mathbf{W}}_{N,\gamma} \mathbf{p}_{\sigma}$ over both $\hat{\boldsymbol{\Sigma}}$ and $\hat{\boldsymbol{\eta}}$, the law of total variance gives

$$\begin{aligned} \text{Var}_{\text{tr}}(N, \gamma, \sigma) &= \text{Var}_{\text{tr}}(N, \sigma) + \Delta \text{Var}_{\text{tr}}^{(\eta)}(N, \sigma), \\ \Delta \text{Var}_{\text{tr}}^{(\eta)}(N, \sigma) &\equiv \gamma^2 \mathbb{E}_{\hat{\boldsymbol{\Sigma}}} \|\hat{\mathbf{P}}_N^+ \mathbf{p}_{\sigma}\|_2^2. \end{aligned} \quad (55)$$

This expression contains the right projection by the physical probability vector $\mathbf{p}_{\sigma} \in \text{im}(\mathbf{P})$ and therefore does not probe the divergent (2,2) block directly. However, it probes the (2,1) block that approaches a constant as $N \rightarrow \infty$. This difference between eqs. (54) and (55) is responsible for their markedly different scaling with N . Combining the two stages, the MSE averaged over training-target noise, training-feature noise, test sampling, and test-target noise is therefore

$$\begin{aligned} \text{MSE}(N, M, \gamma, \sigma) &= \gamma^2 + \text{bias}^2(N, \sigma) + \text{Var}_{\text{tr}}(N, \gamma, \sigma) \\ &\quad + \mathbb{E}_{\hat{\boldsymbol{\Sigma}}, \hat{\boldsymbol{\eta}}}[\text{Var}_{\text{test}}(\hat{\mathbf{W}}_{N,\gamma}, M, \sigma)]. \end{aligned} \quad (56)$$

The leading coefficients can be evaluated explicitly in the homogeneous Gaussian reference model of section V. The derivation is given in section SM5.4. Assuming that the training and test ensembles have the same average categorical covariance, $\mathbb{E}_{\sigma}[\boldsymbol{\Sigma}_{\sigma}] = \bar{\boldsymbol{\Sigma}}$, one obtains for large N

$$\mathbb{E}_{\sigma}[\Delta \text{Var}_{\text{test}}^{(\eta)}(N, M, \sigma)] = \frac{N\gamma^2}{M} R + O\left(\frac{\gamma^2 \sqrt{N}}{M}\right), \quad (57)$$

where we introduced the coefficient R that depends on n_{tr} , n_{out} and d only as:

$$R = \frac{n_{\text{out}} - d^2}{n_{\text{tr}} - n_{\text{out}} - 1} \quad (58)$$

The inverse-Wishart mean in this expression exists for $n_{\text{tr}} > n_{\text{out}} + 1$. Thus, for fixed nonzero γ , target noise produces a test-variance contribution proportional to N/M . Increasing the number of training shots drives

$\hat{P}_N$ closer to the rank-deficient matrix P and makes the unregularized inverse more sensitive to the component of $\hat{\eta}$ in $\ker(P)$.

By contrast, the training-variance correction is

$$\Delta \text{Var}_{\text{tr}}^{(\eta)}(N, \sigma) = \gamma^2 R \|P^{-1} p_\sigma\|_2^2 + O\left(\frac{\gamma^2}{\sqrt{N}}\right). \quad (59)$$

This norm is explicit in the symmetric reference model already used in section SM4.3. Indeed, $P_{11}^{-1} p_{\sigma,1} = \langle \tilde{p}, \sigma \rangle$, and the tight dual frame in eq. (SM122) gives, for any density operator σ ,

$$\|P^{-1} p_\sigma\|_2^2 = \sum_{i=1}^{n_{\text{tr}}} \langle \tilde{p}_i, \sigma \rangle^2 = \frac{d}{n_{\text{tr}}} \left[(d+1) \text{tr}(\sigma^2) - 1 \right]. \quad (60)$$

In particular, for the pure test states considered there this reduces exactly to d^2/n_{tr} . Thus the leading target-noise contribution in eq. (59) is $\gamma^2 d^2 R/n_{\text{tr}}$ in that model. Hence, at leading order, finite-shot training features leave the multiplicative excess

$$\frac{n_{\text{tr}} - d^2 - 1}{n_{\text{tr}} - n_{\text{out}} - 1} = 1 + \frac{n_{\text{out}} - d^2}{n_{\text{tr}} - n_{\text{out}} - 1}. \quad (61)$$

Consequently, target noise adds an $O(\gamma^2/n_{\text{tr}})$ training-variance floor, but no factor growing with N . For $n_{\text{tr}} \gg n_{\text{out}}$ at fixed d , the relative inflation in eq. (61) is only $O(n_{\text{out}}/n_{\text{tr}})$, while the test-variance correction remains $O(N\gamma^2 n_{\text{out}}/(Mn_{\text{tr}}))$. Increasing M or n_{tr} therefore suppresses the amplified test contribution; increasing N alone does not.

WE NEED TO DISCUSS PLOTS HERE, NO TRUE DESCRIPTION In fig. 7, we compare the large-training-set Gaussian prediction in eq. (57) with numerical results for $d = 2$, $n_{\text{out}} = 64$, $n_{\text{tr}} = 2048$, $M = 1$, and $\gamma^2 = 0.04$. The theory gives the mean, whereas the numerical curve reports the median. In this strongly over-determined regime the inverse-Wishart trace is sharply concentrated, so the two statistics are close on the scale shown.

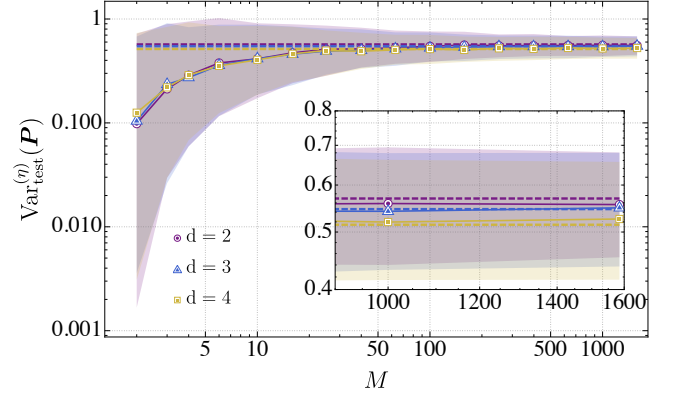


Figure 7. Comparison between numerical simulation and the large- p Gaussian prediction for the label-noise contribution to the test variance as N increases. Here $d = 2$, $M = 1$, $n_{\text{out}} = 64$, $n_{\text{tr}} = 2048$, and $\gamma^2 = 0.04$. The numerical median is taken over 1000 experiments.

VIII. CONCLUSIONS AND PERSPECTIVES

We analyzed how finite measurement statistics affect quantum linear regression with probability features, focusing on the unregularized pseudoinverse training method. Unlike many classical settings, the feature (probability) matrix generated by an informationally complete measurement and a sufficiently rich training set is intrinsically constrained by the geometry of quantum state space: in the ideal, infinite-shot limit, it is rank-deficient, while in realistic experiments its empirical estimate is perturbed by shot noise and becomes effectively full rank. This seemingly innocuous mismatch turns finite sampling into a qualitatively nontrivial source of error, since learning depends on solving a linear inversion problem whose sensitivity is strongly amplified by rank-increasing fluctuations.

To address this, we developed a perturbative approach tailored to the rank-changing regime induced by shot noise. This framework revealed an explicit bias-variance decomposition of the mean-squared prediction error, first at fixed noise realization and then after averaging over noise using standard concentration arguments. A key outcome is that the dominant contributions exhibit distinct and operationally meaningful scalings: the squared bias term can persist even in the quantum analogue of the underparametrized regime ($n_{\text{tr}} > n_{\text{out}}$) and scales as N^{-2} , whereas the variance scales as $(Nn_{\text{tr}})^{-1}$. These behaviors highlight a trade-off that departs from classical intuition: increasing the number of training states is not necessarily beneficial if the per-state measurement budget is too small, because the irreducible bias induced by rank-increasing noise cannot be mitigated by more data alone.

Beyond the specific QELM setting, our results suggest a broader lesson for quantum machine learning:

resource estimates and performance guarantees must explicitly account for finite-shot effects and for constraints imposed by informational completeness. In particular, the analysis motivates training protocols and benchmarking criteria that treat the total experimental budget as a joint allocation problem between the number of training states and the number of shots per state, rather than prioritizing dataset size in isolation.

Several directions follow naturally from this work. First, in a companion paper [XXXXXXXXXXXXX] we will explore additional operating regimes in which finite sampling alone can induce double-descent-like behavior, and we will characterize how the location and sharpness of these transitions depend on the measurement budget and on the structure of the feature matrix. Second, it will be important to extend our analysis to other quantum linear regres-

sion paradigms—most notably fidelity-kernel methods and quantum reservoir computing—where the features have a different statistical origin but are still estimated from finite-shot data. In the same spirit, a systematic comparison of learning strategies based on regularized estimators (e.g., ridge/Tikhonov-type schemes, truncated inverses, early stopping) and alternative linear solvers is left for future work, with the goal of identifying when regularization mitigates shot noise-induced bias without overly inflating variance. Finally, a pressing open question is how these effects propagate beyond linear models: it is natural to ask to what extent rank-mismatch and finite-statistics corrections persist, or are amplified, in more general QML architectures—such as variational circuits, hybrid models, and deep quantum kernels—and whether they can be leveraged to derive resource-aware training principles across quantum machine learning.

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SUPPLEMENTAL MATERIALS

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SM1. PRELIMINARIES

We use the following linear-algebra notation throughout the main text and the Supplemental Material. Lowercase bold letters, such as $\boldsymbol{v}$, denote vectors, with components v_k , whereas uppercase bold letters, such as $\boldsymbol{M}$, denote matrices. This section also fixes the common coordinate system used by the two pseudoinverse analyses later in the SM. [Section SM1.1](#) translates operators and operator collections into real vectors and matrices, [section SM1.2](#) reviews the Landau notation for deterministic sequences and sequences of random variables used in the main text and the appendix, [section SM1.3](#) introduces the signal, kernel, and cokernel blocks of the ideal probability matrix, and [section SM1.4](#) records the pseudoinverse identities used throughout [sections SM2](#) and [SM3](#).

SM1.1. Vectorized notation

This subsection connects the operator language of quantum states, POVMs, and frames with the real matrix coordinates used in all subsequent calculations. In particular, it makes the frame operators of [section SM1.5](#) and the probability-matrix blocks of [section SM1.3](#) part of one consistent notation. Given a Hilbert space $\mathcal{H}$ of dimension d , let $\mathcal{L}(\mathcal{H})$ denote its space of linear operators, equipped with the Hilbert–Schmidt inner product $\langle A, B \rangle = \text{Tr}(A^\dagger B)$. Our coordinate matrices represent the real Hilbert space $\text{Herm}(\mathcal{H}) \subset \mathcal{L}(\mathcal{H})$ using a Hermitian orthonormal basis; hence its dimension is d^2 , its coordinate vectors are real, and matrix adjoints reduce to transposes. Ordinary uppercase Roman and Greek letters denote operators. For a vector $\psi \in \mathcal{H}$, we write $\mathbb{P}(\psi) = \mathbb{P}_\psi \equiv |\psi\rangle\langle\psi|$. We also use the same symbol for the rank-one superoperator generated by $A \in \mathcal{L}(\mathcal{H})$:

$$\mathbb{P}(A) : \mathcal{L}(\mathcal{H}) \rightarrow \mathcal{L}(\mathcal{H}), \quad \mathbb{P}(A)(B) \equiv \langle A, B \rangle A = \text{Tr}(A^\dagger B) A. \quad (\text{SM1})$$

This superoperator is an orthogonal projector when $\langle A, A \rangle = 1$; otherwise it is the corresponding unnormalized rank-one outer product.

Given a Hermitian orthonormal operator basis $(\mathcal{P}_i)_{i=1}^{d^2}$ (for example, normalized Pauli operators), any $\mathcal{O} \in \text{Herm}(\mathcal{H})$ has the real vectorized representation $\boldsymbol{v}_\mathcal{O} = \text{vec}(\mathcal{O})$ and $(\boldsymbol{v}_\mathcal{O})_i = \text{Tr}(\mathcal{P}_i^\dagger \mathcal{O})$. Collections of Hermitian operators are denoted by bold Greek letters, as in the POVM $\boldsymbol{\mu} = (\mu_a)_{a=1}^m$. Such a collection is represented by the matrix

$$\boldsymbol{M}_\boldsymbol{\mu} \equiv \sum_{a=1}^m \boldsymbol{v}_{\mu_a} \boldsymbol{e}_a^T = \sum_{a=1}^m \text{vec}(\mu_a) \boldsymbol{e}_a^T, \quad (\text{SM2})$$

where $\boldsymbol{e}_a$ is the a th canonical basis vector of $\mathbb{R}^m$.

SM1.2. Asymptotic notation

Several results below concern the asymptotic behavior of the shot-noise matrix as $n_{\text{tr}} \rightarrow \infty$. For deterministic quantities, we use the standard Landau notation, with the subscript indicating the variable with respect to which the asymptotic limit is taken. Thus, for instance, $a_n = o_n(b_n)$ means

$$\frac{a_n}{b_n} \longrightarrow 0 \quad \text{as } n \rightarrow \infty,$$

while $a_n = O_n(b_n)$ means that $|a_n/b_n|$ remains bounded as $n \rightarrow \infty$. When the asymptotic variable is clear from the context, we may omit the subscript and simply write $o(b_n)$ and $O(b_n)$.

For a generic sequence of random variables $(\hat{Z}_n)_n$, we write $\hat{Z}_n = o_{\mathbb{P}}(1)$, or $\hat{Z}_n \xrightarrow{\mathbb{P}} 0$, if, for every $\eta > 0$, $\mathbb{P}(|\hat{Z}_n| > \eta) \rightarrow 0$ as $n \rightarrow \infty$. More generally, $\hat{Z}_n = o_{\mathbb{P}}(a_n)$ means $\hat{Z}_n/a_n \xrightarrow{\mathbb{P}} 0$, whereas $\hat{Z}_n = O_{\mathbb{P}}(a_n)$ means that the sequence $(\hat{Z}_n/a_n)_n$ is bounded in probability.

SM1.3. Block decompositions

We decompose noisy matrices with respect to the support and image of the exact probability matrix. These coordinates isolate the rank-preserving and rank-increasing directions and are used both in the formal expansion of

section SM2 and in the direct remainder analysis of section SM3. In particular,

$$\hat{\mathbf{E}} = \begin{pmatrix} \hat{\mathbf{E}}_{11} & \hat{\mathbf{E}}_{12} \\ \hat{\mathbf{E}}_{21} & \hat{\mathbf{E}}_{22} \end{pmatrix}, \quad (\text{SM3})$$

where the block decomposition is induced by the support and image of the noiseless probability matrix P . More explicitly, let

$$\Pi_1 \equiv \Pi_{\text{supp}(P)} = V_1 V_1^T, \quad \Pi_2 \equiv I - \Pi_1 = V_2 V_2^T, \quad \Pi_{\bar{1}} \equiv \Pi_{\text{im}(P)} = U_1 U_1^T, \quad \Pi_{\bar{2}} \equiv I - \Pi_{\bar{1}} = U_2 U_2^T. \quad (\text{SM4})$$

We write $r \equiv \text{rank}(P) = d^2$. Thus $V_1 \in \mathbb{R}^{n_{\text{tr}} \times r}$, $V_2 \in \mathbb{R}^{n_{\text{tr}} \times (n_{\text{tr}} - r)}$, $U_1 \in \mathbb{R}^{n_{\text{out}} \times r}$, and $U_2 \in \mathbb{R}^{n_{\text{out}} \times (n_{\text{out}} - r)}$. The blocks of $\hat{\mathbf{E}}$ are then $\hat{\mathbf{E}}_{ab} \equiv U_a^T \hat{\mathbf{E}} V_b$, where $a, b \in \{1, 2\}$. Here and below the index a labels the image/cokernel block, while b labels the support/kernel block. The normalized kernel-block covariances introduced below require $n_{\text{tr}} - r > 0$. When $n_{\text{tr}} = r$ there is no kernel block, and the corresponding statements are read after deleting every block with second index 2.

Throughout this section, the training states $\rho = (\rho_1, \dots, \rho_{n_{\text{tr}}})$ are fixed and all averages are conditional on them; the only randomness is the training shot noise. We write the columns of $\hat{\mathbf{E}}$ as

$$\hat{\mathbf{E}} = [\hat{\xi}_1, \dots, \hat{\xi}_{n_{\text{tr}}}], \quad \hat{\xi}_i = \sqrt{N}(\hat{p}_i - p_i), \quad \Sigma_i \equiv \Sigma(\rho_i) \equiv \mathbb{E}[\hat{\xi}_i \hat{\xi}_i^T | \rho] = \text{diag}(p_i) - p_i p_i^T, \quad (\text{SM5})$$

where $p_i \equiv \langle \mu, \rho_i \rangle$, $\hat{p}_i = \frac{1}{N} \hat{n}^{(i)}$, $\hat{n}^{(i)} \sim \text{Mult}(N, p_i)$. Conditionally on ρ , the columns $\hat{\xi}_i$ are independent and centered, with covariance matrices Σ_i . They become asymptotically Gaussian as $N \rightarrow \infty$, but no Gaussian approximation is assumed here.

It will be useful to introduce the following normalized covariance blocks. Given a nonzero projection $\Pi \in \mathbb{R}^{n_{\text{tr}} \times n_{\text{tr}}}$ and an isometry $U \in \mathbb{R}^{n_{\text{out}} \times m}$, define $\hat{C}_{U, \Pi} \equiv \frac{1}{\text{tr}(\Pi)} U^T \hat{\mathbf{E}} \Pi \hat{\mathbf{E}}^T U$, and $C_{U, \Pi} \equiv \frac{1}{\text{tr}(\Pi)} \sum_{i=1}^{n_{\text{tr}}} \Pi_{ii} U^T \Sigma_i U$, and note that $\mathbb{E}[\hat{C}_{U, \Pi}] = C_{U, \Pi}$. The off-diagonal entries of Π do not contribute to the expectation because distinct columns of $\hat{\mathbf{E}}$ are independent and centered. For the blocks relevant to the perturbative expansion we use the shorthand

$$\hat{C}_{\alpha\beta} \equiv \frac{U_\alpha^T \hat{\mathbf{E}} \Pi_2 \hat{\mathbf{E}}^T U_\beta}{n_{\text{tr}} - r} = \frac{\hat{\mathbf{E}}_{\alpha 2} \hat{\mathbf{E}}_{\beta 2}^T}{n_{\text{tr}} - r}, \quad C_{\alpha\beta} \equiv \mathbb{E}[\hat{C}_{\alpha\beta}] = \frac{1}{n_{\text{tr}} - r} \sum_i (\Pi_2)_{ii} U_\alpha^T \Sigma_i U_\beta, \quad \alpha, \beta \in \{1, 2\}. \quad (\text{SM6})$$

Thus $\hat{C}_{11} \in \mathbb{R}^{r \times r}$, $\hat{C}_{12} \in \mathbb{R}^{r \times (n_{\text{out}} - r)}$, $\hat{C}_{21} \in \mathbb{R}^{(n_{\text{out}} - r) \times r}$, and $\hat{C}_{22} \in \mathbb{R}^{(n_{\text{out}} - r) \times (n_{\text{out}} - r)}$. Moreover, $C_{11}, C_{22} \succeq 0$ and $\text{tr}(C_{11}) + \text{tr}(C_{22}) \leq 1$. Because Π_2 projects onto $\text{supp}(P)^\perp$, $\text{tr}(\Pi_2) = \sum_i (\Pi_2)_{ii} = n_{\text{tr}} - r$; hence the normalization prevents C_{ab} from acquiring a generic factor of n_{tr} . The diagonal weights can be made explicit. Since

$$\Pi_1 = P^+ P = P^T (P P^T)^+ P = M_\rho^T \mathcal{F}_\rho^{-1} M_\rho,$$

we have $(\Pi_1)_{ii} = \langle \rho_i, \mathcal{F}_\rho^{-1}(\rho_i) \rangle = \langle \rho_i, \tilde{\rho}_i \rangle$ and $(\Pi_2)_{ii} = 1 - \langle \rho_i, \tilde{\rho}_i \rangle$. In the special case where the training states are pure and form a complex-projective 2-design, we get $(\Pi_2)_{ii} = 1 - d^2/n_{\text{tr}} = (n_{\text{tr}} - r)/n_{\text{tr}}$, and thus $C_{ab} = U_a^T \bar{\Sigma} U_b$ where $\bar{\Sigma} \equiv \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \Sigma_i$ is the average covariance matrix. The 2-design property is also sufficient to get an explicit deterministic formula for $\bar{\Sigma}$, which in this case equals

$$\bar{\Sigma}_{ab} = \frac{\text{tr}(\mu_a)}{d} \delta_{ab} - \frac{\text{tr}(\mu_a) \text{tr}(\mu_b) + \text{tr}(\mu_a \mu_b)}{d(d+1)}. \quad (\text{SM7})$$

If we further assume that the POVM is balanced and rank-one, i.e. $\mu_a = \frac{d}{n_{\text{out}}} |\phi_a\rangle\langle\phi_a|$ for some pure states $|\phi_a\rangle$ satisfying $\frac{1}{n_{\text{out}}} \sum_{a=1}^{n_{\text{out}}} |\phi_a\rangle\langle\phi_a| = \frac{I}{d}$, then $C_{12} = 0$ and $C_{22} = \frac{I}{n_{\text{out}}}$. These formulas will be used in section SM4.3 and in the Gaussian consistency check of section SM5.2.

SM1.4. Properties of pseudoinverses

This final notational subsection collects the pseudoinverse facts needed to turn the block decomposition of section SM1.3 into the formal expansion of section SM2. It also fixes the projector identities used later in the direct and probabilistic arguments.

Definition via SVD — The pseudoinverse of a rectangular matrix is most conveniently expressed through its compact singular value decomposition (SVD). Given a matrix A , write $A = \mathbf{U}\mathbf{D}\mathbf{V}^\dagger$, where $\mathbf{U}, \mathbf{V}$ are isometries and $\mathbf{D} > 0$ is diagonal. Its pseudoinverse then reads $A^+ = \mathbf{V}\mathbf{D}^{-1}\mathbf{U}^\dagger$. If A is $n \times m$, then $\mathbf{U}$ is $n \times r$, $\mathbf{D}$ is $r \times r$, and $\mathbf{V}$ is $m \times r$, where $r \equiv \text{rank}(A)$. In dyadic notation, this amounts to $A = \sum_k d_k \mathbf{u}_k \mathbf{v}_k^\dagger$ having pseudoinverse $A^+ = \sum_k d_k^{-1} \mathbf{v}_k \mathbf{u}_k^\dagger$.

Properties of the pseudoinverse — The pseudoinverse A^+ is a true (left and right) inverse iff A is invertible. More generally, A^+ is a left inverse, $A^+A = \mathbf{I}$, if and only if A is injective, and it is a right inverse, $AA^+ = \mathbf{I}$, if and only if A is surjective. If A is not full-rank, then A^+ is neither a left nor a right inverse, but it can still be seen as an inverse on the range of support of A . In general, $A^+A = \Pi_{\text{supp}(A)}$ is the orthogonal projector onto the row space of A , whereas $AA^+ = \Pi_{\text{im}(A)}$ projects onto its image.

Explicit formulas for special cases — The pseudoinverse can be given an explicit formula when A is injective or surjective. For injective A , we have $A^+ = (A^\dagger A)^{-1} A^\dagger$. For surjective A , we have $A^+ = A^\dagger (AA^\dagger)^{-1}$.

Product formulas — It is not generally the case that $(AB)^+ = B^+A^+$. However, it does hold when A is injective and B is surjective. Indeed, the compact SVD of B is $B = \mathbf{U}_B \mathbf{D}_B \mathbf{V}_B^\dagger$, with $\mathbf{U}_B$ unitary, whereas that of A is $A = \mathbf{U}_A \mathbf{D}_A \mathbf{V}_A^\dagger$, with $\mathbf{V}_A$ unitary. Therefore

$$AB = \mathbf{U}_A \mathbf{D}_A \mathbf{V}_A^\dagger \mathbf{U}_B \mathbf{D}_B \mathbf{V}_B^\dagger \quad (\text{SM8})$$

where $\mathbf{V}_A^\dagger \mathbf{U}_B$ is unitary and $\mathbf{D}_A, \mathbf{D}_B$ are square and invertible. Hence $C \equiv \mathbf{D}_A \mathbf{V}_A^\dagger \mathbf{U}_B \mathbf{D}_B$ is also square and invertible. Let $C = \mathbf{U}_C \mathbf{D}_C \mathbf{V}_C^\dagger$ be its SVD. Then

$$AB = (\mathbf{U}_A \mathbf{U}_C) \mathbf{D}_C (\mathbf{V}_B \mathbf{V}_C)^\dagger \quad (\text{SM9})$$

is an SVD for AB , with $\mathbf{U}_C, \mathbf{V}_C$ unitaries, hence

$$(AB)^+ = (\mathbf{V}_B \mathbf{V}_C) \mathbf{D}_C^{-1} (\mathbf{U}_A \mathbf{U}_C)^\dagger = \mathbf{V}_B C^{-1} \mathbf{U}_A^\dagger, \quad (\text{SM10})$$

which equals B^+A^+ .

Block inverses and Schur complements — Given an invertible square matrix $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, its inverse can be written in terms of Schur complements [55] as

$$\begin{aligned} M^{-1} &= \begin{pmatrix} A^{-1} + A^{-1}B(M/A)^{-1}CA^{-1} & -A^{-1}B(M/A)^{-1} \\ -(M/A)^{-1}CA^{-1} & (M/A)^{-1} \end{pmatrix} \\ &= \begin{pmatrix} (M/D)^{-1} & -(M/D)^{-1}BD^{-1} \\ -D^{-1}C(M/D)^{-1} & D^{-1} + D^{-1}C(M/D)^{-1}BD^{-1} \end{pmatrix}, \end{aligned} \quad (\text{SM11})$$

where

$$(M/A) \equiv D - CA^{-1}B, \quad (M/D) \equiv A - BD^{-1}C. \quad (\text{SM12})$$

The first representation assumes that A is invertible, whereas the second assumes that D is invertible. This identity is used by both the complementary structural analysis and the direct proof below.

SM1.5. Frame operators

This section supplies the frame-theoretic bridge between quantum measurements and the regression coefficients used later in [section SM4](#). We introduce frames, their canonical duals, and their frame operators, and then specialize these constructions to state ensembles and POVMs. The symmetric-design identities derived at the end of the section are the ingredients used to evaluate the leading resource coefficients in [section SM4.3](#). We make frequent use of frames and frame operators [56–58]. For a finite-dimensional inner-product space V , a frame is a collection $(v_k)_k \subset V$ for which

$$\alpha \|w\|^2 \leq \sum_k |\langle v_k, w \rangle|^2 \leq \beta \|w\|^2 \quad (\text{SM13})$$

for some $0 < \alpha \leq \beta < \infty$. Frames generalize orthonormal bases by allowing overcompleteness. Given any dual frame $(\tilde{v}_k)_k$, every $w \in V$ admits the reconstructions

$$w = \sum_k \langle v_k, w \rangle \tilde{v}_k = \sum_k \langle \tilde{v}_k, w \rangle v_k. \quad (\text{SM14})$$

The canonical dual is $\tilde{v}_k = \mathcal{F}^{-1}(v_k)$, where the positive, invertible frame operator is $\mathcal{F} = \sum_k v_k v_k^\dagger$. An overcomplete frame has nonunique duals.

The frame operator $\mathcal{F}_\rho \equiv M_\rho M_\rho^T$ associated with the training set ρ and the frame operator $\mathcal{F}_\mu \equiv M_\mu M_\mu^T$ associated with the POVM μ are intimately related to the probability matrix $P \equiv \langle \mu, \rho \rangle \equiv M_\mu^T M_\rho$. In particular, this relation naturally emerges by looking at their singular values. The singular values of the probability matrix are by definition the square roots of the eigenvalues of $PP^T = M_\mu^T M_\rho M_\rho^T M_\mu$ that in turn are the same as the nonzero eigenvalues of $M_\rho M_\rho^T M_\mu M_\mu^T = \mathcal{F}_\rho \mathcal{F}_\mu$. We therefore conclude that the nonzero singular values of P are equal to the square roots of the nonzero eigenvalues of $\mathcal{F}_\mu \mathcal{F}_\rho$.

Case of symmetric states and measurement — In particular, if the training states are pure, $\rho_k = \mathbb{P}(\psi_k)$, and form a complex-projective 2-design, then in superoperator notation

$$\mathcal{F}_\rho \equiv \sum_{k=1}^{n_{\text{tr}}} \mathbb{P}(\rho_k) = n_{\text{tr}} \frac{\text{Id} + \mathbb{P}(\mathbf{I})}{d(d+1)}, \quad (\text{SM15})$$

where Id is the identity superoperator and $\mathbb{P}(\mathbf{I})(X) = \text{Tr}(X)\mathbf{I}$, with $\mathbf{I}$ the identity operator on $\mathcal{H}$. Note that, as a linear operator, this $\mathcal{F}_\rho$ has eigenvalues $\frac{n_{\text{tr}}}{d}$ and $\frac{n_{\text{tr}}}{d(d+1)}$, with degeneracies 1 and $d^2 - 1$, respectively. If the training states are independent Haar-random pure states, then their normalized empirical frame operator approaches this form with high probability as n_{tr} grows. Similarly, symmetric rank-one POVMs have the form $\mu_b = \frac{d}{n_{\text{out}}} \mathbb{P}(\phi_b)$ with the states $|\phi_b\rangle$ such that $\sum_{b=1}^{n_{\text{out}}} \mathbb{P}(\phi_b) = \frac{n_{\text{out}}}{d} \mathbf{I}$. If these states further form a complex-projective 2-design, then the frame superoperator reads

$$\mathcal{F}_\mu \equiv \sum_{b=1}^{n_{\text{out}}} \mathbb{P}(\mu_b) = \frac{d^2}{n_{\text{out}}} \frac{\text{Id} + \mathbb{P}(\mathbf{I})}{d(d+1)}. \quad (\text{SM16})$$

Furthermore, if both states and POVM have these symmetric properties, then we find

$$\mathcal{F}_\rho \circ \mathcal{F}_\mu = \frac{n_{\text{tr}} d^2}{n_{\text{out}} [d(d+1)]^2} (\text{Id} + \mathbb{P}(\mathbf{I}))^2 = \frac{n_{\text{tr}}}{n_{\text{out}} (d+1)^2} (\text{Id} + (2+d)\mathbb{P}(\mathbf{I})), \quad (\text{SM17})$$

whose eigenvalues are

$$\sigma(\mathcal{F}_\rho \circ \mathcal{F}_\mu) \in \left\{ \frac{n_{\text{tr}}}{n_{\text{out}}}, \frac{n_{\text{tr}}}{n_{\text{out}}(d+1)^2} \right\}. \quad (\text{SM18})$$

The eigenvector corresponding to $n_{\text{tr}}/n_{\text{out}}$ is the rescaled identity, $\mathbf{I}/\sqrt{d}$, while the other eigenvalue has degeneracy $d^2 - 1$, and eigenspace spanned by any choice of orthonormal basis for the space of Hermitian traceless $d \times d$ operators. The singular values of P are then the square roots of the eigenvalues of $\mathcal{F}_\rho \circ \mathcal{F}_\mu$. Thus all d^2 nonzero singular values of the $n_{\text{out}} \times n_{\text{tr}}$ probability matrix scale as $\sqrt{n_{\text{tr}}/n_{\text{out}}}$ (with the two dimension-dependent prefactors shown above).

SM2. PERTURBATIVE PSEUDOINVERSE EXPANSION

In this section we present a perturbative analysis of the pseudoinverse of the empirical probability matrix. More generally, we consider the pseudoinverse of a rectangular singular matrix subject to a generally full-rank perturbation. The main results are the expansions in [eqs. \(SM30\) and \(SM32\)](#). Here we focus only on the mathematical structure of the perturbative expansion. This formal calculation explains which blocks become singular when shot noise changes the rank. The independent direct analysis in [section SM3](#) then establishes uniform probabilistic control of the prediction error, and [section SM4](#) translates that rigorous control into training-resource statements.

The main technical difficulty is that matrix inversion is not continuous with respect to perturbations that change the rank, and the same is true for the Moore–Penrose pseudoinverse [33, 34, 52]. Let P be the unperturbed matrix and let $\hat{P}_\tau = P + \tau \hat{\Xi}$. Here τ is a generic parameter; we will use $\tau = N^{-1/2}$ when specializing the discussion to the empirical frequency matrix. If P is rank deficient and the perturbation $\hat{\Xi}$ has components outside the row or column spaces of P , then $\tau \mapsto \hat{P}_\tau^+$ is discontinuous at $\tau = 0$, and in particular $\hat{P}_\tau^+$ can diverge as $\tau \rightarrow 0$. More precisely, when the perturbation changes the rank, one has the lower bound [34]

$$\|\hat{P}_\tau^+ - P^+\|_{\text{op}} \geq \frac{1}{\tau \|\hat{\Xi}\|_{\text{op}}}. \quad (\text{SM19})$$

For finite-dimensional matrices, continuity of the pseudoinverse is equivalent to the rank being stable under the perturbation. However, the stochastic setting considered here generally does cause the rank to increase, being P has rank at most d^2 , while shot noise makes $\hat{P}$ have maximal rank $\min(n_{\text{tr}}, n_{\text{out}})$ with high probability. Here and throughout, $\|\cdot\|_{\text{op}}$ denotes the spectral norm, i.e. the largest singular value.

To illustrate the structural effect of eq. (SM11), suppose now that a symmetric singular matrix M undergoes a rank-increasing perturbation $M_\tau \equiv M + \tau X$, and let $x \in \text{supp}(M)$ and $y \in \text{im}(M)$. In a basis adapted to $\text{supp}(M) \oplus \ker(M)$, we can write these variables more explicitly as

$$M = \begin{pmatrix} M_{11} & 0 \\ 0 & 0 \end{pmatrix}, \quad M_\tau = \begin{pmatrix} M_{11} + \tau X_{11} & \tau X_{12} \\ \tau X_{21} & \tau X_{22} \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ 0 \end{pmatrix}, \quad y = \begin{pmatrix} y_1 \\ 0 \end{pmatrix}. \quad (\text{SM20})$$

The Schur-complement identities in eq. (SM11) then give

$$x^T M_\tau^{-1} y = x_1^T (M_\tau / \tau X_{22})^{-1} y_1 = x_1^T [M_{11} + \tau (X / X_{22})]^{-1} y_1, \quad (\text{SM21})$$

where X_{ij} are the blocks of X and $X / X_{22} \equiv X_{11} - X_{12} X_{22}^{-1} X_{21}$. Thus, to leading order in τ , we have

$$x^T M_\tau^{-1} y = x_1^T M_{11}^+ y_1 - \tau x_1^T M_{11}^{-1} (X / X_{22}) M_{11}^{-1} y_1 + O(\tau^2). \quad (\text{SM22})$$

This can be compared to the naive perturbative expansion, which would have been $x^T M_\tau^{-1} y = x_1^T M_{11}^+ y_1 - \tau x_1^T M_{11}^{-1} X_{11} M_{11}^{-1} y_1 + O(\tau^2)$. Thus, although the full pseudoinverse is discontinuous under a rank-increasing perturbation, the bilinear form $x^T M_\tau^{-1} y$ remains finite for vectors supported on M , but the divergent block of M_τ^{-1} nonetheless changes the order- τ correction by replacing X_{11} with $X_{11} - X_{12} X_{22}^{-1} X_{21}$.

Let us now specialize this observation to the probability matrix. In the block coordinates introduced in section SM1.3, we write $A_{ij} \equiv U_i^T A V_j$, $i, j = 1, 2$, and thus

$$\hat{P}_\tau \equiv \begin{pmatrix} U_1^T \hat{P}_\tau V_1 & U_1^T \hat{P}_\tau V_2 \\ U_2^T \hat{P}_\tau V_1 & U_2^T \hat{P}_\tau V_2 \end{pmatrix} = \begin{pmatrix} P_{11} + \tau \hat{\Xi}_{11} & \tau \hat{\Xi}_{12} \\ \tau \hat{\Xi}_{21} & \tau \hat{\Xi}_{22} \end{pmatrix}. \quad (\text{SM23})$$

We assume throughout that the empirical matrix is wide, $n_{\text{tr}} \geq n_{\text{out}}$, and has full row rank after perturbation. The formulas in this formal expansion are conditional on $\hat{\Xi}_{22}$ having full row rank and on the displayed inverses existing. This holds almost surely in the homogeneous Gaussian reference model when $n_{\text{tr}} > n_{\text{out}}$. For the true multinomial model, the probability of $\hat{P}_N$ and $\hat{\Xi}_{22}$ not having full row rank is technically nonzero, but decreases exponentially fast with n_{tr} . The argument in section SM3 takes this into account showing that even the unlikely events where the empirical matrix is extremely ill-conditioned do not affect the overall averaged MSE. Under these conditions, the Moore–Penrose pseudoinverse is $\hat{P}_\tau^+ = \hat{P}_\tau^T (\hat{P}_\tau \hat{P}_\tau^T)^{-1}$, and we can apply the Schur-complement expansion to the square matrix

$$\hat{M}_\tau \equiv \hat{P}_\tau \hat{P}_\tau^T = P P^T + \tau (P \hat{\Xi}^T + \hat{\Xi} P^T) + \tau^2 \hat{\Xi} \hat{\Xi}^T. \quad (\text{SM24})$$

For notational simplicity, in the block formulas below we suppress the subscript τ on $\hat{M}_\tau$ and write its blocks as $\hat{M}_{ab}$. The block decomposition induced by $\Pi_{\bar{1}}$ and $\Pi_{\bar{2}}$ then gives

$$\begin{aligned} \hat{M}_{11} &= P P^T + \tau (P \hat{\Xi}^T + \hat{\Xi} P^T)_{11} + \tau^2 \sum_{c=1}^2 \hat{\Xi}_{1c} \hat{\Xi}_{1c}^T, & \hat{M}_{12} &= \tau P \hat{\Xi}_{21}^T + \tau^2 \sum_{c=1}^2 \hat{\Xi}_{1c} \hat{\Xi}_{2c}^T, \\ \hat{M}_{21} &= \tau \hat{\Xi}_{21} P^T + \tau^2 \sum_{c=1}^2 \hat{\Xi}_{2c} \hat{\Xi}_{1c}^T, & \hat{M}_{22} &= \tau^2 \sum_{c=1}^2 \hat{\Xi}_{2c} \hat{\Xi}_{2c}^T, \end{aligned} \quad (\text{SM25})$$

where the sums over c arise from the block decomposition of $\hat{\Xi}\hat{\Xi}^T$. As discussed above, a concise expression for the Schur complement of $\hat{M}_\tau$ makes its inverse manageable. For example, the diagonal blocks of this inverse can be written as $(\hat{M}^{-1})_{11} = (\hat{M}/\hat{M}_{22})^{-1}$ and $(\hat{M}^{-1})_{22} = (\hat{M}/\hat{M}_{11})^{-1}$, where

$$\hat{M}/\hat{M}_{11} \equiv \hat{M}_{22} - \hat{M}_{21}\hat{M}_{11}^{-1}\hat{M}_{12}, \quad (\hat{M}/\hat{M}_{22}) \equiv \hat{M}_{11} - \hat{M}_{12}\hat{M}_{22}^{-1}\hat{M}_{21}. \quad (\text{SM26})$$

More generally, we can write $\hat{M}^{-1}$ as

$$\begin{aligned} \hat{M}_\tau^{-1} &= \begin{pmatrix} \hat{M}_{11}^{-1} + \hat{M}_{11}^{-1}\hat{M}_{12}(\hat{M}/\hat{M}_{11})^{-1}\hat{M}_{21}\hat{M}_{11}^{-1} & -\hat{M}_{11}^{-1}\hat{M}_{12}(\hat{M}/\hat{M}_{11})^{-1} \\ -(\hat{M}/\hat{M}_{11})^{-1}\hat{M}_{21}\hat{M}_{11}^{-1} & (\hat{M}/\hat{M}_{11})^{-1} \end{pmatrix} \\ &= \begin{pmatrix} (\hat{M}/\hat{M}_{22})^{-1} & -(\hat{M}/\hat{M}_{22})^{-1}\hat{M}_{12}\hat{M}_{22}^{-1} \\ -\hat{M}_{22}^{-1}\hat{M}_{21}(\hat{M}/\hat{M}_{22})^{-1} & \hat{M}_{22}^{-1} + \hat{M}_{22}^{-1}\hat{M}_{21}(\hat{M}/\hat{M}_{22})^{-1}\hat{M}_{12}\hat{M}_{22}^{-1} \end{pmatrix}. \end{aligned} \quad (\text{SM27})$$

Simplification of Schur complements — We want to simplify the expression for the Schur complements in order to easily calculate the various perturbative orders obtained for small τ . Despite the apparent complexity of the definitions of $\hat{M}/\hat{M}_{11}$ and $(\hat{M}/\hat{M}_{22})$, they can actually be expressed rather concisely in terms of the Schur complement of the perturbation matrix $\hat{\Xi}$ itself. Indeed, we have

$$\begin{aligned} \hat{M}/\hat{M}_{22} &= [P + \tau(\hat{\Xi}/\hat{\Xi}_{22})]\hat{Y}^{-1}[P + \tau(\hat{\Xi}/\hat{\Xi}_{22})]^T + \tau^2\hat{\Xi}_{12}\hat{Q}\hat{\Xi}_{12}^T, \\ \hat{Y} &\equiv I + \hat{\Xi}_{21}^T(\hat{\Xi}_{22}\hat{\Xi}_{22}^T)^{-1}\hat{\Xi}_{21}, \quad \hat{\Xi}/\hat{\Xi}_{22} \equiv \hat{\Xi}_{11} - \hat{\Xi}_{12}\hat{\Xi}_{22}^+\hat{\Xi}_{21}, \quad \hat{Q} \equiv I - \hat{\Xi}_{22}^+\hat{\Xi}_{22}. \end{aligned} \quad (\text{SM28})$$

Here we introduced the generalized Schur complement of the perturbation $\hat{\Xi}/\hat{\Xi}_{22}$, and the projector $\hat{Q}$, which acts on the kernel of $\hat{\Xi}_{22}$, and therefore captures the correction that is present in the rectangular case and vanishes when $\hat{\Xi}_{22}$ is square and full rank. This result is obtained expanding eq. (SM26) and suitably collecting terms. Indeed, collecting for example the τ^0 terms in $\hat{M}/\hat{M}_{22}$, we find

$$\hat{M}/\hat{M}_{22}|_{\tau=0} = PP^T - P\hat{\Xi}_{21}^T(\hat{\Xi}_{21}\hat{\Xi}_{21}^T + \hat{\Xi}_{22}\hat{\Xi}_{22}^T)^{-1}\hat{\Xi}_{21}P^T = P(I + \hat{\Xi}_{21}^T(\hat{\Xi}_{22}\hat{\Xi}_{22}^T)^{-1}\hat{\Xi}_{21})^{-1}P^T, \quad (\text{SM29})$$

where we used the *Woodbury identity* $I - A^T(AA^T + Q)^{-1}A = (I + A^TQ^{-1}A)^{-1}$. Similar manipulations are used to derive the rest of eq. (SM28). Let us now use these expressions to perform a perturbative analysis of the pseudoinverse $\hat{P}_\tau^+ = \hat{P}_\tau^T\hat{M}^{-1}$. We have

$$\begin{aligned} (\hat{P}_\tau^+)_{11} &= (P^T + \tau\hat{\Xi}_{11}^T)(\hat{M}^{-1})_{11} + \tau\hat{\Xi}_{21}^T(\hat{M}^{-1})_{21} = (P^T + \tau\hat{\Xi}_{11}^T - \tau\hat{\Xi}_{21}^T\hat{M}_{22}^{-1}\hat{M}_{21})(\hat{M}/\hat{M}_{22})^{-1} \\ &= P^+ - \tau P^+(\hat{\Xi}/\hat{\Xi}_{22})P^+ + \tau^2 P^+(\hat{\Xi}/\hat{\Xi}_{22})P^+(\hat{\Xi}/\hat{\Xi}_{22})P^+ - \tau^2 P^+\hat{\Xi}_{12}\hat{Q}\hat{\Xi}_{12}^T P^{-T}\hat{Y}P^+ + O(\tau^3). \end{aligned} \quad (\text{SM30})$$

In the square case, when $\hat{Q} = 0$, the expression reduces to $[P + \tau(\hat{\Xi}/\hat{\Xi}_{22})]^{-1}$. Its convergent Neumann expansion is

$$[P + \tau(\hat{\Xi}/\hat{\Xi}_{22})]^{-1} = \sum_{i=0}^{\infty} [-\tau P^{-1}(\hat{\Xi}/\hat{\Xi}_{22})]^i P^{-1}, \quad (\text{SM31})$$

whenever $\tau\|P^{-1}(\hat{\Xi}/\hat{\Xi}_{22})\|_{\text{op}} < 1$. We can similarly compute the expressions for the other components of the pseudoinverse. As explained in the main text, those are not useful for our discussion. However, we collect here the leading orders for completeness:

$$\begin{aligned} (\hat{P}_\tau^+)_{12} &= -\left[P^{-1} - \tau P^{-1}(\hat{\Xi}/\hat{\Xi}_{22})P^{-1}\right]\hat{\Xi}_{12}\hat{\Xi}_{22}^+ + \tau P^{-1}\hat{\Xi}_{12}\hat{Q}\hat{\Xi}_{12}^T P^{-T}\hat{\Xi}_{21}^T(\hat{\Xi}_{22}\hat{\Xi}_{22}^T)^{-1} + O(\tau^2) \\ (\hat{P}_\tau^+)_{21} &= -\hat{\Xi}_{22}^+\hat{\Xi}_{21}\left[P^{-1} - \tau P^{-1}(\hat{\Xi}/\hat{\Xi}_{22})P^{-1}\right] + \tau\hat{Q}\hat{\Xi}_{12}^T P^{-T}\hat{Y}P^{-1} + O(\tau^2), \\ (\hat{P}_\tau^+)_{22} &= \frac{\hat{\Xi}_{22}^+}{\tau} + \hat{\Xi}_{22}^+\hat{\Xi}_{21}P^{-1}\hat{\Xi}_{12}\hat{\Xi}_{22}^+ - \hat{Q}\hat{\Xi}_{12}^T P^{-T}\hat{\Xi}_{21}^T(\hat{\Xi}_{22}\hat{\Xi}_{22}^T)^{-1} + O(\tau). \end{aligned} \quad (\text{SM32})$$

Note the asymmetry in the corrections for $(\hat{P}_\tau^+)_{12}$ and $(\hat{P}_\tau^+)_{21}$. This is caused by the asymmetric nature of the problem, due to the difference in the number of rows and columns of $\hat{P}_N$. The off-diagonal blocks $(\hat{P}_\tau^+)_{12}$ and

$(\hat{P}_\tau^+)_{21}$ have finite limits as $\tau \rightarrow 0$. By contrast, the singular contribution is localized in the $(2,2)$ block, which contains the divergent term $\tau^{-1}\hat{\Xi}_{22}^+$. This is the expected discontinuity of the pseudoinverse when the perturbation populates directions outside the row and column spaces of the unperturbed matrix.

In conclusion, the perturbative expansion of $\hat{P}_\tau^+$ coincides with that of the square case upon replacing $\hat{\Xi}_{22}^{-1}$ with $\hat{\Xi}_{22}^+$ in the Schur complements, up to rectangular corrections whose leading orders are

$$\Delta_{\hat{Q}}\hat{P}_\tau^+ = \begin{pmatrix} -\tau^2 P^{-1}\hat{\Xi}_{12}\hat{Q}\hat{\Xi}_{12}^T P^{-T}\hat{Y}P^{-1} & \tau P^{-1}\hat{\Xi}_{12}\hat{Q}\hat{\Xi}_{12}^T P^{-T}\hat{\Xi}_{21}^T(\hat{\Xi}_{22}\hat{\Xi}_{22}^T)^{-1} \\ \tau\hat{Q}\hat{\Xi}_{12}^T P^{-T}\hat{Y}P^{-1} & -\hat{Q}\hat{\Xi}_{12}^T P^{-T}\hat{\Xi}_{21}^T(\hat{\Xi}_{22}\hat{\Xi}_{22}^T)^{-1} \end{pmatrix} + \text{subleading terms depending on } \hat{Q}. \quad (\text{SM33})$$

When $n_{\text{tr}} = n_{\text{out}}$ and $\hat{\Xi}_{22}$ is nonsingular, $\hat{Q} = 0$ and these corrections vanish. Nonsingularity holds almost surely in the continuous Gaussian reference model; for finite multinomial noise it is an event whose probability must be controlled rather than an identity.

Basic definitions (section SM1)

r	$r = \text{rank}(P) = d^2$
P	$P = \langle \mu, \rho \rangle, \quad P^+ = \langle \tilde{\rho}, \tilde{\mu} \rangle$
y, p_σ	$y = \langle \rho, \mathcal{O} \rangle \in \text{im}(P^T), \quad p_\sigma = \langle \mu, \sigma \rangle \in \text{im}(P)$
$w_{\mathcal{O}}$	$w_{\mathcal{O}} = \langle \tilde{\mu}, \mathcal{O} \rangle$
U_1, U_2	$\text{im}(U_1) = \text{im}(P), \quad \text{im}(U_2) = \text{coker}(P), \quad U_1 \in \mathbb{R}^{n_{\text{out}} \times r}, \quad U_2 \in \mathbb{R}^{n_{\text{out}} \times (n_{\text{out}} - r)}$

Basic stochastic quantities

$\hat{P}, \hat{P}_N, \Delta\hat{P}_N$	$\hat{P}_N = \hat{P} = P + N^{-1/2}\hat{\Xi}, \quad \Delta\hat{P}_N = \hat{P}_N - P$
$\hat{S}, \hat{\ell}$	$\hat{S} = n_{\text{tr}}^{-1}\hat{P}_N\hat{P}_N^T, \quad \hat{\ell} = n_{\text{tr}}^{-1}\hat{P}_N y \quad (\text{eq. (SM34)})$
$\Delta\hat{\delta}$	$\Delta\hat{\delta} = y^T(\Delta\hat{P}_N^+)p_\sigma = (\hat{\ell}^T\hat{S}^+ - \ell^T S^+)p_\sigma = \hat{q}^T\hat{S}_{\text{eff}}^{-1}p_{\sigma,1} \quad (\text{eqs. (SM34) and (SM51)})$

Population quantities

S, ℓ	$S = n_{\text{tr}}^{-1}PP^T, \quad \ell = n_{\text{tr}}^{-1}Py \quad (\text{eq. (SM34)})$
$\bar{\Sigma}, \bar{\Sigma}_{ab}, \Sigma_i$	$\bar{\Sigma} = n_{\text{tr}}^{-1}\sum_{i=1}^{n_{\text{tr}}}\Sigma_i, \quad \bar{\Sigma}_{ab} = U_a^T\bar{\Sigma}U_b, \quad \Sigma_i = \mathbb{E}[\hat{\xi}_i\hat{\xi}_i^T]$

Effective Schur-reduced quantities

S_{eff}	$S_{\text{eff}} = S_{11} + N^{-1}\bar{\Sigma}_{\text{eff}} \quad (\text{eq. (SM37)})$
$\hat{S}_{\text{eff}}$	$\hat{S}_{\text{eff}} = \hat{S}_{11} - \hat{S}_{12}\hat{S}_{22}^{-1}\hat{S}_{21} = S_{\text{eff}} + (\Delta_c\hat{S}_{\text{eff}})_p + (\Delta_c\hat{S}_{\text{eff}})_{\text{nl}} \quad (\text{eq. (SM41)})$
$\bar{\Sigma}_{\text{eff}}$	$\bar{\Sigma}_{\text{eff}} = \bar{\Sigma}_{11} - \bar{\Sigma}_{12}\bar{\Sigma}_{22}^{-1}\bar{\Sigma}_{21}$
$w_{\mathcal{O}}^{\text{eff}}$	$w_{\mathcal{O}}^{\text{eff}} = (I_{n_{\text{out}}} - U_2\bar{\Sigma}_{22}^{-1}\bar{\Sigma}_{21}U_1^T)w_{\mathcal{O}}. \quad (\text{eq. (SM46)})$
$\hat{\ell}_{\text{eff}}$	$\hat{\ell}_{\text{eff}} = \hat{\ell}_1 - \hat{S}_{12}\hat{S}_{22}^{-1}\hat{\ell}_2 = \ell_1 + (\Delta_c\hat{\ell}_{\text{eff}})_p + (\Delta_c\hat{\ell}_{\text{eff}})_{\text{nl}} \quad (\text{eqs. (SM37) and (SM40)})$
$\hat{q}$	$\hat{q} \equiv \hat{\ell}_{\text{eff}} - \hat{S}_{\text{eff}}U_1^T w_{\mathcal{O}} = q_0 + \hat{q}_p + \hat{q}_{\text{nl}} \quad (\text{eq. (SM50)})$

Table SM1. Summary of main definitions. All quantities are conditional on the fixed training states. Stochastic quantities are denoted with a hat, and stochasticity is always understood with respect to the training noise realizations. The symbols d , n_{tr} , n_{out} , and N denote the Hilbert-space dimension, number of training states, number of POVM outcomes, and training statistics, respectively. Expressions involving stochastic inverses are understood on events where the displayed inverses exist; the pseudoinverse formula for $\Delta\hat{\delta}$ is defined for every realization.

SM3. DIRECT PROOF: BIAS, LINEAR FLUCTUATION, AND REMAINDER

This section contains the proof of the MSE and resource-scaling result stated in section VI, and is logically independent of the formal small- τ expansion in section SM2. Both analyses use the exact wide-matrix identity $\hat{P}^+ = \hat{P}(\hat{P}\hat{P}^T)^{-1}$ when $\hat{P}$ has full row rank, and the block inverse formula eq. (SM11), but none of the truncated expansions eqs. (SM30) and (SM32) is used below. Instead, we first eliminate the rank-increasing directions by an exact Schur complement and only then expand the resulting well-conditioned $d^2 \times d^2$ matrix. This order of operations enables a uniform high-probability bound on the remainder.

In section SM3.1, we prove a decomposition $\Delta\hat{\delta} = B + \hat{L} + \hat{R}$ of the infinite-test-statistics error $\Delta\hat{\delta}$ into a deterministic component B , a centered fluctuation $\hat{L}$, and a remainder $\hat{R}$. We identify the deterministic bias component B

in [section SM3.2](#). In [section SM3.3](#) we derive the form of the fluctuation term $\hat{L}$, show that it has zero average, and compute its variance. We then prove in [section SM3.4](#) that the remainder $\hat{R}$ is subleading, conditionally on a stochastic event where the pseudoinverses are sufficiently well-conditioned. In [section SM3.5](#) we then prove that this event has high probability for large n_{tr} . To conclude the bound on the MSE we only need to make sure that the stochastic events in which pseudoinverses are ill-conditioned do not make the MSE explode. We do this in [section SM3.6](#), where we show that the divergence of $\hat{R}$ in the exceptional realizations does not overcome the exponentially small probability of these events. This concludes the proof and provides us with an unconditional MSE bound for large n_{tr} . A summary of the notation used in the proof is given in [table SM1](#).

SM3.1. Deterministic decomposition of infinite-test-statistics error

We study the infinite-test-statistics estimation error $\Delta\hat{\delta} = \mathbf{y}^T(\hat{\mathbf{P}}_N^+ - \mathbf{P}^+)\mathbf{p}_\sigma$, where $\mathbf{y}^T = \langle \mathcal{O}, \boldsymbol{\rho} \rangle$ is the row of training targets and $\mathbf{p}_\sigma = \langle \boldsymbol{\mu}, \sigma \rangle$ is the test-state probability vector. We assume $r \equiv \text{rank}(\mathbf{P}) = d^2 < n_{\text{out}} \leq n_{\text{tr}}$ and, as in the main text, $\mathbf{y} \in \text{im}(\mathbf{P}^T)$ and $\mathbf{p}_\sigma \in \text{im}(\mathbf{P})$. The minimal informationally complete case $n_{\text{out}} = d^2$ has no cokernel block and is handled by the ordinary inverse expansion on $\text{im}(\mathbf{P})$; the Schur-complement analysis below concerns the rank-increasing case $n_{\text{out}} > d^2$. The identity $\mathbf{A}^+ = \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^+$ gives

$$\Delta\hat{\delta} = \hat{\boldsymbol{\ell}}^T \hat{\mathbf{S}}^+ \mathbf{p}_\sigma - \boldsymbol{\ell}^T \mathbf{S}^+ \mathbf{p}_\sigma, \quad \hat{\boldsymbol{\ell}} \equiv \frac{\hat{\mathbf{P}}\mathbf{y}}{n_{\text{tr}}}, \quad \boldsymbol{\ell} \equiv \frac{\mathbf{P}\mathbf{y}}{n_{\text{tr}}}, \quad \hat{\mathbf{S}} \equiv \frac{\hat{\mathbf{P}}\hat{\mathbf{P}}^T}{n_{\text{tr}}}, \quad \mathbf{S} \equiv \frac{\mathbf{P}\mathbf{P}^T}{n_{\text{tr}}}. \quad (\text{SM34})$$

Here $\hat{\mathbf{P}} \equiv \hat{\mathbf{P}}_N = \mathbf{P} + N^{-1/2}\hat{\boldsymbol{\Xi}}$. In this section we omit the N subscript on stochastic quantities for notational brevity. Conditional on the training states, the columns of $\hat{\boldsymbol{\Xi}}$ are independent and centered, with $\mathbb{E}[\hat{\boldsymbol{\xi}}_i \hat{\boldsymbol{\xi}}_i^T] = \boldsymbol{\Sigma}_i$. Consequently,

$$\mathbb{E}[\hat{\boldsymbol{\ell}}] = \boldsymbol{\ell}, \quad \mathbb{E}[\hat{\mathbf{S}}] = \mathbf{S} + \frac{1}{N}\bar{\boldsymbol{\Sigma}}, \quad \bar{\boldsymbol{\Sigma}} \equiv \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \boldsymbol{\Sigma}_i. \quad (\text{SM35})$$

The deterministic $N^{-1}\bar{\boldsymbol{\Sigma}}$ shift is the source of the leading bias. Directly expanding $\hat{\mathbf{S}}^{-1}$ is nevertheless unhelpful: the eigenvalues created outside $\text{im}(\mathbf{P})$ are of order N^{-1} , so that $\|\mathbb{E}[\hat{\mathbf{S}}]^{-1}\|_{\text{op}}$ is of order N . The Schur complement below removes these apparently singular directions before the fluctuation expansion is made.

Let $U = (U_1, U_2)$ be orthogonal, with $\text{im}(U_1) = \text{im}(\mathbf{P})$ and $\text{im}(U_2) = \ker(\mathbf{P})^\perp$, and write $\mathbf{P}_1 \equiv U_1^T \mathbf{P}$, $\mathbf{P}_2 = 0$, $U_a^T \hat{\mathbf{P}} = \hat{\mathbf{P}}_a$, and $U_1^T \mathbf{p}_\sigma = \mathbf{p}_{\sigma,1}$. Thus $\mathbf{P}_1, \hat{\mathbf{P}}_1 \in \mathbb{R}^{r \times n_{\text{tr}}}$ and $\hat{\mathbf{P}}_2 \in \mathbb{R}^{(n_{\text{out}}-r) \times n_{\text{tr}}}$. Denote the i -th column of $\mathbf{P}_1$ by $\mathbf{p}_{i,1}$ and set $\hat{\boldsymbol{\Xi}}_a \equiv U_a^T \hat{\boldsymbol{\Xi}}$, with columns $\hat{\boldsymbol{\xi}}_{i,a} \equiv U_a^T \hat{\boldsymbol{\xi}}_i$. We use

$$\hat{\boldsymbol{\ell}}_a = n_{\text{tr}}^{-1} \hat{\mathbf{P}}_a \mathbf{y}, \quad \boldsymbol{\ell}_a = n_{\text{tr}}^{-1} \mathbf{P}_a \mathbf{y}, \quad \hat{\mathbf{S}}_{ab} = U_a^T \hat{\mathbf{S}} U_b, \quad \mathbf{S}_{ab} = U_a^T \mathbf{S} U_b,$$

where $\mathbf{P}_2 = 0$, and hence $\boldsymbol{\ell}_2 = \mathbf{S}_{12} = \mathbf{S}_{21} = \mathbf{S}_{22} = 0$. When $\hat{\mathbf{S}}_{22}$ and its Schur complement are invertible, we have

$$\hat{\boldsymbol{\ell}}^T \hat{\mathbf{S}}^{-1} \mathbf{p}_\sigma = \frac{1}{n_{\text{tr}}} (\hat{\mathbf{P}}\mathbf{y})^T \hat{\mathbf{S}}^{-1} \mathbf{p}_\sigma = \hat{\boldsymbol{\ell}}_{\text{eff}}^T \hat{\mathbf{S}}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}, \quad \hat{\boldsymbol{\ell}}_{\text{eff}} \equiv \hat{\boldsymbol{\ell}}_1 - \hat{\mathbf{S}}_{12} \hat{\mathbf{S}}_{22}^{-1} \hat{\boldsymbol{\ell}}_2, \quad \hat{\mathbf{S}}_{\text{eff}} \equiv \hat{\mathbf{S}}_{11} - \hat{\mathbf{S}}_{12} \hat{\mathbf{S}}_{22}^{-1} \hat{\mathbf{S}}_{21}, \quad (\text{SM36})$$

Observe that $\mathbf{y} = \mathbf{P}^T \mathbf{w}_\mathcal{O}$ and $\boldsymbol{\ell}_1 = \mathbf{S}_{11} U_1^T \mathbf{w}_\mathcal{O}$. If $\bar{\mathbf{S}}_{22}$ is invertible, define $\bar{\mathbf{S}}_{\text{eff}} = \bar{\mathbf{S}}_{11} - \bar{\mathbf{S}}_{12} \bar{\mathbf{S}}_{22}^{-1} \bar{\mathbf{S}}_{21}$. Introduce

$$\begin{aligned} \mathbf{S}_{\text{eff}} &\equiv \mathbb{E}[\hat{\mathbf{S}}_{11}] - \mathbb{E}[\hat{\mathbf{S}}_{12}] \mathbb{E}[\hat{\mathbf{S}}_{22}]^{-1} \mathbb{E}[\hat{\mathbf{S}}_{21}] = \mathbf{S}_{11} + \frac{1}{N} \bar{\mathbf{S}}_{\text{eff}}, & \Delta \hat{\mathbf{S}}_{ab} &\equiv \hat{\mathbf{S}}_{ab} - \mathbb{E}[\hat{\mathbf{S}}_{ab}], & \Delta_c \hat{\mathbf{S}}_{\text{eff}} &\equiv \hat{\mathbf{S}}_{\text{eff}} - \mathbf{S}_{\text{eff}}, \\ \boldsymbol{\ell}_{\text{eff}} &\equiv \mathbb{E}[\hat{\boldsymbol{\ell}}_1] - \mathbb{E}[\hat{\mathbf{S}}_{12}] \mathbb{E}[\hat{\mathbf{S}}_{22}]^{-1} \mathbb{E}[\hat{\boldsymbol{\ell}}_2] = \boldsymbol{\ell}_1, & \Delta \hat{\boldsymbol{\ell}}_a &\equiv \hat{\boldsymbol{\ell}}_a - \boldsymbol{\ell}_a, & \Delta_c \hat{\boldsymbol{\ell}}_{\text{eff}} &\equiv \hat{\boldsymbol{\ell}}_{\text{eff}} - \boldsymbol{\ell}_{\text{eff}}. \end{aligned} \quad (\text{SM37})$$

where Δ denotes a centered block fluctuation, whereas Δ_c denotes deviation from the deterministic concentration target. Thus $\mathbb{E}[\Delta \hat{\mathbf{S}}_{ab}] = 0$, but in general $\mathbb{E}[\Delta_c \hat{\mathbf{S}}_{\text{eff}}] \neq 0$ because the Schur complement is nonlinear.

We then observe that $\hat{\boldsymbol{\ell}}^T \hat{\mathbf{S}}^{-1} \mathbf{p}_\sigma = \hat{\boldsymbol{\ell}}_{\text{eff}}^T \hat{\mathbf{S}}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}$ and $\boldsymbol{\ell}^T \mathbf{S}^+ \mathbf{p}_\sigma = \boldsymbol{\ell}_1^T \mathbf{S}_{11}^{-1} \mathbf{p}_{\sigma,1}$, and adding and subtracting $\boldsymbol{\ell}_1^T \mathbf{S}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}$ and $\hat{\boldsymbol{\ell}}_{\text{eff}}^T \mathbf{S}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}$ in [eq. \(SM34\)](#), we get

$$\Delta\hat{\delta} = \boldsymbol{\ell}_1^T [\mathbf{S}_{\text{eff}}^{-1} - \mathbf{S}_{11}^{-1}] \mathbf{p}_{\sigma,1} + \hat{\boldsymbol{\ell}}_{\text{eff}}^T (\hat{\mathbf{S}}_{\text{eff}}^{-1} - \mathbf{S}_{\text{eff}}^{-1}) \mathbf{p}_{\sigma,1} + \Delta_c \hat{\boldsymbol{\ell}}_{\text{eff}}^T \mathbf{S}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}. \quad (\text{SM38})$$

Applying the resolvent identity twice,

$$\hat{\ell}_{\text{eff}}^T (\hat{S}_{\text{eff}}^{-1} - S_{\text{eff}}^{-1}) \mathbf{p}_{\sigma,1} = -\hat{\ell}_{\text{eff}}^T S_{\text{eff}}^{-1} \Delta_c \hat{S}_{\text{eff}} S_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1} + \hat{\ell}_{\text{eff}}^T S_{\text{eff}}^{-1} \Delta_c \hat{S}_{\text{eff}} S_{\text{eff}}^{-1} \Delta_c \hat{S}_{\text{eff}} \hat{S}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}, \quad (\text{SM39})$$

We next decompose $\hat{\ell}_{\text{eff}}$ in its deterministic, centered fluctuations, and nonlinear components, as:

$$\begin{aligned} \hat{\ell}_{\text{eff}} &= \ell_1 + (\Delta_c \hat{\ell}_{\text{eff}})_{\text{p}} + (\Delta_c \hat{\ell}_{\text{eff}})_{\text{nl}}, \\ (\Delta_c \hat{\ell}_{\text{eff}})_{\text{p}} &\equiv \Delta \hat{\ell}_1 - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \Delta \hat{\ell}_2 = \frac{1}{n_{\text{tr}} \sqrt{N}} (\hat{\Xi}_1 - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \hat{\Xi}_2) \mathbf{y} = \frac{1}{n_{\text{tr}} \sqrt{N}} \sum_{i=1}^{n_{\text{tr}}} \hat{\xi}_{i,\text{eff}} y_i, \end{aligned} \quad (\text{SM40})$$

where $\hat{\xi}_{i,\text{eff}} \equiv \hat{\xi}_{i,1} - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \hat{\xi}_{i,2}$ denotes the i -th column of $\hat{\Xi}_{\text{eff}} \equiv \hat{\Xi}_1 - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \hat{\Xi}_2$. Similarly,

$$\begin{aligned} \Delta_c \hat{S}_{\text{eff}} &= (\Delta_c \hat{S}_{\text{eff}})_{\text{p}} + (\Delta_c \hat{S}_{\text{eff}})_{\text{nl}}, \\ (\Delta_c \hat{S}_{\text{eff}})_{\text{p}} &\equiv \Delta \hat{S}_{11} - \Delta \hat{S}_{12} \bar{\Sigma}_{22}^{-1} \bar{\Sigma}_{21} - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \Delta \hat{S}_{21} + \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \Delta \hat{S}_{22} \bar{\Sigma}_{22}^{-1} \bar{\Sigma}_{21} \\ &= \frac{1}{n_{\text{tr}} \sqrt{N}} \sum_{i=1}^{n_{\text{tr}}} \left(\mathbf{p}_{i,1} \hat{\xi}_{i,\text{eff}}^T + \hat{\xi}_{i,\text{eff}} \mathbf{p}_{i,1}^T \right) + \frac{1}{N n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \left(\hat{\xi}_{i,\text{eff}} \hat{\xi}_{i,\text{eff}}^T - \mathbb{E}[\hat{\xi}_{i,\text{eff}} \hat{\xi}_{i,\text{eff}}^T] \right). \end{aligned} \quad (\text{SM41})$$

Both first-order terms are centered: $\mathbb{E}[(\Delta_c \hat{\ell}_{\text{eff}})_{\text{p}}] = \mathbb{E}[(\Delta_c \hat{S}_{\text{eff}})_{\text{p}}] = 0$. The subscript “p” denotes terms that are first order in the centered block fluctuations, and can comprise of terms that are both linear and quadratic in $\hat{\Xi}$: e.g. the last term of $(\Delta_c \hat{S}_{\text{eff}})_{\text{p}}$ is quadratic in $\hat{\Xi}$.

Thus eq. (SM38) provides a decomposition of the form

$$\begin{aligned} \Delta \hat{o} &= B + \hat{L} + \hat{R}, \\ B &\equiv \ell_1^T \left(S_{\text{eff}}^{-1} - S_{11}^{-1} \right) \mathbf{p}_{\sigma,1}, \quad \hat{L} \equiv (\Delta_c \hat{\ell}_{\text{eff}})_{\text{p}}^T S_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1} - \ell_1^T S_{\text{eff}}^{-1} (\Delta_c \hat{S}_{\text{eff}})_{\text{p}} S_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}. \end{aligned} \quad (\text{SM42})$$

Here B is deterministic, $\hat{L}$ is centered and first order in the centered block fluctuations, and $\hat{R}$ contains every remaining term.

SM3.2. Deterministic bias

From eq. (SM42), remembering $S_{\text{eff}} = S_{11} + N^{-1} \bar{\Sigma}_{\text{eff}}$ and $\ell_1 = S_{11} U_1^T \mathbf{w}_{\mathcal{O}}$, the resolvent identity gives

$$B = \ell_1^T \left(S_{\text{eff}}^{-1} - S_{11}^{-1} \right) \mathbf{p}_{\sigma,1} = -\frac{1}{N} \mathbf{w}_{\mathcal{O}}^T U_1 \bar{\Sigma}_{\text{eff}} S_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1} = -\frac{1}{N} \mathbf{w}_{\mathcal{O}}^T U_1 \bar{\Sigma}_{\text{eff}} S_{11}^{-1} \mathbf{p}_{\sigma,1} + O\left(\frac{1}{N^2}\right). \quad (\text{SM43})$$

We can therefore write the bias as $B = -b_N/N$ with $b_N = \mathbf{w}_{\mathcal{O}}^T U_1 \bar{\Sigma}_{\text{eff}} S_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}$.

SM3.3. Centered first-order fluctuation

After isolating the deterministic shift in section SM3.2, we now identify the centered term that governs training-shot variance. The goal is to express it as a sum of independent column contributions and compute its second moment before controlling the nonlinear remainder in section SM3.4. From eqs. (SM40) and (SM42) and $\mathbf{y} = \mathbf{P}_1^T \mathbf{w}_{\mathcal{O}}$, we rewrite $\hat{L}$ more explicitly as

$$\hat{L} = \left[\frac{1}{n_{\text{tr}} \sqrt{N}} \mathbf{w}_{\mathcal{O}}^T U_1 \mathbf{P}_1 \hat{\Xi}_{\text{eff}}^T - \ell_1^T S_{\text{eff}}^{-1} (\Delta_c \hat{S}_{\text{eff}})_{\text{p}} \right] S_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}. \quad (\text{SM44})$$

More explicitly, using eq. (SM41), this amounts to $\hat{L} = \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \hat{L}_i$, where

$$\begin{aligned} \hat{L}_i &= \left\{ \frac{1}{\sqrt{N}} \left[y_i \hat{\xi}_{i,\text{eff}}^T - \ell_1^T S_{\text{eff}}^{-1} \left(\mathbf{p}_{i,1} \hat{\xi}_{i,\text{eff}}^T + \hat{\xi}_{i,\text{eff}} \mathbf{p}_{i,1}^T \right) \right] - \frac{1}{N} \ell_1^T S_{\text{eff}}^{-1} \left(\hat{\xi}_{i,\text{eff}} \hat{\xi}_{i,\text{eff}}^T - \mathbb{E}[\hat{\xi}_{i,\text{eff}} \hat{\xi}_{i,\text{eff}}^T] \right) \right\} S_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1} \\ &= \frac{1}{\sqrt{N}} \left[\underbrace{(y_i - \ell_1^T S_{\text{eff}}^{-1} \mathbf{p}_{i,1})}_{O(N^{-1})} \hat{\xi}_{i,\text{eff}}^T - \underbrace{\ell_1^T S_{\text{eff}}^{-1} \hat{\xi}_{i,\text{eff}} \mathbf{p}_{i,1}^T}_{\mathbf{w}_{\mathcal{O}}^T U_1 + O(N^{-1})} \right] S_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1} + O(N^{-1}). \end{aligned} \quad (\text{SM45})$$

The variables $\hat{L}_i$ are independent and centered, therefore $\mathbb{E}[\hat{L}] = 0$ and $\mathbb{E}[\hat{L}^2] = \frac{1}{n_{\text{tr}}^2} \sum_{i=1}^{n_{\text{tr}}} \mathbb{E}[\hat{L}_i^2]$. Let $\mathbf{w}_{\mathcal{O}} = \langle \tilde{\boldsymbol{\mu}}, \mathcal{O} \rangle \in \text{im}(\mathbf{P}) \subset \mathbb{R}^{n_{\text{out}}}$ be the canonical dual-frame outcome weights for the observable $\mathcal{O}$, and define

$$\mathbf{w}_{\mathcal{O}}^{\text{eff}} \equiv (\mathbf{I}_{n_{\text{out}}} - \mathbf{U}_2 \bar{\boldsymbol{\Sigma}}_{22}^{-1} \bar{\boldsymbol{\Sigma}}_{21} \mathbf{U}_1^T) \mathbf{w}_{\mathcal{O}}. \quad (\text{SM46})$$

Interestingly, the two weight vectors $\mathbf{w}_{\mathcal{O}}$ and $\mathbf{w}_{\mathcal{O}}^{\text{eff}}$ have parallel variational interpretations: $\mathbf{w}_{\mathcal{O}}$ solves $\mathbf{P}^T \mathbf{u} = \langle \boldsymbol{\rho}, \mathcal{O} \rangle$ with minimum Euclidean norm, whereas $\mathbf{w}_{\mathcal{O}}^{\text{eff}}$ solves the same linear system but minimizing the weighted quadratic form $\mathbf{u}^T \bar{\boldsymbol{\Sigma}} \mathbf{u}$.

Using $\mathbf{w}_{\mathcal{O}}^T \mathbf{U}_1 \hat{\boldsymbol{\xi}}_{i,\text{eff}} = (\mathbf{w}_{\mathcal{O}}^{\text{eff}})^T \hat{\boldsymbol{\xi}}_i$, eq. (SM45) becomes

$$\hat{L}_i = -\frac{1}{\sqrt{N}} (\mathbf{p}_{i,1}^T \mathbf{S}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}) (\mathbf{w}_{\mathcal{O}}^{\text{eff}})^T \hat{\boldsymbol{\xi}}_i + \hat{r}_i, \quad \mathbb{E}[\hat{r}_i^2] = O\left(\frac{1}{N^2}\right). \quad (\text{SM47})$$

Consider now the second moments $\mathbb{E}[\hat{L}_i^2]$. The leading component of $\hat{L}_i$ is linear in $\hat{\boldsymbol{\xi}}$ and has order $N^{-1/2}$. The centered quadratic component has instead order N^{-1} . Their cross term would thus seem to give a factor of order $N^{-3/2}$; however, this term is cubic in the noise vectors, hence its expectation value contains a third centered moment, which for multinomial noise has the form $\mathbb{E}[\hat{\boldsymbol{\xi}}_{i,a} \hat{\boldsymbol{\xi}}_{i,b} \hat{\boldsymbol{\xi}}_{i,c}] = O(N^{-1/2})$. Thus the cross term actually leads to a term of order N^{-2} , and we can write

$$\mathbb{E}[\hat{L}^2] = \frac{V_N}{N n_{\text{tr}}} + O\left(\frac{1}{N^2 n_{\text{tr}}}\right), \quad V_N = \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} (\mathbf{p}_{i,1}^T \mathbf{S}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1})^2 \text{Var}_i(\mathbf{w}_{\mathcal{O}}^{\text{eff}}), \quad \text{Var}_i(\mathbf{w}_{\mathcal{O}}^{\text{eff}}) \equiv (\mathbf{w}_{\mathcal{O}}^{\text{eff}})^T \boldsymbol{\Sigma}_i \mathbf{w}_{\mathcal{O}}^{\text{eff}}. \quad (\text{SM48})$$

For large N , we can furthermore write

$$V_N = V + O(N^{-1}), \quad V \equiv \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} (\mathbf{p}_{i,1}^T \mathbf{S}_{11}^{-1} \mathbf{p}_{\sigma,1})^2 \text{Var}_i(\mathbf{w}_{\mathcal{O}}^{\text{eff}}) = n_{\text{tr}} \sum_{i=1}^{n_{\text{tr}}} \langle \tilde{\rho}_i, \sigma \rangle^2 \text{Var}_i(\mathbf{w}_{\mathcal{O}}^{\text{eff}}), \quad (\text{SM49})$$

using the identity $\mathbf{p}_{i,1}^T \mathbf{S}_{11}^{-1} \mathbf{p}_{\sigma,1} = n_{\text{tr}} \langle \tilde{\rho}_i, \sigma \rangle$.

SM3.4. Deterministic bound on the remainder

The bias and centered fluctuation are useful leading terms only if all remaining Schur-complement and inverse-expansion contributions are uniformly smaller. This subsection proves that deterministic comparison on a good event; section SM3.5 will then show that multinomial shot noise lies in that event with high probability. The main result of this subsection is eq. (SM74).

Define the residual vector

$$\begin{aligned} \hat{\mathbf{q}} &\equiv \hat{\boldsymbol{\ell}}_{\text{eff}} - \hat{\mathbf{S}}_{\text{eff}} \mathbf{U}_1^T \mathbf{w}_{\mathcal{O}} = \mathbf{q}_0 + \hat{\mathbf{q}}_{\text{p}} + \hat{\mathbf{q}}_{\text{nl}}, & \mathbf{q}_0 &\equiv \boldsymbol{\ell}_1 - \mathbf{S}_{\text{eff}} \mathbf{U}_1^T \mathbf{w}_{\mathcal{O}} = -\frac{1}{N} \bar{\boldsymbol{\Sigma}}_{\text{eff}} \mathbf{U}_1^T \mathbf{w}_{\mathcal{O}}, \\ \hat{\mathbf{q}}_{\text{p}} &\equiv (\Delta_c \hat{\boldsymbol{\ell}}_{\text{eff}})_{\text{p}} - (\Delta_c \hat{\mathbf{S}}_{\text{eff}})_{\text{p}} \mathbf{U}_1^T \mathbf{w}_{\mathcal{O}}, & \hat{\mathbf{q}}_{\text{nl}} &\equiv (\Delta_c \hat{\boldsymbol{\ell}}_{\text{eff}})_{\text{nl}} - (\Delta_c \hat{\mathbf{S}}_{\text{eff}})_{\text{nl}} \mathbf{U}_1^T \mathbf{w}_{\mathcal{O}}. \end{aligned} \quad (\text{SM50})$$

Remember that $\boldsymbol{\ell}_1^T \mathbf{S}_{11}^{-1} \mathbf{p}_{\sigma,1} = \mathbf{w}_{\mathcal{O}}^T \mathbf{U}_1 \mathbf{p}_{\sigma,1}$. Thus we can rewrite eq. (SM38) as

$$\Delta \delta = \hat{\boldsymbol{\ell}}_{\text{eff}}^T \hat{\mathbf{S}}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1} - \boldsymbol{\ell}_1^T \mathbf{S}_{11}^{-1} \mathbf{p}_{\sigma,1} = (\hat{\boldsymbol{\ell}}_{\text{eff}} - \hat{\mathbf{S}}_{\text{eff}} \mathbf{U}_1^T \mathbf{w}_{\mathcal{O}})^T \hat{\mathbf{S}}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1} = \hat{\mathbf{q}}^T \hat{\mathbf{S}}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}. \quad (\text{SM51})$$

This rewriting is useful because $\hat{\boldsymbol{\ell}}_{\text{eff}}$ and $\hat{\mathbf{S}}_{\text{eff}} \mathbf{U}_1^T \mathbf{w}_{\mathcal{O}}$ are $O(1)$, but their difference is $o(1)$. Thus writing $\Delta \delta$ in terms of their difference allows to get tighter bounds to estimate the magnitude of the remainder $\hat{R}$. In this notation,

$$\mathbf{B} = \mathbf{q}_0^T \mathbf{S}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}, \quad \hat{\mathbf{L}} = \left[\hat{\mathbf{q}}_{\text{p}} - (\Delta_c \hat{\mathbf{S}}_{\text{eff}})_{\text{p}} \mathbf{S}_{\text{eff}}^{-1} \mathbf{q}_0 \right]^T \mathbf{S}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}. \quad (\text{SM52})$$

Applying

$$(\mathbf{A} + \mathbf{F})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1} \mathbf{F} \mathbf{A}^{-1} + (\mathbf{A}^{-1} \mathbf{F})^2 (\mathbf{A} + \mathbf{F})^{-1} \quad (\text{SM53})$$

with $A = S_{\text{eff}}$ and $F = \Delta_c \hat{S}_{\text{eff}}$ yields

$$\hat{R} = -q_0^T S_{\text{eff}}^{-1} (\Delta_c \hat{S}_{\text{eff}})_{\text{nl}} S_{\text{eff}}^{-1} p_{\sigma,1} - (\hat{q} - q_0)^T S_{\text{eff}}^{-1} \Delta_c \hat{S}_{\text{eff}} S_{\text{eff}}^{-1} p_{\sigma,1} + \hat{q}^T (S_{\text{eff}}^{-1} \Delta_c \hat{S}_{\text{eff}})^2 \hat{S}_{\text{eff}}^{-1} p_{\sigma,1} + \hat{q}_{\text{nl}}^T S_{\text{eff}}^{-1} p_{\sigma,1}. \quad (\text{SM54})$$

Each term now contains either two centered fluctuations or one nonlinear Schur-complement remainder.

Assume $\lambda_{\min}(S_{11}) \geq \kappa_{11} > 0$ and $\lambda_{\min}(\bar{S}_{22}) \geq \kappa_{22} > 0$. Since $\bar{S} \succeq 0$, its Schur complement satisfies $\bar{S}_{\text{eff}} = \bar{S}_{11} - \bar{S}_{12} \bar{S}_{22}^{-1} \bar{S}_{21} \succeq 0$, and therefore also $S_{\text{eff}} = S_{11} + \frac{1}{N} \bar{S}_{\text{eff}} \succeq S_{11}$, and $\kappa_{\text{eff}} \equiv \lambda_{\min}(S_{\text{eff}}) \geq \kappa_{11} > 0$. These are deterministic conditioning assumptions on the POVM and training states. All constants below may depend on κ_{11}^{-1} , κ_{22}^{-1} , the operator norms of the covariance blocks, and the fixed vectors $w_{\mathcal{O}}, p_{\sigma,1}$, but not on N or n_{tr} . The direct resource theorem in this section is a fixed- d , fixed- n_{out} result. Let $K_B > 0$ be the absolute constant in the quadratic concentration bound of [lemma 3](#), and define

$$\hat{h} \equiv \Delta \hat{S}_{21} U_1^T w_{\mathcal{O}} - \Delta \hat{\ell}_2. \quad (\text{SM55})$$

For $u > 0$, define $B_y \equiv \max_{1 \leq i \leq n_{\text{tr}}} |y_i|$, and let $\mathcal{E}_u$ be the event

$$\mathcal{E}_u \equiv \left\{ \begin{array}{l} \|\Delta \hat{S}_{11}\|_{\text{op}}, \|\Delta \hat{S}_{12}\|_{\text{op}}, \|\Delta \hat{S}_{21}\|_{\text{op}} \leq \frac{u}{\sqrt{N}}, \quad \|\Delta \hat{S}_{22}\|_{\text{op}} \leq \frac{u}{N}, \\ \|\Delta \hat{\ell}_a\|_2 \leq \frac{2B_y u}{\sqrt{N}} \quad (a = 1, 2), \quad \|\hat{h}\|_2 \leq \frac{K_B \|w_{\mathcal{O}}\|_2 u}{N} \end{array} \right\}. \quad (\text{SM56})$$

The constant factors in this definition are chosen to match the simultaneous concentration result proved below. We first work deterministically on $\mathcal{E}_u$:

1. We start with the first term involving $(\Delta_c \hat{S}_{\text{eff}})_{\text{nl}}$.

After some algebra starting from $\hat{S}_{\text{eff}} = \hat{S}_{11} - (\mathbb{E}[\hat{S}_{12}] + \Delta \hat{S}_{12}) \hat{S}_{22}^{-1} (\mathbb{E}[\hat{S}_{21}] + \Delta \hat{S}_{21})$, we find

$$\begin{aligned} (\Delta_c \hat{S}_{\text{eff}})_{\text{nl}} &= \hat{S}_{\text{eff}} - S_{\text{eff}} - (\Delta_c \hat{S}_{\text{eff}})_{\text{p}} \\ &= -\mathbb{E}[\hat{S}_{12}] \hat{T} \mathbb{E}[\hat{S}_{21}] - \Delta \hat{S}_{12} (\hat{S}_{22}^{-1} - \mathbb{E}[\hat{S}_{22}]^{-1}) \mathbb{E}[\hat{S}_{21}] - \mathbb{E}[\hat{S}_{12}] (\hat{S}_{22}^{-1} - \mathbb{E}[\hat{S}_{22}]^{-1}) \Delta \hat{S}_{21} - \Delta \hat{S}_{12} \hat{S}_{22}^{-1} \Delta \hat{S}_{21} \\ &= -\mathbb{E}[\hat{S}_{12}] \hat{T} \mathbb{E}[\hat{S}_{21}] + \Delta \hat{S}_{12} \mathbb{E}[\hat{S}_{22}]^{-1} \Delta \hat{S}_{22} \hat{S}_{22}^{-1} \mathbb{E}[\hat{S}_{21}] + \mathbb{E}[\hat{S}_{12}] \mathbb{E}[\hat{S}_{22}]^{-1} \Delta \hat{S}_{22} \hat{S}_{22}^{-1} \Delta \hat{S}_{21} - \Delta \hat{S}_{12} \hat{S}_{22}^{-1} \Delta \hat{S}_{21}, \end{aligned} \quad (\text{SM57})$$

where we used the identities

$$\begin{aligned} \hat{T} &\equiv \hat{S}_{22}^{-1} - \mathbb{E}[\hat{S}_{22}]^{-1} + \mathbb{E}[\hat{S}_{22}]^{-1} \Delta \hat{S}_{22} \mathbb{E}[\hat{S}_{22}]^{-1} = (\mathbb{E}[\hat{S}_{22}]^{-1} \Delta \hat{S}_{22})^2 \hat{S}_{22}^{-1}, \\ (\hat{S}_{22}^{-1} - \mathbb{E}[\hat{S}_{22}]^{-1}) &= -\mathbb{E}[\hat{S}_{22}]^{-1} \Delta \hat{S}_{22} \hat{S}_{22}^{-1}. \end{aligned} \quad (\text{SM58})$$

If furthermore $u \leq \kappa_{22}/2$, then $\lambda_{\min}(\hat{S}_{22}) \geq \frac{\kappa_{22}-u}{N} \geq \frac{\kappa_{22}}{2N}$ and thus

$$\|\mathbb{E}[\hat{S}_{12}]\|_{\text{op}} \leq \frac{\|\bar{S}_{12}\|_{\text{op}}}{N}, \quad \|\mathbb{E}[\hat{S}_{21}]\|_{\text{op}} \leq \frac{\|\bar{S}_{21}\|_{\text{op}}}{N}, \quad \|\mathbb{E}[\hat{S}_{22}]^{-1}\|_{\text{op}} \leq \frac{N}{\kappa_{22}}, \quad \|\hat{S}_{22}^{-1}\|_{\text{op}} \leq \frac{2N}{\kappa_{22}} \quad (\text{SM59})$$

$$\|\hat{T}\|_{\text{op}} \leq \|\mathbb{E}[\hat{S}_{22}]^{-1}\|_{\text{op}}^2 \|\Delta \hat{S}_{22}\|_{\text{op}}^2 \|\hat{S}_{22}^{-1}\|_{\text{op}} \leq \left(\frac{N}{\kappa_{22}}\right)^2 \left(\frac{u}{N}\right)^2 \frac{2N}{\kappa_{22}} = \frac{2Nu^2}{\kappa_{22}^3}. \quad (\text{SM60})$$

Applying these bounds to [eq. \(SM57\)](#) we get

$$\|(\Delta_c \hat{S}_{\text{eff}})_{\text{nl}}\|_{\text{op}} \leq K_S^{\text{nl}} u^2, \quad K_S^{\text{nl}} = \frac{2\|\bar{S}_{12}\|_{\text{op}} \|\bar{S}_{21}\|_{\text{op}}}{\kappa_{22}^3} + \frac{2(\|\bar{S}_{12}\|_{\text{op}} + \|\bar{S}_{21}\|_{\text{op}})}{\kappa_{22}^2} + \frac{2}{\kappa_{22}}, \quad (\text{SM61})$$

where K_S^{nl} is deterministic and independent of N and n_{tr} (and is uniform in n_{out} whenever the displayed conditioning quantities are uniform). Thus

$$|q_0^T S_{\text{eff}}^{-1} (\Delta_c \hat{S}_{\text{eff}})_{\text{nl}} S_{\text{eff}}^{-1} p_{\sigma,1}| \leq \frac{\|\bar{S}_{\text{eff}}\|_{\text{op}} \|w_{\mathcal{O}}\|_2 \|p_{\sigma,1}\|_2 K_S^{\text{nl}} u^2}{\kappa_{\text{eff}}^2 N} = K_{R,1} \frac{u^2}{N}, \quad (\text{SM62})$$

with $K_{R,1}$ deterministic and independent of N, n_{tr} , assuming $u \leq \kappa_{22}/2$.

2. Consider now the second term of eq. (SM54). Combining eqs. (SM41) and (SM56), we have

$$\|(\Delta_c \hat{\mathbf{S}}_{\text{eff}})_p\|_{\text{op}} \leq K_S^p \frac{u}{\sqrt{N}}, \quad K_S^p \equiv 1 + \|\bar{\mathbf{\Sigma}}_{22}^{-1} \bar{\mathbf{\Sigma}}_{21}\|_{\text{op}} + \|\bar{\mathbf{\Sigma}}_{12} \bar{\mathbf{\Sigma}}_{22}^{-1}\|_{\text{op}} + \frac{\|\bar{\mathbf{\Sigma}}_{12} \bar{\mathbf{\Sigma}}_{22}^{-1}\|_{\text{op}} \|\bar{\mathbf{\Sigma}}_{22}^{-1} \bar{\mathbf{\Sigma}}_{21}\|_{\text{op}}}{\sqrt{N}}. \quad (\text{SM63})$$

Combining this with eq. (SM61) we get

$$\|\Delta_c \hat{\mathbf{S}}_{\text{eff}}\|_{\text{op}} \leq K_S^p \frac{u}{\sqrt{N}} + K_S^{\text{nl}} u^2 \leq (K_S^p + K_S^{\text{nl}}) u. \quad (\text{SM64})$$

We now need to get bounds for the components of the $\hat{\mathbf{q}}$ vector. We first observe that

$$\|\hat{\mathbf{q}}_p\|_2 \leq K_q^p \frac{u}{\sqrt{N}}, \quad K_q^p \equiv 2B_y(1 + \|\bar{\mathbf{\Sigma}}_{12} \bar{\mathbf{\Sigma}}_{22}^{-1}\|_{\text{op}}) + K_S^p \|\mathbf{w}_{\mathcal{O}}\|_2. \quad (\text{SM65})$$

The last condition in $\mathcal{E}_u$ controls the cancellation specific to $\hat{\mathbf{q}}_{\text{nl}}$. Indeed,

$$\hat{\mathbf{h}} = \frac{1}{N n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \left(\hat{\xi}_{i,2} \hat{\xi}_{i,1}^T - \mathbb{E}[\hat{\xi}_{i,2} \hat{\xi}_{i,1}^T] \right) U_1^T \mathbf{w}_{\mathcal{O}}. \quad (\text{SM66})$$

The terms of order $N^{-1/2}$ cancel exactly because $\mathbf{y} = \mathbf{P}_1^T \mathbf{w}_{\mathcal{O}}$. Thus the required bound on $\hat{\mathbf{h}}$ is a quadratic concentration estimate; it is part of the definition of $\mathcal{E}_u$, and will be proved in corollary 1.

Expanding the definitions given in eq. (SM40) we get the more explicit expressions

$$\begin{aligned} (\Delta_c \hat{\ell}_{\text{eff}})_p &= \Delta \hat{\ell}_1 - \mathbb{E}[\hat{\mathbf{S}}_{12}] \mathbb{E}[\hat{\mathbf{S}}_{22}]^{-1} \Delta \hat{\ell}_2 = \Delta \hat{\ell}_1 - \bar{\mathbf{\Sigma}}_{12} \bar{\mathbf{\Sigma}}_{22}^{-1} \Delta \hat{\ell}_2, \\ (\Delta_c \hat{\ell}_{\text{eff}})_{\text{nl}} &= \hat{\ell}_{\text{eff}} - \ell_1 - (\Delta_c \hat{\ell}_{\text{eff}})_p = -\mathbb{E}[\hat{\mathbf{S}}_{12}] (\hat{\mathbf{S}}_{22}^{-1} - \mathbb{E}[\hat{\mathbf{S}}_{22}]^{-1}) \Delta \hat{\ell}_2 - \Delta \hat{\mathbf{S}}_{12} \hat{\mathbf{S}}_{22}^{-1} \Delta \hat{\ell}_2. \end{aligned} \quad (\text{SM67})$$

Substitution of eqs. (SM57) and (SM67) into eq. (SM50) gives

$$\hat{\mathbf{q}}_{\text{nl}} = \mathbb{E}[\hat{\mathbf{S}}_{12}] \hat{\mathbf{T}} \mathbb{E}[\hat{\mathbf{S}}_{21}] U_1^T \mathbf{w}_{\mathcal{O}} + \mathbb{E}[\hat{\mathbf{S}}_{12}] (\hat{\mathbf{S}}_{22}^{-1} - \mathbb{E}[\hat{\mathbf{S}}_{22}]^{-1}) \hat{\mathbf{h}} + \Delta \hat{\mathbf{S}}_{12} (\hat{\mathbf{S}}_{22}^{-1} - \mathbb{E}[\hat{\mathbf{S}}_{22}]^{-1}) \mathbb{E}[\hat{\mathbf{S}}_{21}] U_1^T \mathbf{w}_{\mathcal{O}} + \Delta \hat{\mathbf{S}}_{12} \hat{\mathbf{S}}_{22}^{-1} \hat{\mathbf{h}}. \quad (\text{SM68})$$

Using eqs. (SM56) and (SM59) in this expansion and $N^{-1} \leq N^{-1/2}$ gives

$$\|\hat{\mathbf{q}}_{\text{nl}}\|_2 \leq K_q^{\text{nl}} \frac{u^2}{\sqrt{N}}. \quad (\text{SM69})$$

Here K_q^{nl} is deterministic and may depend on K_B , κ_{22}^{-1} , $\|\bar{\mathbf{\Sigma}}_{12}\|_{\text{op}}$, $\|\bar{\mathbf{\Sigma}}_{21}\|_{\text{op}}$, and $\|\mathbf{w}_{\mathcal{O}}\|_2$, but not on N or n_{tr} . Finally, we have $\|\mathbf{q}_0\|_2 \leq K_q^0/N$, and thus

$$\|\hat{\mathbf{q}}\|_2 \leq \frac{K_q}{\sqrt{N}}, \quad \|\hat{\mathbf{q}} - \mathbf{q}_0\|_2 \leq K'_q \frac{u}{\sqrt{N}}. \quad (\text{SM70})$$

This is sufficient to bound the second term of eq. (SM54) as:

$$|(\hat{\mathbf{q}} - \mathbf{q}_0)^T \mathbf{S}_{\text{eff}}^{-1} \Delta_c \hat{\mathbf{S}}_{\text{eff}} \mathbf{S}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}| \leq K_{R,2} \frac{u^2}{\sqrt{N}}. \quad (\text{SM71})$$

3. For the third term in eq. (SM54) we also need $\|\hat{\mathbf{S}}_{\text{eff}}^{-1}\|_{\text{op}}$. If the last condition in eq. (SM74) below holds, then eq. (SM64) and Weyl's inequality give $\|\hat{\mathbf{S}}_{\text{eff}}^{-1}\|_{\text{op}} \leq 2/\kappa_{\text{eff}}$. We now only need to use the already derived bounds in eqs. (SM59), (SM64) and (SM70), to conclude

$$|\hat{\mathbf{q}}^T (\mathbf{S}_{\text{eff}}^{-1} \Delta_c \hat{\mathbf{S}}_{\text{eff}})^2 \hat{\mathbf{S}}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}| \leq K_{R,3} \frac{u^2}{\sqrt{N}}. \quad (\text{SM72})$$

4. The final term in eq. (SM54) is now also straightforwardly bounded:

$$|\hat{\mathbf{q}}_{\text{nl}}^T \mathbf{S}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}| \leq K_{R,4} \frac{u^2}{\sqrt{N}}. \quad (\text{SM73})$$

Combining eqs. (SM62) and (SM71) to (SM73), we conclude that

$$\left[\hat{\mathbf{E}} \in \mathcal{E}_u \text{ and } \left(u \leq 1, u \leq \kappa_{22}/2, K_S^p \frac{u}{\sqrt{N}} + K_S^{\text{nl}} u^2 \leq \frac{\kappa_{\text{eff}}}{2} \right) \right] \implies |\hat{R}| \leq K_R \frac{u^2}{\sqrt{N}}, \quad (\text{SM74})$$

for some deterministic constant $K_R = \max(K_{R,1}, K_{R,2}, K_{R,3}, K_{R,4})$. In words, we proved the $|\hat{R}|$ is small for well-conditioned events $\hat{\mathbf{E}} \in \mathcal{E}_u$, assuming a sufficiently small u . This further condition on u is important because we will show in section SM3.5 that u is small with high probability, but these probability statements involve a confidence coefficients, and requiring the confidence $1 - \delta$ to be very close to 1 necessarily increases the error bound u . In other words, this condition prevents us from using probability bounds with too high a confidence $1 - \delta$, as that would make u too large and violate the invertibility conditions needed to derive the result. In particular, the small u is required to ensure the invertibility of the stochastic matrices $\hat{\mathbf{S}}_{22}$ and $\hat{\mathbf{S}}_{\text{eff}}$, via the bounds

$$\lambda_{\min}(\hat{\mathbf{S}}_{22}) \geq \frac{\kappa_{22}}{2N}, \quad \lambda_{\min}(\hat{\mathbf{S}}_{\text{eff}}) \geq \frac{\kappa_{\text{eff}}}{2}. \quad (\text{SM75})$$

It remains to show that $\mathcal{E}_u$ has high probability.

SM3.5. High-probability bounds

This subsection converts the deterministic remainder estimate of section SM3.4 into a probabilistic statement for the actual multinomial experiment. We establish simultaneous concentration of the required block and vector fluctuations and then deduce theorem 3, which is the probabilistic input to the MSE analysis in section SM3.6.

For each training state $i \in \{1, \dots, n_{\text{tr}}\}$, let $Z_{i,1}, \dots, Z_{i,N}$ be independent categorical random variables with probability vector $\mathbf{p}_i \in \mathbb{R}^{n_{\text{out}}}$. Assume also that the samples associated with distinct training states are independent. Define

$$\hat{\mathbf{p}}_i = \frac{1}{N} \sum_{s=1}^N \mathbf{e}_{Z_{i,s}}, \quad \hat{\boldsymbol{\xi}}_i = \sqrt{N}(\hat{\mathbf{p}}_i - \mathbf{p}_i), \quad (\text{SM76})$$

where $\mathbf{e}_{Z_{i,s}}$ denotes the canonical vectors. The two block dimensions are $m_1 = d^2$ and $m_2 = n_{\text{out}} - d^2$, so that $m_1 + m_2 = n_{\text{out}}$. We denote by $\hat{\boldsymbol{\xi}}_{i,a} \in \mathbb{R}^{m_a}$ the a -th block of $\hat{\boldsymbol{\xi}}_i$. Recall that $\mathbf{y}_i = \mathbf{p}_{i,1}^T \mathbf{w}_{\mathcal{O}}$ and $B_y = \max_i |y_i|$. For $0 < \delta < 1$, define

$$q_\delta = 2n_{\text{out}} \log 9 + \log \left(\frac{32}{\delta} \right), \quad r_\delta = \sqrt{\frac{q_\delta}{n_{\text{tr}}}} + \frac{q_\delta}{n_{\text{tr}}}. \quad (\text{SM77})$$

The numerical factors in q_δ are chosen so that the vector and matrix bounds below hold simultaneously.

Lemma 1 (Subgaussian multinomial error). *For every $\mathbf{v} \in \mathbb{R}^{n_{\text{out}}}$ and $\lambda \in \mathbb{R}$,*

$$\mathbb{E} \exp \left(\lambda \mathbf{v}^T (\hat{\mathbf{p}}_i - \mathbf{p}_i) \right) \leq \exp \left(\frac{\lambda^2 \|\mathbf{v}\|_2^2}{4N} \right). \quad (\text{SM78})$$

Equivalently, in the convention $\mathbb{E} e^{\lambda X} \leq e^{\lambda^2 \sigma^2 / 2}$, the random variable $\mathbf{v}^T \hat{\boldsymbol{\xi}}_i$ is subgaussian with parameter $\sigma^2 \leq \|\mathbf{v}\|_2^2 / 2$.

Proof. Set $Y_s = \mathbf{v}^T (\mathbf{e}_{Z_{i,s}} - \mathbf{p}_i)$. Then $\mathbb{E}[Y_s] = 0$, and the range of Y_s has length $\max_j v_j - \min_j v_j \leq \sqrt{2} \|\mathbf{v}\|_2$. Hoeffding's lemma therefore implies $\mathbb{E} e^{t Y_s} \leq \exp(t^2 \|\mathbf{v}\|_2^2 / 4)$. Independence and $\mathbf{v}^T (\hat{\mathbf{p}}_i - \mathbf{p}_i) = N^{-1} \sum_{s=1}^N Y_s$ now give

$$\mathbb{E} \exp \left(\lambda \mathbf{v}^T (\hat{\mathbf{p}}_i - \mathbf{p}_i) \right) = \prod_{s=1}^N \mathbb{E} \exp \left(\frac{\lambda}{N} Y_s \right) \leq \exp \left(\frac{\lambda^2 \|\mathbf{v}\|_2^2}{4N} \right). \quad (\text{SM79})$$

□

Lemma 2 (Concentration of $\Delta \hat{\boldsymbol{\ell}}_a$). *With probability at least $1 - \delta$, simultaneously for $a \in \{1, 2\}$,*

$$\|\Delta \hat{\boldsymbol{\ell}}_a\|_2 \leq 2B_y \sqrt{\frac{q_\delta}{N n_{\text{tr}}}} \leq \frac{2B_y r_\delta}{\sqrt{N}}. \quad (\text{SM80})$$

Proof. Because $\Delta \hat{\ell}_a = (n_{\text{tr}} \sqrt{N})^{-1} \sum_{i=1}^{n_{\text{tr}}} y_i \hat{\xi}_{i,a}$, [lemma 1](#) and independence across i imply, for every unit vector $v \in \mathbb{R}^{m_a}$,

$$\mathbb{E} \exp(\lambda v^T \Delta \hat{\ell}_a) \leq \exp\left(\frac{\lambda^2}{4Nn_{\text{tr}}^2} \sum_{i=1}^{n_{\text{tr}}} y_i^2\right) \leq \exp\left(\frac{B_y^2 \lambda^2}{4Nn_{\text{tr}}}\right). \quad (\text{SM81})$$

If $B_y = 0$, then every $y_i = 0$ and $\Delta \hat{\ell}_a = 0$, so the claim is immediate. Assume henceforth that $B_y > 0$. The Chernoff bound therefore yields

$$\mathbb{P}\left(|v^T \Delta \hat{\ell}_a| > t\right) \leq 2 \exp\left(-\frac{Nn_{\text{tr}} t^2}{B_y^2}\right). \quad (\text{SM82})$$

Let $\mathcal{N}_a$ be a $1/4$ -net of the unit sphere in $\mathbb{R}^{m_a}$ with $|\mathcal{N}_a| \leq 9^{m_a}$. For any nonzero $z \in \mathbb{R}^{m_a}$, choose $v \in \mathcal{N}_a$ within distance $1/4$ of $z/\|z\|_2$; the case $z = 0$ is immediate. Then

$$\|z\|_2 = \frac{z^T z}{\|z\|_2} \leq |v^T z| + \frac{1}{4} \|z\|_2, \quad (\text{SM83})$$

so $\|z\|_2 \leq (4/3) \max_{v \in \mathcal{N}_a} |v^T z|$; we use the looser factor 2 below. Taking $t = B_y \sqrt{q_\delta / (Nn_{\text{tr}})}$ in [eq. \(SM82\)](#) and applying a union bound gives

$$\mathbb{P}\left(\|\Delta \hat{\ell}_a\|_2 > 2B_y \sqrt{\frac{q_\delta}{Nn_{\text{tr}}}}\right) \leq 2 \cdot 9^{m_a} e^{-q_\delta} \leq \frac{\delta}{16}. \quad (\text{SM84})$$

A second union bound over $a \in \{1, 2\}$ proves the simultaneous statement. $\square$

Lemma 3 (Quadratic concentration). *For $a, b \in \{1, 2\}$, define*

$$\hat{Q}_{ab} = \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \left(\hat{\xi}_{i,a} \hat{\xi}_{i,b}^T - \mathbb{E}[\hat{\xi}_{i,a} \hat{\xi}_{i,b}^T] \right). \quad (\text{SM85})$$

There is an absolute constant $K_B > 0$ such that, simultaneously for all four pairs (a, b) , with probability at least $1 - \delta$,

$$\|\hat{Q}_{ab}\|_{\text{op}} \leq K_B r_\delta. \quad (\text{SM86})$$

Equivalently,

$$\left\| \frac{1}{Nn_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \left(\hat{\xi}_{i,a} \hat{\xi}_{i,b}^T - \mathbb{E}[\hat{\xi}_{i,a} \hat{\xi}_{i,b}^T] \right) \right\|_{\text{op}} \leq \frac{K_B r_\delta}{N}. \quad (\text{SM87})$$

Proof. Fix unit vectors $u \in \mathbb{R}^{m_a}$ and $v \in \mathbb{R}^{m_b}$, and let

$$X_i = (u^T \hat{\xi}_{i,a})(v^T \hat{\xi}_{i,b}) - \mathbb{E}[(u^T \hat{\xi}_{i,a})(v^T \hat{\xi}_{i,b})]. \quad (\text{SM88})$$

Both factors have uniformly bounded subgaussian norm by [lemma 1](#). They need not be independent: the standard product inequality for Orlicz norms still gives a uniform subexponential bound for their product, and centering changes that bound by at most a factor of two. Thus $\|X_i\|_{\psi_1} \leq K_X$ for one absolute constant K_X . Bernstein's inequality then gives an absolute constant K_B such that, for every $s > 0$,

$$\mathbb{P}\left(\left| \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} X_i \right| > K_B \left(\sqrt{\frac{s}{n_{\text{tr}}}} + \frac{s}{n_{\text{tr}}} \right)\right) \leq 2e^{-s}. \quad (\text{SM89})$$

Let $\mathcal{N}_a$ and $\mathcal{N}_b$ be $1/4$ -nets of the corresponding unit spheres. Their product has cardinality at most $9^{m_a+m_b}$, and any $B \in \mathbb{R}^{m_a \times m_b}$ satisfies $\|B\|_{\text{op}} \leq 2 \max_{u \in \mathcal{N}_a, v \in \mathcal{N}_b} |u^T B v|$. Apply [eq. \(SM89\)](#) with $s = q_\delta$. Since $m_a + m_b \leq 2n_{\text{out}}$, a union bound over the two nets and all four block pairs has failure probability at most $8 \cdot 9^{2n_{\text{out}}} e^{-q_\delta} = \delta/4$. Absorbing the net factor 2 into K_B proves [eq. \(SM86\)](#). $\square$

Lemma 4 (Concentration of the linear matrix terms). *For $a \in \{1, 2\}$, define $\hat{A}_a = n_{\text{tr}}^{-1} \sum_{i=1}^{n_{\text{tr}}} \mathbf{p}_{i,1} \hat{\xi}_{i,a}^T$, where $\mathbf{p}_{i,1} \in \mathbb{R}^{m_1}$ is the first block of $\mathbf{p}_i$. Then, simultaneously for $a \in \{1, 2\}$, with probability at least $1 - \delta$,*

$$\|\hat{A}_a\|_{\text{op}} \leq 2\sqrt{\frac{q_\delta}{n_{\text{tr}}}} \leq 2r_\delta. \quad (\text{SM90})$$

Consequently,

$$\left\| \frac{1}{n_{\text{tr}} \sqrt{N}} \sum_{i=1}^{n_{\text{tr}}} \mathbf{p}_{i,1} \hat{\xi}_{i,a}^T \right\|_{\text{op}} \leq \frac{2r_\delta}{\sqrt{N}}. \quad (\text{SM91})$$

Proof. Fix unit vectors $\mathbf{u} \in \mathbb{R}^{m_1}$ and $\mathbf{v} \in \mathbb{R}^{m_a}$. The subgaussian moment-generating-function bound gives

$$\mathbb{E} \exp(\lambda \mathbf{u}^T \hat{A}_a \mathbf{v}) \leq \exp\left(\frac{\lambda^2}{4n_{\text{tr}}^2} \sum_{i=1}^{n_{\text{tr}}} (\mathbf{u}^T \mathbf{p}_{i,1})^2\right) \leq \exp\left(\frac{\lambda^2}{4n_{\text{tr}}}\right), \quad (\text{SM92})$$

where we used $\|\mathbf{p}_{i,1}\|_2 \leq \|\mathbf{p}_i\|_2 \leq 1$. Hence $\mathbb{P}(|\mathbf{u}^T \hat{A}_a \mathbf{v}| > t) \leq 2e^{-n_{\text{tr}} t^2}$. Applying the same 1/4-net argument as above with $t = \sqrt{q_\delta / n_{\text{tr}}}$ gives, for each a ,

$$\mathbb{P}\left(\|\hat{A}_a\|_{\text{op}} > 2\sqrt{\frac{q_\delta}{n_{\text{tr}}}}\right) \leq 29^{m_1+m_a} e^{-q_\delta} \leq \frac{\delta}{16}. \quad (\text{SM93})$$

A union bound over the two values of a completes the proof. $\square$

Corollary 1 (Simultaneous basic event). *There is a fixed absolute constant $K_S > 0$ such that, with probability at least $1 - \delta$, all of the following bounds hold simultaneously, where r_δ is defined in eq. (SM77):*

$$\begin{aligned} \|\Delta \hat{\mathbf{S}}_{11}\|_{\text{op}}, \|\Delta \hat{\mathbf{S}}_{12}\|_{\text{op}}, \|\Delta \hat{\mathbf{S}}_{21}\|_{\text{op}} &\leq \frac{K_S r_\delta}{\sqrt{N}}, & \|\hat{\mathbf{h}}\|_2 &\leq K_B \|\mathbf{w}_\mathcal{O}\|_2 \frac{r_\delta}{N}, \\ \|\Delta \hat{\mathbf{S}}_{22}\|_{\text{op}} &\leq \frac{K_S r_\delta}{N}, & \|\Delta \hat{\ell}_a\|_2 &\leq \frac{2B_y r_\delta}{\sqrt{N}}, \quad a \in \{1, 2\}. \end{aligned} \quad (\text{SM94})$$

Thus, for some $C_0 \geq \max(1, K_S)$, we have $\mathbb{P}(\mathcal{E}_{C_0 r_\delta}) \geq 1 - \delta$.

Proof. The block fluctuations have the exact decompositions

$$\begin{aligned} \Delta \hat{\mathbf{S}}_{11} &= \frac{\hat{\mathbf{A}}_1 + \hat{\mathbf{A}}_1^T}{\sqrt{N}} + \frac{\hat{\mathbf{Q}}_{11}}{N}, & \Delta \hat{\mathbf{S}}_{12} &= \frac{\hat{\mathbf{A}}_2}{\sqrt{N}} + \frac{\hat{\mathbf{Q}}_{12}}{N}, \\ \Delta \hat{\mathbf{S}}_{21} &= \frac{\hat{\mathbf{A}}_2^T}{\sqrt{N}} + \frac{\hat{\mathbf{Q}}_{21}}{N}, & \Delta \hat{\mathbf{S}}_{22} &= \frac{\hat{\mathbf{Q}}_{22}}{N}. \end{aligned} \quad (\text{SM95})$$

Combining lemmas 2 to 4 and using $N^{-1} \leq N^{-1/2}$ proves the asserted block and vector bounds. The common choice of q_δ makes their total failure probability smaller than δ . Finally, the bound on $\hat{\mathbf{h}}$ follows from eqs. (SM66) and (SM86) with $(a, b) = (2, 1)$. $\square$

Theorem 3 (High-probability remainder bound). *For fixed d and n_{out} , there are deterministic constants $u_\star, C_R, C_0 > 0$, independent of N and n_{tr} , such that for any δ satisfying $C_0 r_\delta \leq u_\star$, with probability at least $1 - \delta$, we have $|\hat{R}| \leq C_R \frac{r_\delta^2}{\sqrt{N}}$.*

Proof. The proof combines the probabilistic control of the basic fluctuations with the deterministic remainder estimate. Specifically, corollary 1 places all the required blocks and vectors in an event of the form $\mathcal{E}_u$ with probability at least $1 - \delta$. Once its tolerance u is below a fixed threshold, eqs. (SM74) to (SM75) guarantee both invertibility and the quadratic remainder bound.

Choose $C_0 \geq \max\{1, K_S\}$ as in corollary 1, and recall the definition of r_δ from eq. (SM77). Thus the event $\mathcal{E}_{C_0 r_\delta}$ happens with probability at least $1 - \delta$. Choose $u_\star > 0$, independent on N , such that $u_\star \leq 1$, $u_\star \leq \kappa_{22}/2$, and

$K_S^p u_* + K_S^l u_*^2 \leq \kappa_{\text{eff}}/2$. Then, for all $N \geq 1$ and $u \leq u_*$, the conditions of [eq. \(SM74\)](#) are satisfied, and thus $|\hat{R}| \leq K_R u^2 / \sqrt{N}$ holds with probability at least $1 - \delta$.

Suppose now that δ is such that $C_0 r_\delta \leq u_*$. Then $\mathbb{P}(\mathcal{E}_{C_0 r_\delta}) \geq 1 - \delta$, and also the conditions of [eq. \(SM74\)](#) are satisfied with $u = C_0 r_\delta$. Thus $|\hat{R}| \leq K_R (C_0 r_\delta)^2 / \sqrt{N} = C_R r_\delta^2 / \sqrt{N}$, where $C_R \equiv K_R C_0^2$. $\square$

To convert the condition $C_0 r_\delta \leq u_*$ into a condition on n_{tr} , observe that

Unpacking the definition of r_δ with [eq. \(SM77\)](#), we see that $C_0 r_\delta \leq u_*$ is equivalent to $n_{\text{tr}} \geq q_\delta / c_*$ where $c_* \equiv (\sqrt{1 + 4(u_*/C_0)} - 1)^2/4$. And plugging in q_δ , this amounts to $n_{\text{tr}} \geq (2n_{\text{out}} \log 9 + \log(32/\delta)) / c_*$. Thus, the high probability bound of [theorem 3](#) holds for $n_{\text{tr}} \gtrsim n_{\text{out}} + \log(1/\delta)$. For every fixed confidence level $1 - \delta$ and fixed n_{out} , this condition therefore holds for all sufficiently large n_{tr} . At fixed n_{tr} , taking δ arbitrarily close to zero increases q_δ and can violate the condition; this makes the bound inconclusive, but does not assert that the empirical matrices are singular.

For fixed n_{out} and fixed confidence level, as $n_{\text{tr}} \rightarrow \infty$ one has $r_\delta^2 = O(n_{\text{tr}}^{-1})$, and hence

$$\hat{R} = O_P\left(\frac{1}{n_{\text{tr}} \sqrt{N}}\right). \quad (\text{SM96})$$

The constant implicit in $O(n_{\text{tr}}^{-1})$ may depend on δ and on the fixed problem dimensions and conditioning data.

Combining [eqs. \(SM48\)](#) and [\(SM96\)](#), we see that, if $V_N = O(1)$, $\sqrt{\mathbb{E}[\hat{L}^2]} = O((N n_{\text{tr}})^{-1/2})$. [Equation \(SM96\)](#) shows that the remainder is smaller than the root-mean-square leading fluctuation by a factor $n_{\text{tr}}^{-1/2}$. Likewise, the remainder is subleading to the deterministic bias when $n_{\text{tr}} \gg \sqrt{N}$.

SM3.6. Mean-squared error bound

This final subsection assembles the deterministic bias of [section SM3.2](#), the fluctuation variance of [section SM3.3](#), and the remainder control of [sections SM3.4](#) and [SM3.5](#). Controlling both the high-probability event and its exceptional complement yields the rigorous MSE expansion that [section SM4](#) interprets as a resource tradeoff. The confidence condition in ?? is needed only for the fixed- δ high-probability statement. The second-moment argument below does not attempt to send an arbitrarily chosen δ to zero. Instead it uses the confidence-independent event $\mathcal{A} = \{\epsilon \leq e_0\}$ and controls its complement with the everywhere-defined Moore–Penrose pseudoinverse.

By [section SM3.2](#), the deterministic bias term is $B = -b_N/N$, where $b_N = \mathbf{w}_O^T U_1 \bar{\Sigma}_{\text{eff}}^{-1} \mathbf{p}_{\sigma,1}$. Likewise, [eq. \(SM48\)](#) defines V_N and gives $\mathbb{E}[\hat{L}^2] = \frac{V_N}{N n_{\text{tr}}} + O(\frac{1}{N^2 n_{\text{tr}}})$. Although [theorem 3](#) shows that the remainder is subleading with high probability, that statement alone does not control $\mathbb{E}[\Delta \delta^2]$. It bounds $\hat{R}$ only on a good event; on its complement the empirical matrix can be nearly singular and $\|\hat{\mathbf{P}}^+\|_{\text{op}}$, and hence $|\hat{R}|$, can be very large. Even an event of very small probability can therefore make a non-negligible contribution to the second moment unless the size of $\hat{R}$ on that event is also controlled.

We therefore use $\hat{R} = \Delta \hat{\delta} - B - \hat{L}$ on the entire probability space, with $\Delta \hat{\delta}$ given by the pseudoinverse expression [eq. \(SM34\)](#); this definition is meaningful for every realization because the Moore–Penrose pseudoinverse always exists. On any event $\mathcal{E}_u$ for which [eq. \(SM74\)](#) holds, [eq. \(SM75\)](#) ensures that all inverses in [eq. \(SM54\)](#) exist. Hence, on that event, this everywhere-defined remainder agrees with the inverse-based algebraic expression in [eq. \(SM54\)](#).

Defining ϵ as

$$\epsilon \equiv \max \left(\sqrt{N} \max_{(a,b) \neq (2,2)} \|\Delta \hat{\mathbf{S}}_{ab}\|_{\text{op}}, N \|\Delta \hat{\mathbf{S}}_{22}\|_{\text{op}}, \sqrt{N} (1 + B_y)^{-1} \max_{a \in \{1,2\}} \|\Delta \hat{\ell}_a\|_2, N (1 + \|\mathbf{w}_O\|_2)^{-1} \|\hat{\mathbf{h}}\|_2 \right), \quad (\text{SM97})$$

the bounds in [corollary 1](#) imply that, for

$$K_\epsilon \equiv \max\{K_S, 2, K_B\}, \quad (\text{SM98})$$

$\mathbb{P}(\epsilon > K_\epsilon r_\delta) \leq \delta$. Set

$$C_q \equiv 2 \log 9 + \log 2 + 1, \quad K_{\text{tail}} \equiv K_\epsilon \max\{\sqrt{C_q}, C_q\}. \quad (\text{SM99})$$

For $t \geq \log 16$, choose $\delta = 16e^{-t}$ in eq. (SM77). Then

$$q_\delta = 2n_{\text{out}} \log 9 + t + \log 2 \leq C_q(n_{\text{out}} + t),$$

and therefore $K_\epsilon r_\delta \leq K_{\text{tail}}[\sqrt{(n_{\text{out}} + t)/n_{\text{tr}}} + (n_{\text{out}} + t)/n_{\text{tr}}]$. For $0 \leq t < \log 16$, the right-hand side of the probability bound below is at least one, so the statement is automatic. Thus, for every $t \geq 0$,

$$\mathbb{P}\left(\epsilon > K_{\text{tail}} \left[\sqrt{\frac{n_{\text{out}} + t}{n_{\text{tr}}}} + \frac{n_{\text{out}} + t}{n_{\text{tr}}} \right]\right) \leq 16e^{-t}, \quad t \geq 0. \quad (\text{SM100})$$

We next relate ϵ to the deterministic event eq. (SM56). Define

$$c_y \equiv \begin{cases} (1 + B_y)/(2B_y), & B_y > 0, \\ 0, & B_y = 0, \end{cases} \quad c_w \equiv \begin{cases} (1 + \|\mathbf{w}_\mathcal{O}\|_2)/(K_B \|\mathbf{w}_\mathcal{O}\|_2), & \|\mathbf{w}_\mathcal{O}\|_2 > 0, \\ 0, & \mathbf{w}_\mathcal{O} = 0, \end{cases} \quad (\text{SM101})$$

Here $B_y = \max_i |y_i|$ is the uniform target bound defined in ???. Set $C_\mathcal{E} \equiv \max\{1, c_y, c_w\}$, where the corresponding random vector vanishes identically in either zero case. Comparing eq. (SM97) with the componentwise bounds defining $\mathcal{E}_u$ in eq. (SM56) gives

$$\epsilon \leq u \implies \mathcal{E}_{C_\mathcal{E} u}. \quad (\text{SM102})$$

Indeed, $\epsilon \leq u$ gives the three non-(2, 2) block bounds with scale $u/\sqrt{N}$ and the (2, 2) bound with scale u/N directly. For $B_y > 0$, the definition of c_y gives $(1 + B_y)u \leq 2B_y C_\mathcal{E} u$; if $B_y = 0$, then $\Delta \hat{\ell}_a = 0$. Similarly, for $\|\mathbf{w}_\mathcal{O}\|_2 > 0$, the definition of c_w gives $(1 + \|\mathbf{w}_\mathcal{O}\|_2)u \leq K_B \|\mathbf{w}_\mathcal{O}\|_2 C_\mathcal{E} u$; if $\mathbf{w}_\mathcal{O} = 0$, then $\hat{\mathbf{h}} = 0$. These are exactly the remaining vector bounds in $\mathcal{E}_{C_\mathcal{E} u}$. Let $u_\star$ be the deterministic smallness threshold from theorem 3, and set

$$e_0 \equiv \frac{u_\star}{C_\mathcal{E}}, \quad \mathcal{A} \equiv \{\epsilon \leq e_0\}. \quad (\text{SM103})$$

Then eqs. (SM74) and (SM102) give, on $\mathcal{A}$,

$$|\hat{R}| \leq K_R^A \frac{\epsilon^2}{\sqrt{N}}, \quad K_R^A \equiv K_R C_\mathcal{E}^2. \quad (\text{SM104})$$

This reformulation prepares the second-moment argument. The fixed-confidence estimate eq. (SM96) controls $\hat{R}$ with high probability, but it cannot by itself bound the contribution of rare, nearly singular realizations to $\mathbb{E}[\hat{R}^2]$. The scalar ϵ packages all normalized block fluctuations into one variable whose tail can be integrated. Moreover, eq. (SM102) shows that $\mathcal{A} \subseteq \mathcal{E}_{u_\star}$, so $\mathcal{A}$ is a fixed, confidence-independent subset of the deterministic good events used above. On $\mathcal{A}$, the same deterministic remainder estimate now takes the pointwise form eq. (SM104), which can be combined with a fourth-moment bound for ϵ . The complement $\mathcal{A}^c$ is handled separately by pairing a deterministic worst-case bound with its exponentially small probability.

Lemma 5 (Fourth moment and remainder on the good event). *There are deterministic constants $c_\epsilon, K_4, K_{\text{good}} > 0$ such that, whenever $n_{\text{tr}} \geq c_\epsilon n_{\text{out}}$,*

$$\mathbb{E}[\epsilon^4] \leq K_4 \left(\frac{n_{\text{out}}}{n_{\text{tr}}} \right)^2, \quad \mathbb{E}[\hat{R}^2 \mathbf{1}_\mathcal{A}] \leq K_{\text{good}} \frac{n_{\text{out}}^2}{N n_{\text{tr}}^2}. \quad (\text{SM105})$$

Proof. Substitute $t = s + \log 16$ in eq. (SM100) and define the increasing function

$$g(s) \equiv K_{\text{tail}} \left[\sqrt{\frac{n_{\text{out}} + \log 16 + s}{n_{\text{tr}}}} + \frac{n_{\text{out}} + \log 16 + s}{n_{\text{tr}}} \right], \quad s \geq 0.$$

Then $\mathbb{P}(\epsilon > g(s)) \leq e^{-s}$. We convert this tail estimate directly into a fourth-moment estimate. The layer-cake identity and the change of variables $z = g(s)$ give

$$\mathbb{E}[\epsilon^4] = 4 \int_0^\infty z^3 \mathbb{P}(\epsilon > z) dz \leq g(0)^4 + \int_0^\infty e^{-s} d[g(s)^4] = \int_0^\infty e^{-s} g(s)^4 ds.$$

The last equality is integration by parts; its boundary term at infinity vanishes because g grows only linearly. Using $(a+b)^4 \leq 8(a^4 + b^4)$ and, for $j = 2, 4$ and $n_{\text{out}} \geq 1$,

$$\int_0^\infty e^{-s} (n_{\text{out}} + \log 16 + s)^j ds \leq C_j n_{\text{out}}^j,$$

we obtain

$$\mathbb{E}[\epsilon^4] \leq C \left[\frac{n_{\text{out}}^2}{n_{\text{tr}}^2} + \frac{n_{\text{out}}^4}{n_{\text{tr}}^4} \right] \leq K_4 \left(\frac{n_{\text{out}}}{n_{\text{tr}}} \right)^2. \quad (\text{SM106})$$

The second claim follows from eq. (SM104) with $K_{\text{good}} = (K_{\text{R}}^A)^2 K_4$. $\square$

Lemma 6 (Deterministic and probabilistic control of the exceptional event). *For every realization of the frequency matrix,*

$$\|\hat{\mathbf{P}}^+\|_{\text{op}} \leq n_{\text{out}}^{n_{\text{out}}-1} N^{n_{\text{out}}}, \quad |\hat{R}|^2 \leq K_{\text{bad}} n_{\text{tr}} (N n_{\text{out}})^{2n_{\text{out}}}. \quad (\text{SM107})$$

Moreover, define $u_0 \equiv \min \left\{ 1, \left(\frac{e_0}{2K_{\text{tail}}} \right)^2, \frac{e_0}{2K_{\text{tail}}} \right\}$. Then, whenever $u_0 n_{\text{tr}} \geq n_{\text{out}}$, we have

$$\mathbb{P}(\mathcal{A}^c) \leq 16e^{-u_0 n_{\text{tr}} + n_{\text{out}}}, \quad \mathbb{E}[\hat{R}^2 \mathbf{1}_{\mathcal{A}^c}] \leq 16K_{\text{bad}} n_{\text{tr}} (N n_{\text{out}})^{2n_{\text{out}}} e^{-u_0 n_{\text{tr}} + n_{\text{out}}}. \quad (\text{SM108})$$

Proof. Let $\mathbf{A} = N\hat{\mathbf{P}}$. Its entries are integer shot counts between 0 and N . If $s = \text{rank}(\mathbf{A}) > 0$, there is a nonsingular $s \times s$ minor $\mathbf{B}$. Since $\det \mathbf{B}$ is a nonzero integer, $|\det \mathbf{B}| \geq 1$, while $\|\mathbf{B}\|_{\text{op}} \leq \|\mathbf{B}\|_{\text{F}} \leq sN$. Writing the determinant as the product of singular values gives

$$1 \leq \prod_{j=1}^s \sigma_j(\mathbf{B}) \leq (sN)^{s-1} \sigma_s(\mathbf{B}),$$

and hence $\sigma_s(\mathbf{B}) \geq (sN)^{-(s-1)}$. Because $\mathbf{B}$ is a row-and-column compression of $\mathbf{A}$, singular-value interlacing gives $\sigma_s(\mathbf{A}) \geq \sigma_s(\mathbf{B})$. Rescaling by N therefore yields

$$\sigma_s(\hat{\mathbf{P}}) \geq s^{-(s-1)} N^{-s}, \quad \|\hat{\mathbf{P}}^+\|_{\text{op}} = \sigma_s(\hat{\mathbf{P}})^{-1} \leq n_{\text{out}}^{n_{\text{out}}-1} N^{n_{\text{out}}}.$$

The rank-zero case has $\hat{\mathbf{P}}^+ = 0$ and satisfies the same bound.

Next, $\|\mathbf{y}\|_2 \leq B_y \sqrt{n_{\text{tr}}}$ and

$$|\Delta\delta| \leq \|\mathbf{y}\|_2 (\|\hat{\mathbf{P}}^+\|_{\text{op}} + \|\mathbf{P}^+\|_{\text{op}}) \|\mathbf{p}_\sigma\|_2.$$

The deterministic conditioning assumption on $\mathbf{S}_{11}$ bounds $\|\mathbf{P}^+\|_{\text{op}} \leq (\kappa_{11} n_{\text{tr}})^{-1/2}$. To make the remaining deterministic estimate explicit, let $C_{\text{eff}} = 1 + \|\bar{\mathbf{S}}_{12} \bar{\mathbf{S}}_{22}^{-1}\|_{\text{op}}$. Every normalized multinomial column satisfies $\|\hat{\xi}_i\|_2 \leq \sqrt{2N}$, and therefore

$$\|\hat{\mathbf{e}}_{\text{eff}}\|_{\text{op}} \leq \|\hat{\mathbf{e}}_{\text{eff}}\|_{\text{F}} \leq C_{\text{eff}} \sqrt{2N n_{\text{tr}}}, \quad \|\mathbf{P}_1\|_{\text{op}} \leq \|\mathbf{P}_1\|_{\text{F}} \leq \sqrt{n_{\text{tr}}}.$$

Using these inequalities in eqs. (SM40) and (SM41) shows that both $\|(\Delta_c \hat{\ell}_{\text{eff}})_{\text{p}}\|_2$ and $\|(\Delta_c \hat{\mathbf{S}}_{\text{eff}})_{\text{p}}\|_{\text{op}}$ are bounded by deterministic constants independent of N and n_{tr} . The fixed inverse bounds in eq. (SM42) then give $|B| + |\hat{L}| \leq K_L$. Combining this with the displayed bound on $|\Delta\delta|$ and $\hat{R} = \Delta\delta - B - \hat{L}$ proves the second inequality in eq. (SM107), after enlarging K_{bad} .

Finally, take $t = u_0 n_{\text{tr}} - n_{\text{out}} \geq 0$ in eq. (SM100). The corresponding threshold is $K_{\text{tail}}(\sqrt{u_0} + u_0) \leq e_0$, by the definition of u_0 . Thus $\mathcal{A}^c = \{\epsilon > e_0\}$ is contained in the tail event, which proves the probability bound in eq. (SM108). Multiplying that probability by eq. (SM107) proves the second-moment bound. $\square$

Within the fixed- n_{out} regime, the exceptional-event term in eq. (SM108) has logarithm

$$-u_0 n_{\text{tr}} + 2n_{\text{out}} \log N + \log n_{\text{tr}} + O(1).$$

Thus $n_{\text{tr}}/\log N \rightarrow \infty$ makes the negative exponential dominate every displayed polynomial factor at fixed n_{out} ; in particular, the exceptional-event contribution is then smaller than any fixed inverse power of N .

Combining the good- and exceptional-event estimates eqs. (SM105) and (SM108) gives

$$\mathbb{E}[\hat{R}^2] = \mathbb{E}[\hat{R}^2 \mathbf{1}_{\mathcal{A}}] + \mathbb{E}[\hat{R}^2 \mathbf{1}_{\mathcal{A}^c}] \leq \mathcal{R}_{N, n_{\text{tr}}}^{(2)}, \quad (\text{SM109})$$

where the scalar upper bound is

$$\mathcal{R}_{N, n_{\text{tr}}}^{(2)} \equiv \frac{K_{\text{good}} n_{\text{out}}^2}{N n_{\text{tr}}^2} + 16 K_{\text{bad}} n_{\text{tr}} (N n_{\text{out}})^{2n_{\text{out}}} e^{-u_0 n_{\text{tr}} + n_{\text{out}}}.$$

To propagate this remainder bound to the prediction MSE, expand $\Delta\hat{\delta} = B + \hat{L} + \hat{R}$. Since $\mathbb{E}[\hat{L}] = 0$,

$$\mathbb{E}[\Delta\hat{\delta}^2] - B^2 - \mathbb{E}[\hat{L}^2] = 2B\mathbb{E}[\hat{R}] + 2\mathbb{E}[\hat{L}\hat{R}] + \mathbb{E}[\hat{R}^2].$$

Cauchy-Schwarz gives $|\mathbb{E}[\hat{R}]| \leq \sqrt{\mathbb{E}[\hat{R}^2]}$ and $|\mathbb{E}[\hat{L}\hat{R}]| \leq \sqrt{\mathbb{E}[\hat{L}^2]\mathbb{E}[\hat{R}^2]}$; hence

$$\left| \mathbb{E}[\Delta\hat{\delta}^2] - B^2 - \mathbb{E}[\hat{L}^2] \right| \leq 2 \left(|B| + \sqrt{\mathbb{E}[\hat{L}^2]} \right) \sqrt{\mathcal{R}_{N, n_{\text{tr}}}^{(2)}} + \mathcal{R}_{N, n_{\text{tr}}}^{(2)}. \quad (\text{SM110})$$

For fixed d and n_{out} , under the stated conditioning and boundedness assumptions and in a regime in which the exponentially small term is negligible relative to the following polynomial error, define

$$\text{Err}_{N, n_{\text{tr}}} \equiv \frac{1}{N^2 n_{\text{tr}}} + \frac{n_{\text{out}}}{N^{3/2} n_{\text{tr}}} + \frac{n_{\text{out}}}{N n_{\text{tr}}^{3/2}} + \frac{n_{\text{out}}^2}{N n_{\text{tr}}^2}. \quad (\text{SM111})$$

The four summands have distinct origins. The term $N^{-2} n_{\text{tr}}^{-1}$ is the subleading variance term in eq. (SM48). Using $|B| = O(N^{-1})$, $\sqrt{\mathbb{E}[\hat{L}^2]} = O((N n_{\text{tr}})^{-1/2})$, and $\sqrt{\mathbb{E}[\hat{R}^2]} = O(n_{\text{out}}/(\sqrt{N} n_{\text{tr}}))$ in eq. (SM110) gives, respectively, $n_{\text{out}}/(N^{3/2} n_{\text{tr}})$ from $|B|\sqrt{\mathbb{E}[\hat{R}^2]}$, $n_{\text{out}}/(N n_{\text{tr}}^{3/2})$ from $\sqrt{\mathbb{E}[\hat{L}^2]\mathbb{E}[\hat{R}^2]}$, and $n_{\text{out}}^2/(N n_{\text{tr}}^2)$ from $\mathbb{E}[\hat{R}^2]$. Then

$$\mathbb{E}[\Delta\hat{\delta}^2] = \frac{b_N^2}{N^2} + \frac{V_N}{N n_{\text{tr}}} + O(\text{Err}_{N, n_{\text{tr}}}). \quad (\text{SM112})$$

For fixed n_{out} , the two leading contributions are the squared deterministic bias, of order N^{-2} , and the sampling variance, of order $(N n_{\text{tr}})^{-1}$. Provided b_N and V_N remain bounded away from zero, the crossover occurs at $n_{\text{tr}} \asymp N$.

SM4. COST SCALING OF TRAINING A QELM

The goal of this section is to translate the rigorous error expansion of section SM3 into a resource-cost statement for QELM training, focusing on the number of training states and shots per state. Section SM4.1 rewrites the exact bias and variance coefficients in frame language, section SM4.2 converts their asymptotic limits into the dataset-size/shot-budget tradeoff, and section SM4.3 evaluates the resulting leading coefficients for symmetric tight designs. Thus this section is the application layer connecting the direct probabilistic theorem to the resource statements highlighted in the main text.

Problem setting and notation — Suppose that every state in the training set is measured N times. Because shot noise generally makes the target rows inconsistent with the row space of the observed frequency matrix, QELM training is naturally formulated as the least-squares problem

$$\hat{W}_{\mathcal{O}} \in \arg \min_{\mathbf{W}} \left\| \mathbf{W} \hat{\mathbf{P}}_N - \mathbf{M}_{\mathcal{O}}^T \mathbf{M}_{\rho} \right\|_{\text{F}}^2. \quad (\text{SM113})$$

Here $\hat{\mathbf{P}}_N$ is the noisy probability matrix, whose k -th column contains the frequencies obtained from N measurements of ρ_k , and $\mathbf{P} = \mathbb{E}[\hat{\mathbf{P}}_N] = \langle \boldsymbol{\mu}, \boldsymbol{\rho} \rangle = \mathbf{M}_{\mu}^T \mathbf{M}_{\rho}$. Furthermore, $\mathbf{M}_{\mathcal{O}} = \sum_{i=1}^{n_{\text{obs}}} \text{vec}(\mathcal{O}_i) \langle i|$ is the matrix whose columns are the

vectorized target observables $\mathcal{O} \equiv \{\mathcal{O}_i\}_{i=1}^{n_{\text{obs}}}$, and similarly $M_\rho = \sum_{i=1}^{n_{\text{tr}}} \text{vec}(\rho_i) \langle i|$ describes the (vectorized) training states $\rho \equiv \{\rho_i\}_{i=1}^{n_{\text{tr}}}$. The canonical least-squares solution is $\hat{W}_\mathcal{O} = \langle \mathcal{O}, \rho \rangle \hat{P}_N^+$.

Training with infinite statistics — At infinite training statistics, $\hat{P}_N$ is replaced by its expectation P . Under the completeness assumptions stated below, the noiseless least-squares residual vanishes, and the canonical solution is

$$W_\mathcal{O} = (M_\mathcal{O}^T M_\rho)(M_\mu^T M_\rho)^+ = M_\mathcal{O}^T M_\mu^{T+} = \langle \mathcal{O}, \tilde{\mu} \rangle. \quad (\text{SM114})$$

This result assumes that both M_ρ and M_μ have full row rank: physically this amounts to the training states being complete — i.e., spanning the space of all Hermitian operators — and the POVM being informationally complete. Under this assumption, the pseudoinverse of the product also factorises as $(M_\mu^T M_\rho)^+ = M_\rho^+ M_\mu^{T+}$. Furthermore, it is a general fact that the pseudoinverse of a surjective matrix can be interpreted as the transpose of the matrix representing the *canonical dual frame* of the frame given by the columns of the original matrix. Formally, this means that $M_\mu^+ = (M_{\tilde{\mu}})^T$. For more details on this perspective, we refer to the discussion in [15, 59]. In conclusion, training with infinite statistics amounts to recovering a description of the POVM μ through its dual $\tilde{\mu}$.

Using $\hat{W}$ rather than $\hat{W}_\mathcal{O}$ — From eq. (SM113), the canonical solution depends linearly on $\mathcal{O}$ through $M_\mathcal{O}$. This means that the training is not really about the training observables $\mathcal{O}$ *per se*, but rather provides a reconstruction of the POVM itself. For the cost analysis it is therefore convenient to use the canonical solution:

$$\hat{W} = \min_W \|W \hat{P}_N - M_\rho\|^2 = M_\rho \hat{P}_N^+. \quad (\text{SM115})$$

This amounts to using the training to find an estimator for the states themselves, rather than for any specific observable. Estimators for target observables can then be recovered afterwards by simply multiplying by the observables we are interested in: $\hat{W}_\mathcal{O} = M_\mathcal{O}^T \hat{W}$. This perspective also highlights the fact that the cost of training does not depend on the number of target observables, *per se*. Specializing the training to a specific target observable is a simple matter of post-processing and does not affect the measurement stage at all.

SM4.1. MSE at infinite test statistics

We now translate the rigorous decomposition of section SM3 into the frame notation used for the QELM resource analysis.

For any $u, v \in \mathbb{R}^{n_{\text{out}}}$, define the conditional covariance and variance as $\text{Cov}_i(u, v) \equiv u^T \Sigma_i v$, and $\text{Var}_i(u) \equiv \text{Cov}_i(u, u)$. The finite- N coefficients in eq. (SM112) can then be written as

$$\begin{aligned} b_N &= \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \text{Cov}_i(w_\mathcal{O}^{\text{eff}}, U_1 S_{\text{eff}}^{-1} p_{\sigma,1}) = (w_\mathcal{O}^{\text{eff}})^T \bar{\Sigma} U_1 S_{\text{eff}}^{-1} p_{\sigma,1} = w_\mathcal{O}^T \bar{\Sigma} \left(S + \frac{1}{N} \bar{\Sigma} \right)^{-1} p_\sigma, \\ V_N &= \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} (p_{i,1}^T S_{\text{eff}}^{-1} p_{\sigma,1})^2 \text{Var}_i(w_\mathcal{O}^{\text{eff}}) = \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \left(p_i^T \left(S + \frac{1}{N} \bar{\Sigma} \right)^{-1} p_\sigma \right)^2 \text{Var}_i(w_\mathcal{O}^{\text{eff}}). \end{aligned} \quad (\text{SM116})$$

Notice in particular how b_N only depends on the averaged covariance matrix $\bar{\Sigma}$, whereas this is not exactly the case for V_N . The term $\text{Var}_i(w_\mathcal{O}^{\text{eff}})$ in V_N can be written more explicitly but does not simplify as much:

$$\text{Var}_i(w_\mathcal{O}^{\text{eff}}) = (w_\mathcal{O}^{\text{eff}})^T \Sigma_i w_\mathcal{O}^{\text{eff}} = w_\mathcal{O}^T (I - U_2 \bar{\Sigma}_{22}^{-1} \bar{\Sigma}_{21} U_1^T)^T \Sigma_i (I - U_2 \bar{\Sigma}_{22}^{-1} \bar{\Sigma}_{21} U_1^T) w_\mathcal{O}. \quad (\text{SM117})$$

Note that $S_{\text{eff}} = S_{11} + O(N^{-1})$, and

$$U_1 S_{11}^{-1} p_{\sigma,1} = n_{\text{tr}} \langle \tilde{\mu}, \tilde{\sigma} \rangle, \quad p_{i,1}^T S_{11}^{-1} p_{\sigma,1} = n_{\text{tr}} \langle \tilde{\rho}_i, \tilde{\sigma} \rangle, \quad (\text{SM118})$$

where $\tilde{\rho}_i = \mathcal{F}_\rho^{-1}(\rho_i)$ and $\tilde{\sigma} = \mathcal{F}_\rho^{-1}(\sigma)$. Substitution in eq. (SM116) therefore yields $b_N = b + O(N^{-1})$ and $V_N = V + O(N^{-1})$, with

$$b = \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \text{Cov}_i(w_\mathcal{O}^{\text{eff}}, n_{\text{tr}} \langle \tilde{\mu}, \tilde{\sigma} \rangle), \quad V = \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} (n_{\text{tr}} \langle \tilde{\rho}_i, \tilde{\sigma} \rangle)^2 \text{Var}_i(w_\mathcal{O}^{\text{eff}}). \quad (\text{SM119})$$

By eq. (SM43), the deterministic bias amplitude is $B = -b_N/N$, and the full mean error is $B + \mathbb{E}[\hat{R}]$, with the moment bound eq. (SM110) making $\mathbb{E}[\hat{R}]$ subleading.

SM4.2. Resource scaling

Section SM4.1 identifies the finite- N frame coefficients and their $N \rightarrow \infty$ limits. We now insert them into the rigorous MSE expansion to expose which error terms are controlled by shots per state and which are controlled by the total number of measured training copies. This is the resource conclusion used in the main text. First, substituting the finite- N coefficients from eq. (SM116) into eq. (SM112) expresses the two leading terms as b_N^2/N^2 and $V_N/(Nn_{\text{tr}})$. Then using $b_N = b + O(N^{-1})$ and $V_N = V + O(N^{-1})$, with the limiting coefficients in eq. (SM119), gives

$$\mathbb{E}_{\hat{\Sigma}}[\Delta\hat{\sigma}^2] = \frac{b_N^2}{N^2} + \frac{V_N}{Nn_{\text{tr}}} + O(\text{Err}_{N,n_{\text{tr}}}) = \frac{b^2}{N^2} + \frac{V}{Nn_{\text{tr}}} + O\left(\text{Err}_{N,n_{\text{tr}}} + \frac{1}{N^3}\right), \quad (\text{SM120})$$

with $\text{Err}_{N,n_{\text{tr}}}$ defined in eq. (SM111). The additional N^{-3} term comes from replacing b_N^2 by b^2 ; the N -dependence of V_N is already absorbed by $\text{Err}_{N,n_{\text{tr}}}$. Note that the N^{-3} correction is retained because $\text{Err}_{N,n_{\text{tr}}}$ is $O(n_{\text{tr}}^{-1})$ for large n_{tr} , and N^{-3} need not be smaller than terms like $N^{-2}n_{\text{tr}}^{-1}$ when $n_{\text{tr}} > N$. At fixed d and n_{out} , the leading variance is controlled by the total number $N_{\text{tot}} = Nn_{\text{tr}}$ of measured training copies, whereas the squared bias is controlled by the number of shots per state. If $|b|$ and V remain bounded above and away from zero, increasing n_{tr} at fixed N suppresses the variance but leaves the bias floor b^2/N^2 , and the crossover occurs at $n_{\text{tr}} \asymp N$. The bias originates from the deterministic $N^{-1}\bar{\Sigma}_{\text{eff}}$ shift induced by the rank-increasing shot noise; the variance originates from the centered first-order fluctuation.

SM4.3. Tight-design average of the leading coefficients

The scaling laws of section SM4.2 are conditional on a fixed target and test state. This final subsection makes their typical size explicit in a symmetric reference model by averaging the two leading coefficients over targets and test states. We use a model for which the frame identities are exact. The training states form a fixed complex projective two-design, and the POVM is a fixed balanced rank-one tight measurement frame,

$$\mu_a = \frac{d}{n_{\text{out}}} \mathbb{P}(\psi_a), \quad a = 1, \dots, n_{\text{out}}, \quad (\text{SM121})$$

where $(\psi_a)_a$ is a complex projective two-design. The test state σ and target observable $\mathcal{O}$ are independent Haar-random rank-one projectors. Thus the POVM is not sampled from the random Haar-isometry model considered later in ?? . An i.i.d. Haar training sample approaches the same population formulas when its empirical frame is well conditioned, but it is not an exact two-design at finite n_{tr} . These coefficient averages are finite-dimensional identities. The resource expansion applies directly for fixed d ; if d grows, its conditioning constants and the dimension dependence of the remainder must also be controlled uniformly.

For the measurement frame and training frame, respectively,

$$\tilde{\mu}_a = (d+1)\mathbb{P}(\psi_a) - \mathbf{I}, \quad \tilde{\rho}_i = \frac{d}{n_{\text{tr}}} [(d+1)\rho_i - \mathbf{I}]. \quad (\text{SM122})$$

The balanced POVM also gives $\bar{\Sigma}_{12} = 0$, so $w_{\mathcal{O}}^{\text{eff}} = w_{\mathcal{O}}$. Furthermore, the symmetric conditions give

$$\begin{aligned} n_{\text{tr}} \langle \tilde{\rho}_i, \sigma \rangle &= d [(d+1) \text{tr}(\rho_i \sigma) - 1], \quad \mathbb{E}_{\sigma} \left[(n_{\text{tr}} \langle \tilde{\rho}_i, \sigma \rangle)^2 \right] = d^2, \\ \mathbb{E}_{\mathcal{O}} \text{Var}_i(w_{\mathcal{O}}) &= 1 - \mathbb{E}_{\mathcal{O}} [\text{tr}(\mathcal{O} \rho_i)^2] = \frac{(d-1)(d+2)}{d(d+1)}, \end{aligned} \quad (\text{SM123})$$

Independence of $\mathcal{O}$ and σ , together with eq. (SM119), therefore gives

$$\mathbb{E}_{\sigma, \mathcal{O}}[V] = \frac{d(d-1)(d+2)}{d+1}. \quad (\text{SM124})$$

For the bias, let $w_A = \langle \tilde{\mu}, A \rangle \in \mathbb{R}^{n_{\text{out}}}$ be the dual-frame outcome weights of a Hermitian operator A . The measurement and training two-design identities give

$$\frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \text{Cov}_i(w_{\mathcal{O}}, w_A) = \frac{d+2}{d+1} \left[\text{tr}(\mathcal{O} A) - \frac{\text{tr}(\mathcal{O}) \text{tr}(A)}{d} \right]. \quad (\text{SM125})$$

Taking $A = n_{\text{tr}}\tilde{\sigma} = d[(d+1)\sigma - \mathbf{I}]$ in eq. (SM119) yields

$$b = (d+2) [d \operatorname{tr}(\mathcal{O}\sigma) - \operatorname{tr}(\mathcal{O})] = (d+2) [d \operatorname{tr}(\mathcal{O}\sigma) - 1], \quad (\text{SM126})$$

where the second equality uses that $\mathcal{O}$ is a rank-one projector. For independent Haar-random pure states,

$$\mathbb{E}[\operatorname{tr}(\mathcal{O}\sigma)] = \frac{1}{d}, \quad \mathbb{E}[\operatorname{tr}(\mathcal{O}\sigma)^2] = \frac{2}{d(d+1)}. \quad (\text{SM127})$$

Consequently,

$$\mathbb{E}_{\sigma, \mathcal{O}}[b^2] = \frac{(d-1)(d+2)^2}{d+1}, \quad \mathbb{E}_{\sigma, \mathcal{O}}[V] = \frac{d(d-1)(d+2)}{d+1}. \quad (\text{SM128})$$

SM5. STATISTICAL PROPERTIES OF THE SHOT-NOISE RANDOM MATRIX

This section evaluates the random coefficients in the perturbative prediction error under the homogeneous Gaussian reference model used in section V. We first derive the conditional laws of the generalized Schur complement and of the term containing $\hat{\mathbf{Q}}$. We then use these laws to compute the mean and variance of the two coefficients in eq. (30). Conditional on the training states, we assume

$$\hat{\Xi} \sim \mathcal{MN}(0, \bar{\Sigma}, \mathbf{I}_{n_{\text{tr}}}), \quad \bar{\Sigma} \equiv \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \Sigma_i. \quad (\text{SM129})$$

Thus the columns are independent Gaussian vectors with a common covariance. Homogeneity of the column covariance defines the reference model. The true multinomial columns have state-dependent covariances and are analyzed in section SM3.

We keep d and n_{out} fixed and let $n_{\text{tr}} \rightarrow \infty$, and assume the deterministic conditioning bounds

$$\lambda_{\min}(\bar{\Sigma}_{22}) \geq \kappa_{22} > 0, \quad \lambda_{\min}(\mathbf{S}_{11}) \geq \kappa_{11} > 0, \quad \mathbf{S}_{11} = \frac{\mathbf{P}_{11}\mathbf{P}_{11}^T}{n_{\text{tr}}}, \quad (\text{SM130})$$

uniformly in n_{tr} . The first bound concerns the covariance of the noise in the kernel block. The second concerns the noiseless feature matrix and is the frame-conditioning assumption used below. Since the eigenvalues of $\mathbf{P}_{11}\mathbf{P}_{11}^T$ are the nonzero squared singular values of $\mathbf{P}$, the frame-operator identity in section SM1.5 gives the sufficient bound

$$\lambda_{\min}(\mathbf{S}_{11}) \geq \lambda_{\min}(\mathcal{F}_{\mu}) \lambda_{\min}\left(\frac{\mathcal{F}_{\rho}}{n_{\text{tr}}}\right). \quad (\text{SM131})$$

Thus a fixed informationally complete POVM together with a family of training frames satisfying $\lambda_{\min}(\mathcal{F}_{\rho}/n_{\text{tr}}) > 0$ uniformly in n_{tr} implies the second condition in eq. (SM130). Completeness of every finite training set alone would not be sufficient, since its lower frame bound could still tend to zero. For the balanced tight designs of eqs. (SM15) and (SM16), one obtains explicitly

$$\lambda_{\min}(\mathbf{S}_{11}) = \frac{1}{n_{\text{out}}(d+1)^2}, \quad (\text{SM132})$$

which is independent of n_{tr} . The bound also holds with probability tending to one for independent Haar-random training states. Indeed, the rank-one superoperators $\mathbb{P}(\rho_i)$ are uniformly bounded and the operator space has fixed dimension d^2 . The entrywise law of large numbers and norm equivalence in fixed dimension therefore give, in operator norm,

$$\frac{\mathcal{F}_{\rho}}{n_{\text{tr}}} = \frac{\text{Id} + \mathbb{P}(\mathbf{I})}{d(d+1)} + o_{\mathbb{P}}(1). \quad (\text{SM133})$$

The smallest eigenvalue of the limiting operator is $1/[d(d+1)]$. Consequently, for example, $\lambda_{\min}(\mathcal{F}_{\rho}/n_{\text{tr}}) \geq 1/[2d(d+1)]$ with probability tending to one. Together with eq. (SM131), this proves the required uniform lower-frame bound for Haar-random training designs with probability tending to one. The block dimensions and the identity $r = d^2$ are those of section SM1.3.

SM5.1. Conditional Gaussian laws

The nonlinear blocks in eq. (30) can be evaluated from their exact conditional distributions. We recall the two quantities that will appear in those laws:

$$\bar{\Sigma}_{\text{eff}} = \bar{\Sigma}_{11} - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \bar{\Sigma}_{21}, \quad \hat{Y} = I_r + \hat{\Sigma}_{21}^T (\hat{\Sigma}_{22} \hat{\Sigma}_{22}^T)^{-1} \hat{\Sigma}_{21}.$$

The first is the covariance Schur complement used above, and the second is the matrix introduced in eq. (SM28). Moreover, $\hat{\Sigma}_{22} = \bar{\Sigma}_{22}^{1/2} \hat{G}$, where $\hat{G} \in \mathbb{R}^{(n_{\text{out}}-r) \times (n_{\text{tr}}-r)}$ has independent standard normal entries. Hence $\hat{\Sigma}_{22}$ has full row rank almost surely when $n_{\text{tr}} \geq n_{\text{out}}$. It follows that $\hat{\Sigma}_{22}^+ \hat{\Sigma}_{22}$ is the orthogonal projector onto the row space of $\hat{\Sigma}_{22}$, whereas

$$\hat{Q} = I_{n_{\text{tr}}-r} - \hat{\Sigma}_{22}^+ \hat{\Sigma}_{22} \quad (\text{SM134})$$

is the orthogonal projector onto $\ker(\hat{\Sigma}_{22})$. Since $\hat{\Sigma}_{22}$ has $n_{\text{out}} - r$ rows and $n_{\text{tr}} - r$ columns, the rank-nullity theorem gives $\text{rank}(\hat{Q}) = n_{\text{tr}} - n_{\text{out}}$. Conditioning on the lower blocks $(\hat{\Sigma}_{21}, \hat{\Sigma}_{22})$ fixes both $\hat{Q}$ and $\hat{Y}$, while leaving the Gaussian fluctuations in the upper blocks to be averaged.

Lemma 7 (Conditional laws of the Schur and Q terms). *Assume eq. (SM129) and $n_{\text{tr}} \geq n_{\text{out}}$. Conditional on $(\hat{\Sigma}_{21}, \hat{\Sigma}_{22})$,*

$$\hat{\Sigma} / \hat{\Sigma}_{22} \sim \mathcal{MN}(0, \bar{\Sigma}_{\text{eff}}, \hat{Y}), \quad (\text{SM135})$$

whereas $\hat{\Sigma}_{12} \hat{Q} \hat{\Sigma}_{12}^T$ has a Wishart distribution with scale $\bar{\Sigma}_{\text{eff}}$ and $n_{\text{tr}} - n_{\text{out}}$ degrees of freedom. The two random matrices in this statement are conditionally independent. Consequently,

$$\mathbb{E} \left[\hat{\Sigma}_{12} \hat{Q} \hat{\Sigma}_{12}^T \mid \hat{\Sigma}_{21}, \hat{\Sigma}_{22} \right] = (n_{\text{tr}} - n_{\text{out}}) \bar{\Sigma}_{\text{eff}}, \quad (\text{SM136})$$

$$\mathbb{E}[\hat{Y}] = \frac{n_{\text{tr}} - r - 1}{n_{\text{tr}} - n_{\text{out}} - 1} I_r, \quad n_{\text{tr}} > n_{\text{out}} + 1. \quad (\text{SM137})$$

Proof. The conditional-normal formula, applied separately to the first r and last $n_{\text{tr}} - r$ columns, gives [60]:

$$\begin{aligned} \hat{\Sigma}_{11} \mid \hat{\Sigma}_{21} &\sim \mathcal{MN}(\bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \hat{\Sigma}_{21}, \bar{\Sigma}_{\text{eff}}, I_r), \\ \hat{\Sigma}_{12} \mid \hat{\Sigma}_{22} &\sim \mathcal{MN}(\bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \hat{\Sigma}_{22}, \bar{\Sigma}_{\text{eff}}, I_{n_{\text{tr}}-r}). \end{aligned}$$

For a jointly Gaussian pair, the residual after subtracting the conditional mean is independent of the variable on which one conditions. Hence

$$\hat{\Sigma}_{12} - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \hat{\Sigma}_{22}$$

is independent of $\hat{\Sigma}_{22}$ and has conditional mean zero. This can be seen directly, column by column, from

$$\text{Cov}(\hat{\Sigma}_{12} - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \hat{\Sigma}_{22}, \hat{\Sigma}_{22}) = \bar{\Sigma}_{12} - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \bar{\Sigma}_{22} = 0 :$$

zero cross-covariance implies independence because the pair is jointly Gaussian. The analogous statement holds for the residual formed from $\hat{\Sigma}_{11}$ and $\hat{\Sigma}_{21}$. Moreover, the two residuals are independent of each other because the first r columns of $\hat{\Sigma}$ are independent of the last $n_{\text{tr}} - r$ columns. Consequently, even after conditioning on both lower blocks, the second term below has conditional mean

$$\mathbb{E} \left[(\hat{\Sigma}_{12} - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \hat{\Sigma}_{22}) \hat{\Sigma}_{22}^+ \hat{\Sigma}_{21} \mid \hat{\Sigma}_{21}, \hat{\Sigma}_{22} \right] = \mathbb{E} \left[\hat{\Sigma}_{12} - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \hat{\Sigma}_{22} \mid \hat{\Sigma}_{22} \right] \hat{\Sigma}_{22}^+ \hat{\Sigma}_{21} = 0.$$

Here $\hat{\Sigma}_{22}^+ \hat{\Sigma}_{21}$ is fixed under the conditioning.

The conditional mean of the generalized Schur complement can also be checked directly:

$$\mathbb{E}[\hat{\Sigma} / \hat{\Sigma}_{22} \mid \hat{\Sigma}_{21}, \hat{\Sigma}_{22}] = \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \hat{\Sigma}_{21} - \bar{\Sigma}_{12} \bar{\Sigma}_{22}^{-1} \hat{\Sigma}_{22} \hat{\Sigma}_{22}^+ \hat{\Sigma}_{21} = 0,$$

where the last equality uses $\hat{\mathbf{\Sigma}}_{22}\hat{\mathbf{\Sigma}}_{22}^+ = \mathbf{I}_{n_{\text{out}}-r}$. Equivalently, subtracting the two conditional means gives the centered representation

$$\hat{\mathbf{\Sigma}}/\hat{\mathbf{\Sigma}}_{22} = \left(\hat{\mathbf{\Sigma}}_{11} - \bar{\mathbf{\Sigma}}_{12}\bar{\mathbf{\Sigma}}_{22}^{-1}\hat{\mathbf{\Sigma}}_{21} \right) - \left(\hat{\mathbf{\Sigma}}_{12} - \bar{\mathbf{\Sigma}}_{12}\bar{\mathbf{\Sigma}}_{22}^{-1}\hat{\mathbf{\Sigma}}_{22} \right) \hat{\mathbf{\Sigma}}_{22}^+ \hat{\mathbf{\Sigma}}_{21}.$$

The first centered matrix in the last display has column covariance $\mathbf{I}_r$; the second is independent of it and, after right multiplication by $\hat{\mathbf{\Sigma}}_{22}^+ \hat{\mathbf{\Sigma}}_{21}$, has column covariance $(\hat{\mathbf{\Sigma}}_{22}^+ \hat{\mathbf{\Sigma}}_{21})^T (\hat{\mathbf{\Sigma}}_{22}^+ \hat{\mathbf{\Sigma}}_{21})$. Therefore the column covariance of their difference is

$$\mathbf{I}_r + (\hat{\mathbf{\Sigma}}_{22}^+ \hat{\mathbf{\Sigma}}_{21})^T (\hat{\mathbf{\Sigma}}_{22}^+ \hat{\mathbf{\Sigma}}_{21}) = \mathbf{I}_r + \hat{\mathbf{\Sigma}}_{21}^T (\hat{\mathbf{\Sigma}}_{22} \hat{\mathbf{\Sigma}}_{22}^T)^{-1} \hat{\mathbf{\Sigma}}_{21} = \hat{\mathbf{Y}}.$$

The row covariance remains $\bar{\mathbf{\Sigma}}_{\text{eff}}$, which proves [eq. \(SM135\)](#).

The same centered representation also exposes the $\hat{\mathbf{Q}}$ term. Since $\hat{\mathbf{\Sigma}}_{22}\hat{\mathbf{Q}} = 0$ and $\hat{\mathbf{Q}}^2 = \hat{\mathbf{Q}}$, we have

$$\hat{\mathbf{\Sigma}}_{12}\hat{\mathbf{Q}}\hat{\mathbf{\Sigma}}_{12}^T = \left(\hat{\mathbf{\Sigma}}_{12} - \bar{\mathbf{\Sigma}}_{12}\bar{\mathbf{\Sigma}}_{22}^{-1}\hat{\mathbf{\Sigma}}_{22} \right) \hat{\mathbf{Q}} \left(\hat{\mathbf{\Sigma}}_{12} - \bar{\mathbf{\Sigma}}_{12}\bar{\mathbf{\Sigma}}_{22}^{-1}\hat{\mathbf{\Sigma}}_{22} \right)^T.$$

In an orthonormal basis that diagonalizes $\hat{\mathbf{Q}}$, the nonzero part of the centered matrix in this display consists of $\text{rank}(\hat{\mathbf{Q}}) = n_{\text{tr}} - n_{\text{out}}$ independent Gaussian columns, each with covariance $\bar{\mathbf{\Sigma}}_{\text{eff}}$. Their sum of outer products is therefore Wishart with the scale and degrees of freedom stated in the lemma.

It remains to justify the claimed conditional independence. In the Schur complement, the contribution from the centered $\hat{\mathbf{\Sigma}}_{12}$ block is right-multiplied by $\hat{\mathbf{\Sigma}}_{22}^+ \hat{\mathbf{\Sigma}}_{21}$ and hence depends only on its projection onto the row space of $\hat{\mathbf{\Sigma}}_{22}$. By contrast, the Wishart matrix in the preceding display depends only on its projection onto $\ker(\hat{\mathbf{\Sigma}}_{22})$. These two projections are orthogonal, since

$$(\hat{\mathbf{\Sigma}}_{22}^+ \hat{\mathbf{\Sigma}}_{22}) \hat{\mathbf{Q}} = 0.$$

The centered $\hat{\mathbf{\Sigma}}_{12}$ block is matrix normal with identity column covariance, so its two orthogonal projections have zero cross-covariance and, being jointly Gaussian, are independent. The centered $\hat{\mathbf{\Sigma}}_{11}$ block appearing in the Schur complement is independent of both because it consists of different columns. This proves the conditional independence of the two matrices in the lemma. Finally, the mean of a Wishart matrix is its number of degrees of freedom times its scale, which gives [eq. \(SM136\)](#).

For [eq. \(SM137\)](#), the two lower blocks are independent and $\hat{\mathbf{\Sigma}}_{22}\hat{\mathbf{\Sigma}}_{22}^T$ is Wishart with scale $\bar{\mathbf{\Sigma}}_{22}$ and $n_{\text{tr}} - r$ degrees of freedom. Conditional on $\hat{\mathbf{\Sigma}}_{22}$, the r columns of $\hat{\mathbf{\Sigma}}_{21}$ are independent with covariance $\bar{\mathbf{\Sigma}}_{22}$. Consequently,

$$\mathbb{E}[\hat{\mathbf{Y}} - \mathbf{I}_r \mid \hat{\mathbf{\Sigma}}_{22}] = \text{Tr} \left[(\hat{\mathbf{\Sigma}}_{22} \hat{\mathbf{\Sigma}}_{22}^T)^{-1} \bar{\mathbf{\Sigma}}_{22} \right] \mathbf{I}_r.$$

The inverse-Wishart identity

$$\mathbb{E}[(\hat{\mathbf{\Sigma}}_{22} \hat{\mathbf{\Sigma}}_{22}^T)^{-1}] = \frac{\bar{\mathbf{\Sigma}}_{22}^{-1}}{n_{\text{tr}} - n_{\text{out}} - 1} \Rightarrow \mathbb{E}[\hat{\mathbf{Y}} - \mathbf{I}_r] = \frac{n_{\text{out}} - r}{n_{\text{tr}} - n_{\text{out}} - 1} \mathbf{I}_r.$$

Here the denominator is $(n_{\text{tr}} - r) - (n_{\text{out}} - r) - 1 = n_{\text{tr}} - n_{\text{out}} - 1$, and the numerator comes from $\text{Tr}(\bar{\mathbf{\Sigma}}_{22}^{-1} \bar{\mathbf{\Sigma}}_{22}) = n_{\text{out}} - r$. Adding $\mathbf{I}_r$ and combining the two terms gives $(n_{\text{tr}} - r - 1)/(n_{\text{tr}} - n_{\text{out}} - 1)$, as in [eq. \(SM137\)](#); see also [\[54, Chapter 3\]](#). $\square$

SM5.2. Leading resource scalings

We now apply the preceding results to the two coefficients in [eq. \(29\)](#). Since $\mathbf{S}_{11} = \mathbf{P}_{11} \mathbf{P}_{11}^T / n_{\text{tr}}$, the second condition in [eq. \(SM130\)](#) implies $\sigma_{\min}(\mathbf{P}_{11})^2 \geq \kappa_{11} n_{\text{tr}}$. Therefore

$$\|\mathbf{P}_{11}^{-1}\|_{\text{op}} = \frac{1}{\sigma_{\min}(\mathbf{P}_{11})} \leq \frac{1}{\sqrt{\kappa_{11} n_{\text{tr}}}} = O(n_{\text{tr}}^{-1/2}), \quad \|V_1^T \langle \tilde{\rho}, \sigma \rangle\|_2 \leq \frac{\|\mathbf{p}_{\sigma,1}\|_2}{\sqrt{\kappa_{11} n_{\text{tr}}}} \leq \frac{1}{\sqrt{\kappa_{11} n_{\text{tr}}}} = O(n_{\text{tr}}^{-1/2}), \quad (\text{SM138})$$

because $V_1^T \langle \tilde{\rho}, \sigma \rangle = P_{11}^{-1} p_{\sigma,1}$ and $\|p_{\sigma,1}\|_2 \leq \|p_\sigma\|_2 \leq 1$. The last inequality uses that p_σ is a probability vector. Thus one factor of P_{11}^{-1} contributes $O(n_{\text{tr}}^{-1/2})$, as does the dual-frame vector $V_1^T \langle \tilde{\rho}, \sigma \rangle$. The homogeneous specializations of the coefficients in eq. (SM119) are therefore

$$\begin{aligned} b &\equiv n_{\text{tr}} w_{\mathcal{O}}^T U_1 \bar{\Sigma}_{\text{eff}} P_{11}^{-T} V_1^T \langle \tilde{\rho}, \sigma \rangle, \\ V &\equiv n_{\text{tr}} \|V_1^T \langle \tilde{\rho}, \sigma \rangle\|_2^2 w_{\mathcal{O}}^T U_1 \bar{\Sigma}_{\text{eff}} U_1^T w_{\mathcal{O}}. \end{aligned} \quad (\text{SM139})$$

The explicit factor n_{tr} in b is canceled by the two $O(n_{\text{tr}}^{-1/2})$ factors just identified. In V , it is canceled by the squared norm of the dual-frame vector. Since d and n_{out} are fixed, the remaining contractions are bounded under the stated assumptions, and hence $b, V = O(1)$.

Lemma 8 (Moments of the two perturbative coefficients). *For $n_{\text{tr}} > n_{\text{out}} + 3$, the contractions appearing in eq. (29) satisfy*

$$\mathbb{E} \left[w_{\mathcal{O}}^T U_1 A_1 V_1^T \langle \tilde{\rho}, \sigma \rangle \right] = 0, \quad (\text{SM140})$$

$$\text{Var} \left[w_{\mathcal{O}}^T U_1 A_1 V_1^T \langle \tilde{\rho}, \sigma \rangle \right] = \frac{V}{n_{\text{tr}}} \frac{n_{\text{tr}} - r - 1}{n_{\text{tr}} - n_{\text{out}} - 1} = \frac{V}{n_{\text{tr}}} + O(n_{\text{tr}}^{-2}), \quad (\text{SM141})$$

$$\mathbb{E} \left[w_{\mathcal{O}}^T U_1 A_2 V_1^T \langle \tilde{\rho}, \sigma \rangle \right] = - \left(1 - \frac{r+1}{n_{\text{tr}}} \right) b = -b + O(n_{\text{tr}}^{-1}), \quad (\text{SM142})$$

$$\text{Var} \left[w_{\mathcal{O}}^T U_1 A_2 V_1^T \langle \tilde{\rho}, \sigma \rangle \right] = O(n_{\text{tr}}^{-1}). \quad (\text{SM143})$$

The covariance between the two displayed contractions is zero. The condition $n_{\text{tr}} > n_{\text{out}} + 3$ ensures that the second inverse-Wishart moment entering eq. (SM143) is finite; the mean formulas only require $n_{\text{tr}} > n_{\text{out}} + 1$. In particular, the two contractions are respectively $O_{\mathbb{P}}(n_{\text{tr}}^{-1/2})$ and $-b + O_{\mathbb{P}}(n_{\text{tr}}^{-1/2})$.

Proof. By eq. (30), A_1 is minus the generalized Schur complement. Its conditional law in eq. (SM135) is centered, which proves eq. (SM140) after averaging over the lower blocks. The matrix-normal covariance rule gives, under the same conditioning,

$$\text{Var} \left[w_{\mathcal{O}}^T U_1 A_1 V_1^T \langle \tilde{\rho}, \sigma \rangle \mid \hat{\Sigma}_{21}, \hat{\Sigma}_{22} \right] = w_{\mathcal{O}}^T U_1 \bar{\Sigma}_{\text{eff}} U_1^T w_{\mathcal{O}} (V_1^T \langle \tilde{\rho}, \sigma \rangle)^T \hat{Y} (V_1^T \langle \tilde{\rho}, \sigma \rangle).$$

Averaging this identity and inserting eq. (SM137) replaces $\hat{Y}$ by a scalar multiple of I_r . The definition of V in eq. (SM139) then gives eq. (SM141).

For the second coefficient, condition again on $(\hat{\Sigma}_{21}, \hat{\Sigma}_{22})$. If a matrix is matrix normal with row covariance $\bar{\Sigma}_{\text{eff}}$ and column covariance $\hat{Y}$, its entries obey

$$\mathbb{E} \left[(\hat{\Sigma}/\hat{\Sigma}_{22})_{ia} (\hat{\Sigma}/\hat{\Sigma}_{22})_{bj} \mid \hat{\Sigma}_{21}, \hat{\Sigma}_{22} \right] = (\bar{\Sigma}_{\text{eff}})_{ib} (\hat{Y})_{aj}.$$

Contracting the two inner indices therefore gives

$$\mathbb{E} \left[(\hat{\Sigma}/\hat{\Sigma}_{22}) P_{11}^{-1} (\hat{\Sigma}/\hat{\Sigma}_{22}) \mid \hat{\Sigma}_{21}, \hat{\Sigma}_{22} \right] = \bar{\Sigma}_{\text{eff}} P_{11}^{-T} \hat{Y}.$$

On the other hand, the conditional Wishart law gives

$$\mathbb{E} \left[\hat{\Sigma}_{12} \hat{Q} \hat{\Sigma}_{12}^T P_{11}^{-T} \hat{Y} \mid \hat{\Sigma}_{21}, \hat{\Sigma}_{22} \right] = (n_{\text{tr}} - n_{\text{out}}) \bar{\Sigma}_{\text{eff}} P_{11}^{-T} \hat{Y}.$$

The matrix $\hat{Y}$ is fixed under the conditioning, so subtracting the two conditional means according to eq. (30) gives

$$\mathbb{E} [A_2 \mid \hat{\Sigma}_{21}, \hat{\Sigma}_{22}] = [1 - (n_{\text{tr}} - n_{\text{out}})] \bar{\Sigma}_{\text{eff}} P_{11}^{-T} \hat{Y} = -(n_{\text{tr}} - n_{\text{out}} - 1) \bar{\Sigma}_{\text{eff}} P_{11}^{-T} \hat{Y}.$$

Now eq. (SM137) cancels the denominator $n_{\text{tr}} - n_{\text{out}} - 1$ and yields

$$\mathbb{E} [A_2] = -(n_{\text{tr}} - r - 1) \bar{\Sigma}_{\text{eff}} P_{11}^{-T},$$

which is eq. (SM142) after contraction and use of eq. (SM139): dividing $n_{\text{tr}} - r - 1$ by the factor n_{tr} contained in b produces $1 - (r+1)/n_{\text{tr}}$. This calculation also shows explicitly how the dependence on n_{out} cancels between the Wishart degrees of freedom and the inverse-Wishart mean of $\hat{Y}$.

We next spell out the variance estimate. The second inverse-Wishart moment, which is finite for $n_{\text{tr}} > n_{\text{out}} + 3$, implies

$$\mathbb{E}[\|\hat{\mathbf{Y}}\|_{\text{op}}^2] = O(1), \quad \mathbb{E}[\|\hat{\mathbf{Y}} - \mathbf{I}_r\|_{\text{op}}^2] = O(n_{\text{tr}}^{-2}).$$

Together with eq. (SM138), the vector multiplying the Wishart matrix on its right satisfies the second-moment bound

$$\mathbb{E}\left[\left\|\mathbf{P}_{11}^{-T}\hat{\mathbf{Y}}\mathbf{V}_1^T\langle\tilde{\rho},\sigma\rangle\right\|_2^2\right] = O(n_{\text{tr}}^{-2}).$$

In fixed dimension, a Wishart matrix with $n_{\text{tr}} - n_{\text{out}}$ degrees of freedom has variance proportional to $n_{\text{tr}} - n_{\text{out}}$ in every fixed scalar contraction. After the preceding right contraction, its centered part therefore contributes $O(n_{\text{tr}} \cdot n_{\text{tr}}^{-2}) = O(n_{\text{tr}}^{-1})$ to the variance. The fluctuation of its conditional mean caused by $\hat{\mathbf{Y}} - \mathbf{I}_r$ contributes only $O(n_{\text{tr}}^{-2})$. For the first, Schur-complement term of A_2 , the conditional fourth Gaussian moment is bounded because $\mathbb{E}[\|\hat{\mathbf{Y}}\|_{\text{op}}^2] = O(1)$. Its two deterministic contractions in eq. (SM138) supply a total factor $O(n_{\text{tr}}^{-1})$ in amplitude, so its second moment is $O(n_{\text{tr}}^{-2})$. The Wishart fluctuation is therefore the leading random contribution, and eq. (SM143) follows.

Finally, condition on the lower blocks once more. The A_1 contraction is linear in the centered Gaussian Schur complement, whereas the first term of the A_2 contraction is quadratic in that same centered matrix. Their product has odd total degree and hence zero conditional expectation. The second term of A_2 depends on the orthogonal Gaussian projection that produces the Wishart matrix; by lemma 7 it is independent of the Schur complement. Since the conditional mean of the A_1 contraction is zero, this product also has zero conditional expectation. Averaging over the lower blocks proves that the two contractions have zero covariance. $\square$

The two terms retained in eq. (29) therefore obey

$$\begin{aligned} \mathbb{E}\left[\left\{w_{\mathcal{O}}^T U_1 \left(\frac{A_1}{\sqrt{N}} + \frac{A_2}{N}\right) V_1^T \langle\tilde{\rho}, \sigma\rangle\right\}^2\right] &= \left(1 - \frac{r+1}{n_{\text{tr}}}\right)^2 \frac{b^2}{N^2} + \frac{V}{N n_{\text{tr}}} \frac{n_{\text{tr}} - r - 1}{n_{\text{tr}} - n_{\text{out}} - 1} + O\left(\frac{1}{N^2 n_{\text{tr}}}\right) \\ &= \frac{b^2}{N^2} + \frac{V}{N n_{\text{tr}}} + O\left(\frac{1}{N^2 n_{\text{tr}}} + \frac{1}{N n_{\text{tr}}^2}\right). \end{aligned} \quad (\text{SM144})$$

In the first equality, the term proportional to b^2 is the squared mean of the $N^{-1}A_2$ contribution, while the term proportional to V is the variance of the centered $N^{-1/2}A_1$ contribution. The $O(N^{-2}n_{\text{tr}}^{-1})$ term contains the variance of the A_2 contraction, and no mixed term remains because the two contractions have zero covariance. The second equality uses fixed d and n_{out} and bounded b, V : both rational prefactors differ from one by $O(n_{\text{tr}}^{-1})$. This moment expansion concerns the first two coefficients of the fixed-realization expansion in $N^{-1/2}$. Standard inverse-Wishart moment bounds ensure integrability of every coefficient at any prescribed fixed order once $n_{\text{tr}} - n_{\text{out}}$ is sufficiently large. Uniform control of the expansion remainder, as well as the result for the true multinomial model, is established in section SM3.

SM5.3. Comparison with the multinomial coefficients

We can now compare the coefficients extracted from the homogeneous Gaussian model in eq. (SM139), with those obtained directly for the multinomial model. This comparison concerns the leading large- N coefficients b and V , rather than the finite- N coefficients b_N, V_N or the subleading terms in the two expansions. For the multinomial model, eq. (SM119) gives

$$b = \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \text{Cov}_i(w_{\mathcal{O}}^{\text{eff}}, n_{\text{tr}} \langle \tilde{\mu}, \tilde{\sigma} \rangle), \quad V = \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} (n_{\text{tr}} \langle \tilde{\rho}_i, \sigma \rangle)^2 \text{Var}_i(w_{\mathcal{O}}^{\text{eff}}). \quad (\text{SM145})$$

Here every covariance is computed with the state-dependent multinomial covariance Σ_i , while $w_{\mathcal{O}}^{\text{eff}}$ is defined from their average $\bar{\Sigma}$ as in eq. (SM46).

The bias comparison does not in fact require homogeneous column covariances. Since the second argument of the covariance in eq. (SM145) is independent of i , the sum over the training states gives

$$b = (w_{\mathcal{O}}^{\text{eff}})^T \bar{\Sigma} U_1 S_{11}^{-1} p_{\sigma,1} = n_{\text{tr}} w_{\mathcal{O}}^T U_1 \bar{\Sigma}_{\text{eff}} P_{11}^{-T} V_1^T \langle \tilde{\rho}, \sigma \rangle. \quad (\text{SM146})$$

The second equality follows from two elementary block identities:

1. $(\mathbf{w}_\mathcal{O}^{\text{eff}})^T \bar{\Sigma} \mathbf{U}_1 = \mathbf{w}_\mathcal{O}^T \mathbf{U}_1 \bar{\Sigma}_{\text{eff}}$ holds because in the $(\mathbf{U}_1, \mathbf{U}_2)$ basis the two blocks of $\mathbf{w}_\mathcal{O}^{\text{eff}}$ are $\mathbf{U}_1^T \mathbf{w}_\mathcal{O}$ and $-\bar{\Sigma}_{22}^{-1} \bar{\Sigma}_{21} \mathbf{U}_1^T \mathbf{w}_\mathcal{O}$; multiplying them by the two block rows of $\bar{\Sigma} \mathbf{U}_1$ produces precisely its Schur complement.
2. $\mathbf{S}_{11}^{-1} \mathbf{p}_{\sigma,1} = n_{\text{tr}} \mathbf{P}_{11}^{-T} \mathbf{V}_1^T \langle \tilde{\rho}, \sigma \rangle$ holds because $\mathbf{S}_{11} = \mathbf{P}_{11} \mathbf{P}_{11}^T / n_{\text{tr}}$ and $\mathbf{P}_{11}^{-1} \mathbf{p}_{\sigma,1} = \mathbf{V}_1^T \langle \tilde{\rho}, \sigma \rangle$.

The right-hand side of eq. (SM146) is exactly the Gaussian coefficient in eq. (SM139). Thus replacing the state-dependent multinomial covariances by their average leaves the leading bias coefficient unchanged.

For the variance, suppose now that the multinomial column covariances are homogeneous, that is, $\Sigma_i = \bar{\Sigma}$, $i = 1, \dots, n_{\text{tr}}$. Under this approximation,

$$\text{Var}_i(\mathbf{w}_\mathcal{O}^{\text{eff}}) = (\mathbf{w}_\mathcal{O}^{\text{eff}})^T \bar{\Sigma} \mathbf{w}_\mathcal{O}^{\text{eff}} = \mathbf{w}_\mathcal{O}^T \mathbf{U}_1 \bar{\Sigma}_{\text{eff}} \mathbf{U}_1^T \mathbf{w}_\mathcal{O}, \quad (\text{SM147})$$

independently of i . Moreover, $\langle \tilde{\rho}, \sigma \rangle$ belongs to the row space of $\mathbf{P}$, and hence

$$\frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} (n_{\text{tr}} \langle \tilde{\rho}_i, \sigma \rangle)^2 = n_{\text{tr}} \|\mathbf{V}_1^T \langle \tilde{\rho}, \sigma \rangle\|_2^2. \quad (\text{SM148})$$

Substitution of eqs. (SM147) and (SM148) into the second line of eq. (SM145) gives

$$V = n_{\text{tr}} \|\mathbf{V}_1^T \langle \tilde{\rho}, \sigma \rangle\|_2^2 \mathbf{w}_\mathcal{O}^T \mathbf{U}_1 \bar{\Sigma}_{\text{eff}} \mathbf{U}_1^T \mathbf{w}_\mathcal{O}, \quad (\text{SM149})$$

which is exactly the Gaussian expression in eq. (SM139).

Consequently, the homogeneous Gaussian model reproduces both leading multinomial coefficients when the multinomial covariances are uniform.

SM5.4. Target-noise corrections to the test and training variances

We now compute the two target-noise corrections introduced in eqs. (54) and (55). The calculation uses the homogeneous Gaussian model eq. (SM129) and the block coordinates of section SM1.3. We write

$$\Sigma_{\sigma,22} = \mathbf{U}_2^T \Sigma_\sigma \mathbf{U}_2, \quad \mathbf{p}_{\sigma,1} = \mathbf{U}_1^T \mathbf{p}_\sigma. \quad (\text{SM150})$$

Lemma 9 (Gaussian target-noise corrections). *Assume eqs. (SM129) and (SM130), $n_{\text{tr}} > n_{\text{out}} + 3$, and the independent Gaussian target-noise model in eq. (51). For every fixed physical test vector $\mathbf{p}_\sigma \in \text{im}(\mathbf{P})$, the large- N corrections are*

$$\Delta \text{Var}_{\text{test}}^{(\eta)}(N, M, \sigma) = \frac{N\gamma^2}{M} \frac{\text{Tr}(\bar{\Sigma}_{22}^{-1} \Sigma_{\sigma,22})}{n_{\text{tr}} - n_{\text{out}} - 1} + O\left(\frac{\gamma^2 \sqrt{N}}{M}\right), \quad (\text{SM151})$$

$$\Delta \text{Var}_{\text{tr}}^{(\eta)}(N, \sigma) = \gamma^2 \frac{n_{\text{tr}} - r - 1}{n_{\text{tr}} - n_{\text{out}} - 1} \|\mathbf{P}_{11}^{-1} \mathbf{p}_{\sigma,1}\|_2^2 + O\left(\frac{\gamma^2}{\sqrt{N}}\right). \quad (\text{SM152})$$

The leading coefficients themselves require only $n_{\text{tr}} > n_{\text{out}} + 1$; the stronger condition stated above ensures that the moments entering the remainders are finite. The $O(\cdot)$ terms are taken at fixed $\mathbf{P}$, n_{tr} , and n_{out} . If the test ensemble satisfies $\mathbb{E}_\sigma[\Sigma_\sigma] = \bar{\Sigma}$, then

$$\mathbb{E}_\sigma[\Delta \text{Var}_{\text{test}}^{(\eta)}(N, M, \sigma)] = \frac{N\gamma^2}{M} \frac{n_{\text{out}} - r}{n_{\text{tr}} - n_{\text{out}} - 1} + O\left(\frac{\gamma^2 \sqrt{N}}{M}\right). \quad (\text{SM153})$$

Proof. We first separate the target fluctuations from the feature fluctuations. For any deterministic matrix $\mathbf{A}$ of compatible size, the conditional Gaussian covariance in eq. (51) gives

$$\mathbb{E}_{\hat{\eta}}[\hat{\eta}^T \mathbf{A} \hat{\eta} \mid \hat{\Xi}] = \gamma^2 \text{Tr}(\mathbf{A}). \quad (\text{SM154})$$

Taking $\mathbf{A} = \hat{\mathbf{P}}_N^+ \Sigma_\sigma (\hat{\mathbf{P}}_N^+)^T / M$ gives the last line of eq. (54). Taking instead $\mathbf{A} = \hat{\mathbf{P}}_N^+ \mathbf{p}_\sigma \mathbf{p}_\sigma^T (\hat{\mathbf{P}}_N^+)^T$ and applying the law of total variance gives eq. (55). The cross terms with the exact-target estimator vanish because $\mathbb{E}[\hat{\eta} \mid \hat{\Xi}] = 0$.

We now insert the large- N block expansion of the pseudoinverse already used in [section IV](#). In particular, [eqs. \(26\)](#) and [\(SM32\)](#) give

$$\hat{\mathbf{P}}_N^+ = \sqrt{N} V_2 \hat{\mathbf{\Xi}}_{22}^+ U_2^T + O(1). \quad (\text{SM155})$$

Substitution in [eq. \(54\)](#) yields

$$\Delta \text{Var}_{\text{test}}^{(\eta)}(N, M, \sigma) = \frac{N\gamma^2}{M} \mathbb{E}_{\hat{\mathbf{\Xi}}} \text{Tr}[\hat{\mathbf{\Xi}}_{22}^+ \Sigma_{\sigma,22} (\hat{\mathbf{\Xi}}_{22}^+)^T] + O\left(\frac{\gamma^2 \sqrt{N}}{M}\right). \quad (\text{SM156})$$

The remainder comes from the cross products between the singular term in [eq. \(SM155\)](#) and the $O(1)$ blocks. Its expectation is finite under the moment condition in the lemma.

To compute the leading coefficient, use the Gaussian factorization from [section SM5.1](#),

$$\hat{\mathbf{\Xi}}_{22} = \bar{\Sigma}_{22}^{-1/2} \hat{\mathbf{G}}, \quad \hat{\mathbf{G}} \in \mathbb{R}^{(n_{\text{out}}-r) \times (n_{\text{tr}}-r)}, \quad (\text{SM157})$$

where the entries of $\hat{\mathbf{G}}$ are independent standard normal variables. The matrix $\hat{\mathbf{G}}$ has full row rank almost surely and

$$\text{Tr}[\hat{\mathbf{\Xi}}_{22}^+ \Sigma_{\sigma,22} (\hat{\mathbf{\Xi}}_{22}^+)^T] = \text{Tr}\left[\bar{\Sigma}_{22}^{-1/2} \Sigma_{\sigma,22} \bar{\Sigma}_{22}^{-1/2} (\hat{\mathbf{G}} \hat{\mathbf{G}}^T)^{-1}\right]. \quad (\text{SM158})$$

The matrix $\hat{\mathbf{G}} \hat{\mathbf{G}}^T$ is Wishart with $n_{\text{tr}} - r$ degrees of freedom and identity scale. Its inverse mean is [\[54\]](#)

$$\mathbb{E}[(\hat{\mathbf{G}} \hat{\mathbf{G}}^T)^{-1}] = \frac{\mathbf{I}_{n_{\text{out}}-r}}{n_{\text{tr}} - n_{\text{out}} - 1}. \quad (\text{SM159})$$

Substitution in [eq. \(SM156\)](#) proves [eq. \(SM151\)](#). Averaging over σ replaces $\Sigma_{\sigma,22}$ by $\bar{\Sigma}_{22}$, whose contraction with its inverse has trace $n_{\text{out}} - r$, and proves [eq. \(SM153\)](#).

For the training variance, the right multiplication by the physical vector $\mathbf{p}_\sigma = U_1 \mathbf{p}_{\sigma,1}$ projects out the singular $(2, 2)$ block. Defining $\mathbf{a} = \mathbf{P}_{11}^{-1} \mathbf{p}_{\sigma,1}$, the $(1, 1)$ and $(2, 1)$ blocks of [eqs. \(SM30\)](#) and [\(SM32\)](#) give

$$\hat{\mathbf{P}}_N^+ \mathbf{p}_\sigma = V_1 \mathbf{a} - V_2 \hat{\mathbf{\Xi}}_{22}^+ \hat{\mathbf{\Xi}}_{21} \mathbf{a} + O(N^{-1/2}). \quad (\text{SM160})$$

Since the two displayed terms belong to the orthogonal subspaces spanned by V_1 and V_2 , respectively, taking the squared norm and averaging gives

$$\mathbb{E}_{\hat{\mathbf{\Xi}}} \|\hat{\mathbf{P}}_N^+ \mathbf{p}_\sigma\|_2^2 = \|\mathbf{a}\|_2^2 + \mathbb{E}_{\hat{\mathbf{\Xi}}} \|\hat{\mathbf{\Xi}}_{22}^+ \hat{\mathbf{\Xi}}_{21} \mathbf{a}\|_2^2 + O(N^{-1/2}). \quad (\text{SM161})$$

The remaining average is already contained in the matrix $\hat{\mathbf{Y}}$ introduced in [eq. \(SM28\)](#). Indeed,

$$\|\hat{\mathbf{\Xi}}_{22}^+ \hat{\mathbf{\Xi}}_{21} \mathbf{a}\|_2^2 = \mathbf{a}^T (\hat{\mathbf{Y}} - \mathbf{I}_r) \mathbf{a}. \quad (\text{SM162})$$

Using the exact Gaussian mean [eq. \(SM137\)](#) in [eq. \(SM162\)](#) gives

$$\mathbb{E}_{\hat{\mathbf{\Xi}}} \|\hat{\mathbf{\Xi}}_{22}^+ \hat{\mathbf{\Xi}}_{21} \mathbf{a}\|_2^2 = \frac{n_{\text{out}} - r}{n_{\text{tr}} - n_{\text{out}} - 1} \|\mathbf{a}\|_2^2. \quad (\text{SM163})$$

Substituting this identity in [eq. \(SM161\)](#) yields

$$\begin{aligned} \mathbb{E}_{\hat{\mathbf{\Xi}}} \|\hat{\mathbf{P}}_N^+ \mathbf{p}_\sigma\|_2^2 &= \left(1 + \frac{n_{\text{out}} - r}{n_{\text{tr}} - n_{\text{out}} - 1}\right) \|\mathbf{a}\|_2^2 + O(N^{-1/2}) \\ &= \frac{n_{\text{tr}} - r - 1}{n_{\text{tr}} - n_{\text{out}} - 1} \|\mathbf{P}_{11}^{-1} \mathbf{p}_{\sigma,1}\|_2^2 + O(N^{-1/2}), \end{aligned} \quad (\text{SM164})$$

which proves [eq. \(SM152\)](#) after multiplication by γ^2 . $\square$

For comparison, if the feature matrix is known exactly, then $\mathbf{P}^+ \mathbf{p}_\sigma = V_1 \mathbf{P}_{11}^{-1} \mathbf{p}_{\sigma,1}$ and target noise adds

$$\Delta \text{Var}_{\text{tr,exact}}^{(\eta)}(\sigma) = \gamma^2 \|\mathbf{P}_{11}^{-1} \mathbf{p}_{\sigma,1}\|_2^2. \quad (\text{SM165})$$

At leading order, the rank-increasing feature noise therefore multiplies the exact-feature answer by $(n_{\text{tr}} - r - 1)/(n_{\text{tr}} - n_{\text{out}} - 1)$. The frame bound in [eq. \(SM138\)](#) gives $\|\mathbf{P}_{11}^{-1} \mathbf{p}_{\sigma,1}\|_2^2 = O(n_{\text{tr}}^{-1})$, so this contribution is $O(\gamma^2/n_{\text{tr}})$ for fixed d and n_{out} . This is to be contrasted with [eq. \(SM153\)](#), whose unnormalized test-variance contribution is $O(N\gamma^2 n_{\text{out}}/(M n_{\text{tr}}))$ when $n_{\text{tr}} \gg n_{\text{out}}$.

SM6. NUMERICAL RESULTS

In this section, we present several numerical simulations supporting the theoretical discussion in the manuscript. All simulations estimate the training bias and variance numerically using a finite number of realizations of the shot noise arising during training or testing. Achieving full agreement between the simulations and the theoretical results therefore requires unbiased estimators of the bias and variance. This subtlety is discussed in detail in [section SM6.1](#).

The numerical simulations serve two distinct purposes. First, we explore a range of settings to demonstrate the broad applicability of the manuscript's theoretical findings. For example, in [section SM6.3](#), we reproduce the predicted scaling of the MSE, bias, and variance for randomly generated POVMs. Second, the simulations in [section SM6.2](#) provide empirical evidence supporting the validity of the Gaussian approximation discussed in the main text.

SM6.1. Unbiased estimators for the training bias and variance

The bias–variance decomposition in [eqs. \(12\) and \(18\)](#) is defined in terms of exact averages over the shot noise. In the simulations, the average over $\hat{\mathbf{p}}_\sigma$ is performed analytically using [eq. \(12\)](#) while the average over the training noise is replaced by a finite sample mean. We derive the resulting estimators at fixed training set, POVM, observable $\mathcal{O}$, and test state σ ; any further average over these objects is performed only afterwards.

For each realization of the training shot-noise, let

$$\hat{y} = \hat{\mathbf{W}}_N^{(r)} \mathbf{p}_\sigma, \quad y = \mathbf{W} \mathbf{p}_\sigma = \langle \mathcal{O}, \sigma \rangle. \quad (\text{SM166})$$

Thus $\hat{y}$ is the infinite-test-statistics prediction obtained from a realization training shot-noise $\hat{\Xi}$, whereas y is the exact target. If $\bar{y}_{\text{tr}} \equiv \mathbb{E}_{\hat{\Xi}}[\hat{y}]$, the quantities defined in [eq. \(18\)](#) may also written as:

$$\text{bias}^2(N, \sigma) = (\bar{y}_{\text{tr}} - y)^2, \quad \text{Var}_{\text{tr}}(N, \sigma) = \mathbb{E}_{\hat{\Xi}}[(\hat{y} - \bar{y}_{\text{tr}})^2]. \quad (\text{SM167})$$

Let $R \geq 2$ be the number of independent training-noise realizations and let introduce the empirical mean and sample variance

$$\hat{y}_R \equiv \frac{1}{R} \sum_{r=1}^R \hat{y}_r, \quad \hat{S}_R^2 \equiv \frac{1}{R-1} \sum_{r=1}^R (\hat{y}_r - \hat{y}_R)^2. \quad (\text{SM168})$$

where $\{\hat{y}_r\}_{r=1}^R$ are R independent realization of $\hat{y}$ over the shot-noise. Using the independence of the R training realization, it is simple to show that:

$$\mathbb{E}[\hat{y}_R] = \bar{y}_{\text{tr}}, \quad \text{Var}[\hat{y}_R] = \frac{\text{Var}_{\text{tr}}(N, \sigma)}{R}, \quad \mathbb{E}[\hat{S}_R^2] = \text{Var}_{\text{tr}}(N, \sigma). \quad (\text{SM169})$$

In particular, using the last identity in [eq. \(SM169\)](#), it is immediate to recognize that the sample variance is actually an unbiased estimator of the true training variance:

$$\widehat{\text{Var}}_{\text{tr}}(N, \sigma) \equiv \hat{S}_R^2 = \frac{1}{R-1} \sum_{r=1}^R (\hat{y}_r - \hat{y}_R)^2 \quad (\text{SM170})$$

is actually an unbiased estimator of the training variance.

On the other hand, the estimator of the square bias is more subtle. A naive guess would be to use $(\hat{y}_R - y)^2$ to estimate the actual bias. However, using again the independence of the noise realizations, it is easy to check that this would not be an unbiased estimator. In fact:

$$\mathbb{E}[(\hat{y}_R - y)^2] = (y - \mathbb{E}[\hat{y}_R])^2 + \text{Var}[\hat{y}_R] = \text{bias}(N, \sigma)^2 + \frac{1}{R} \text{Var}(N, \sigma) \quad (\text{SM171})$$

The term on the left is therefore positively biased at finite R . Unbiased estimator of the squared bias is instead

$$\widehat{\text{bias}}^2(N, \sigma) \equiv (\hat{y}_R - y)^2 - \frac{1}{R} \hat{S}_R^2. \quad (\text{SM172})$$

SM6.2. Numerical tests of the Gaussian approximation

SM6.3. Simulations with random POVMs